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Contents

About this document	7
1 Monitoring network health	9
Network Health view	9
1.1 What is the Network Health view?	9
1.2 How do I get a quick view of my network health?	9
1.3 How do I list all objects for a KPI?	11
1.4 How do I list misaligned services?	11
1.5 How do I check network alarms?	12
1.6 How do I check KPI trending?	12
1.7 How do I cross-launch to another GUI?	13
1.8 How do I set my network monitoring time range?	13
1.9 How do I view a smaller portion of my network?	14
1.10 How do I track unhealthy service levels?	15
1.11 How do I list service-affecting network objects?	16
1.12 How do I set KPI preferences?	17
News Feed	18
1.13 How do I view alarms in the News Feed?	18
1.14 How do I stop/start automatic updates in the News Feed?	18
1.15 How do I change the News Feed sort order?	19
Watchlist	20
1.16 How do I monitor network objects in the Watchlist?	20
1.17 How do I set the trending KPI for NEs in the Watchlist?	21
1.18 How do I stop/start automatic updates in the Watchlist?	21
1.19 How do I adjust the Watchlist contents?	22
Network Map	24
1.20 How do I check network health in the Network Map?	24
1.21 How do I view multiple network layers in the Network Map?	26
1.22 How do I search for an object in the map?	26
1.23 How do I view the NEs in a map region or zone?	26
1.24 How do I view information about an object in the map?	27
1.25 How do I filter the map?	27
1.26 How do I customize my map view?	28
1.27 How do I check link utilization in the map?	30
1.28 How do I highlight links by type in the map?	33

1.29	How do I access cross-domain links in the Network Map?	33
1.30	How do I optimize map performance?	34
1.31	How do I set Utilization map preferences?	34
1.32	How are utilization statistics collected?	35
1.33	What is a Simplified Microwave Router?	37
1.34	How do I open a CLI session with an NE?	38
	Network Search and Inventory	40
1.35	What does Network Search and Inventory show me?	40
1.36	How do I manage advanced filters in Network Search and Inventory?	41
1.37	How do I view statistics plots for inventory objects?	43
1.38	How do I open an inventory object in a different NSP GUI?	44
1.39	How do I configure manual links for the Topology Map?	45
1.40	How do I delete a link from Network Search and Inventory?	46
1.41	How do I modify the Network Search and Inventory view?	47
1.42	How do I manage LSPs in Network Search and Inventory?	48
	Subscriber Monitoring	49
1.43	What is the Subscriber Monitoring view?	49
2	Troubleshooting network objects	51
2.1	What is the Object Troubleshooting dashboard?	51
2.2	How do I troubleshoot a network object?	51
2.3	What can an NE show me in the Troubleshooting Summary dashboard?	52
2.4	Subscriber Troubleshooting view	54
2.5	What can a service show me in the Troubleshooting Summary dashboard?	54
2.6	What can a composite service show me in the Troubleshooting Summary dashboard?	56
2.7	What can a port show me in the Troubleshooting Summary dashboard?	56
2.8	What can a link show me in the Troubleshooting Summary dashboard?	57
2.9	How do I check alarms for a KPI?	58
2.10	How do I view the services contained in a composite service?	58
2.11	How do I open a link in IP Optical Coordination?	59
2.12	What is the Troubleshooting map?	60
2.13	What is the Service Troubleshooting map?	61
2.14	What is the multi-layer map?	62
2.15	How do I view information about an object in the Service Troubleshooting map?	66
2.16	How do I view past events on an object?	67
2.17	How do I set the event time frame?	68
2.18	How do I display comprehensive event data?	69

2.19	How are NSP assurance events retrieved and recorded?	70
2.20	How do I manage assurance event recording?	74
2.21	How do I execute a workflow on an object?	75
2.22	How do I run an OAM test on a service?	76
2.23	How do I view historical OAM test results for a service?	77
2.24	How do I plot performance statistics for an object?	78
3	Network health alarm views	79
3.1	How does the NSP manage alarms?	79
3.2	How do I view a list of alarms in my network?	81
3.3	How do I view root cause distribution in the network?	82
3.4	How do I view network alarms as a chart?	83
3.5	How do I view which NEs have the most alarms?	83
3.6	How do I view which alarms are occurring the most?	84
3.7	Fault management API and tools support	85
4	Managing network alarms	87
	Displaying alarms	87
4.1	How do I configure current alarm list settings?	87
4.2	How do I configure historical alarm list settings?	88
4.3	How do I configure which columns are displayed in an alarm list?	89
4.4	How do I show alarm severity using row color?	89
4.5	How do I pause the current alarm list?	90
4.6	How do I apply a quick alarms filter?	90
4.7	How do I view current alarms of a specific severity?	91
4.8	How do I trigger a sound alert when a new alarm is raised in my current view?	91
4.9	How do I configure a custom sound for audible alerts?	92
4.10	How do I view current alarms based on their root cause analysis status?	93
4.11	How do I create an advanced alarms filter?	94
4.12	How do I delete a saved alarms filter?	95
4.13	How do I combine an advanced filter and a quick filter?	95
4.14	How do I add filters to the watched filters list?	96
4.15	How do I export alarms to a file?	97
	Investigating alarms	99
4.16	How do I find the object affected by an alarm?	99
4.17	How do I list other objects impacted by an alarm?	99
4.18	How do I display the root cause of an alarm?	99
4.19	How do I configure e-mail notifications for alarms?	100

4.20	How do I stop e-mail notifications for alarms?	102
	Managing alarms	104
4.21	How do I configure global alarm settings?	104
4.22	How do I acknowledge an alarm?	105
4.23	How do I delete or clear an alarm?	105
4.24	How do I edit alarm custom text?	106
4.25	How do I automate alarm management using a policy?	106
4.26	How do I suppress all alarms raised on a port, NE, or resource group?	108
4.27	How do I open an SSH or Telnet session with an NE?	109
4.28	How do I manually escalate or de-escalate an alarm?	110
4.29	How do I automate escalating or de-escalating alarms?	111
A	Assurance application evolution	113
A.1	How has assurance changed in NSP?	113
A.2	Managing alarms	113
A.3	Monitoring the network and services	123
A.4	Combining views	129

About this document

Purpose

The *NSP Network and Service Assurance Guide* shows you how to monitor and troubleshoot your network for optimal performance. It introduces the Network Services Platform, or NSP, to technology officers and network operators by describing the tools used for network performance monitoring, including NE and service KPIs, alarm management, OAM testing, performance plots, and map views.

Scope

The NSP Network and Service Assurance Guide information primarily describes elements that are common to all NSP deployments, but may also include high-level information about optional NSP functions that are separately licensed and deployed.

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1 Monitoring network health

Network Health view

1.1 What is the Network Health view?

1.1.1 What do I see in the Network Health view?

The Network Health view provides a dashboard of essential information relating to the proper function of your network. It presents an abbreviated view of equipment and service alarms, root cause alarms, graphical plots of service-affecting network object counts, and network object status.

You can cross-launch from objects in the dashboard to a variety of NSP functions. The function that is launched depends on the object context. For example, you can open the alarm list from an alarm object. Cross-launched functions open in a separate GUI.

Clicking on certain objects in the Network Health view takes you to a different location within the view. For example, clicking on the Unhealthy Services KPI icon in the Service Health dashlet takes you to a detailed list of services with problems in Network Search and Inventory.

The data in all views of the Network Map and Health view is updated every 30 seconds.

1.2 How do I get a quick view of my network health?

1.2.1 Network KPIs

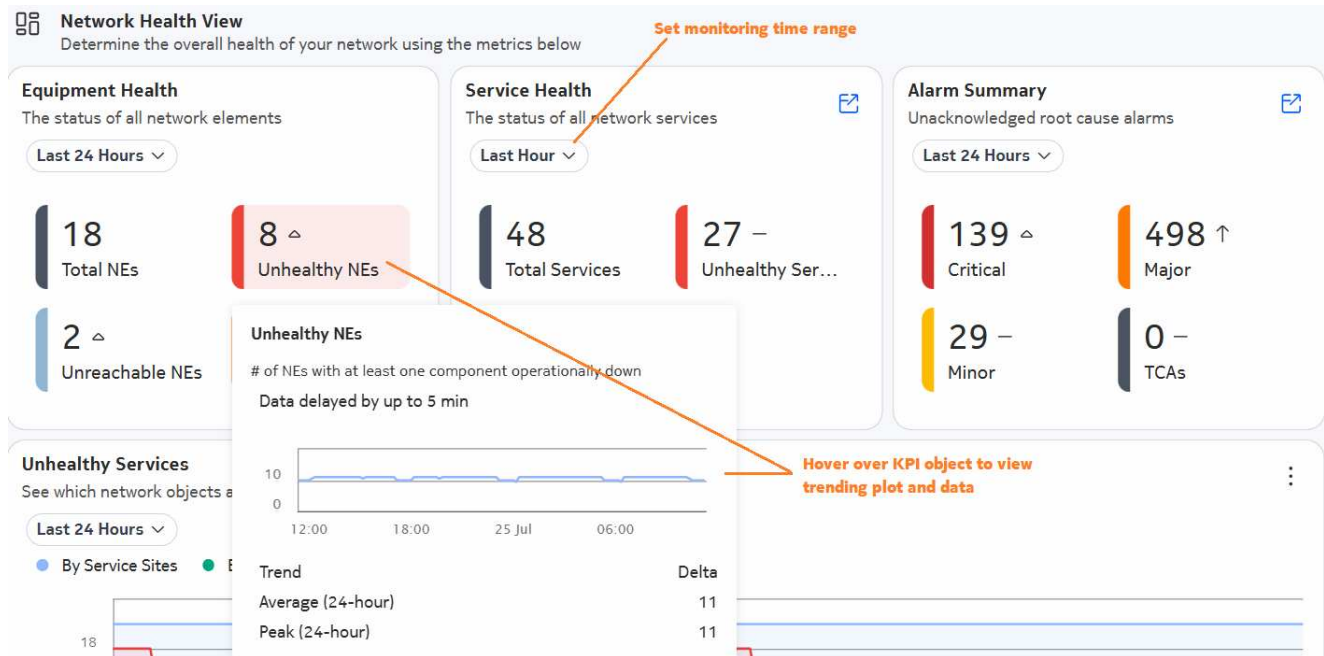
The Network Health view pulls KPI and alarm information from various NSP components to show you the status of your network equipment and services.

A selection of NE and service KPIs are displayed in dashlets:

- The **Total NEs** KPI is a count of the number of NEs in your network.
- The **Total Services** KPI is a count of the number of services configured in your network. Click on this KPI to go to the Services list.
- The **Unhealthy NEs** KPI refers to the number of NEs that have one or more components that are operationally down. Click on this KPI to go to the Network Elements list, filtered by the Affected Objects count.
- The **Unhealthy Services** KPI refers to the number of services that have one or more components that are operationally down. Click on this KPI to go to the Services list, filtered by the Affected Objects count.
- The **Misaligned Services** KPI refers to the number of services which are functioning, but not entirely as intended. Click on this KPI to go to the Services list in Service Management. [1](#)
- The **Misaligned Deployments** KPI refers to the number of NEs which are functioning, but have configuration problems. Click on this KPI to go to the Configured Deployments list in Device Management. [2](#)
- The **Unreachable NEs** KPI refers to the number of NEs whose communication state is set to

Partial or Down. Click on this count to go to the Managed Network Elements list in Device Management, filtered by Reachability set to Unreachable.

Information in the Network Health view is refreshed every 30 seconds.



¹ The Misaligned Services KPI is only displayed if the IBSF component of the serviceActivationAndConfiguration-intentBasedServiceFulfillment feature pack is deployed in NSP.

² The Misaligned Deployments KPI is only displayed if the ICM component of the networkInfrastructureManagement-deviceConfig feature pack is deployed in NSP.

1.2.2 Network monitoring pathway

Use the following dashboard features to expand on your network health investigation:

- **List network objects:** View all Network Elements for a KPI in Network Search and Inventory; see 1.3 "How do I list all objects for a KPI?" (p. 11).
- **List misaligned services:** See 1.4 "How do I list misaligned services?" (p. 11)
- **List alarms:** Investigate alarm KPIs; see 1.5 "How do I check network alarms?" (p. 12).
- **View KPI trending:** View a graphic plot of a KPI; see 1.6 "How do I check KPI trending?" (p. 12).
- **Cross-launch to another GUI:** See 1.7 "How do I cross-launch to another GUI?" (p. 13)

1.2.3 How does user access control affect objects in the Network Map and Health view?

An operator's visibility of network equipment is based on user access control settings, which are configured by an administrator. Depending on the access settings for your user account, some equipment may not be visible. Users must have an access level above "none" to NEs and services in order to see view them in the Network Map and Health view. Configuring user access control settings is described in the *NSP Network Administrator Guide*.

1.3 How do I list all objects for a KPI?

1.3.1 Purpose

You can list all of the network objects associated with a KPI indicator in a dashlet in the Network Health view.

1.3.2 Steps

- 1 _____
Open Network Map and Health, Network Health View.
- 2 _____
In the Equipment Health dashlet, click on a KPI icon.
You are taken to an expanded list of objects related to the KPI in Network Search and Inventory.
- 3 _____
Return to the Network Health view by selecting Overview from the View menu at the top of the page.

END OF STEPS _____

1.4 How do I list misaligned services?

1.4.1 Purpose

You can list all network services whose configuration in NSP is different from what is configured on NEs. The list open in Service Management.

1.4.2 Steps

- 1 _____
Open Network Map and Health, Network Health View.

2

In the Service Configuration Health dashlet, click the **Misaligned Services** KPI icon to list misaligned services.

A list of misaligned services opens in Service Management.

3

Return to the Network Health view by selecting Overview from the View menu at the top of the page.

END OF STEPS

1.5 How do I check network alarms?

1.5.1 Purpose

You can cross-launch from the Alarm Summary dashlet to the Alarm List.

1.5.2 Steps

You can list network root cause alarms, filtered by severity.

1

Open Network Map and Health, Network Health View.

2

In the Alarm Summary dashlet, click an alarm KPI icon.

The alarm list opens, filtered by the KPI you clicked.

END OF STEPS

1.6 How do I check KPI trending?

1.6.1 Purpose

You can check the Trend value for KPI behavior from the present time, looking back over the period specified in the Time Range drop-down list.

1.6.2 Steps

1

Open Network Map and Health, Network Health View.

2

In the Equipment Health, Service Health, or Alarm Summary dashlet, hover over a KPI icon.

A graphic plot displays the KPI over the specified time range, along with KPI Trend, Average, and Peak values.

The Trend value can be:

- Increasing: more objects/resources have been affected over time
- Decreasing: fewer objects/resources have been affected over time
- Delta: the number of affected objects/resources has fluctuated over time, but is currently the same as the initial value (at the beginning of the time range)

Because the plot is meant to display average KPI values, the Peak value may not always appear.

END OF STEPS

1.7 How do I cross-launch to another GUI?

1.7.1 Purpose

Some Network Health dashlets include a cross-launch link to open the dashlet information in the NSP GUI it is sourced from.

1.7.2 Steps

1

Open Network Map and Health, Network Health View.

2

In a dashlet, click the **View in...** cross-launch link.

The dashlet data is displayed in expanded format in an external NSP GUI.

END OF STEPS

1.8 How do I set my network monitoring time range?

1.8.1 Purpose

You can specify the time range over which each Network Health dashlet gathers network data.

1.8.2 Steps

1

Open Network Map and Health, Network Health View.

2

Click on a dashlet  **Time Range** filter and select a time range from the drop-down list.

END OF STEPS

1.9 How do I view a smaller portion of my network?

1.9.1 Purpose

By default, the Network Health View displays data for your entire network. You can reduce the scope of network data displayed in the Network Health View by selecting a smaller group of NEs, based on a map region or an NE resource group. The Scope feature supports up to 10 users at a time. When a user Scopes their Network Health View, the Scope feature remains active for all NSP sessions logged in as that user. Consider clearing the scope feature when you have finished with it, or before you log out.

Map regions and NE resource groups are configured by an administrator under Map Layouts and Groups.



Note: The Network Map View option must be set to Operational to access the Scope feature.

1.9.2 Steps

1

Open Network Map and Health, Network Health View.

2

Click **Change Scope** at the top of the dashboard.

3

In the Change Scope form, disable the **Scope dashboard content to the entire network** check box to reduce the scope of the dashboard.

4

From the **Group Type** menu, select Region or NE Group.

5

Select a region or NE group from the **Groups** list.

The Groups list displays a maximum of 50 groups or regions. If you don't see the item you are looking for, you can filter the list to reduce its contents.

6

Click **Apply**. The dashboard displays data for the selected region or group. It may take up to 30 seconds for the dashlets to fully update.

END OF STEPS

1.10 How do I track unhealthy service levels?

1.10.1 Unhealthy Services plot

The Unhealthy Services dashlet tells you which network objects are affecting the function of your services. Service sites, service endpoints, and tunnel bindings are plotted separately against the number of services they are affecting over the specified time range.

You can display or hide plots by clicking on the **By Service Sites**, **By Service Endpoints**, or **By Tunnel Bindings** options to display or hide their plots on the graph.

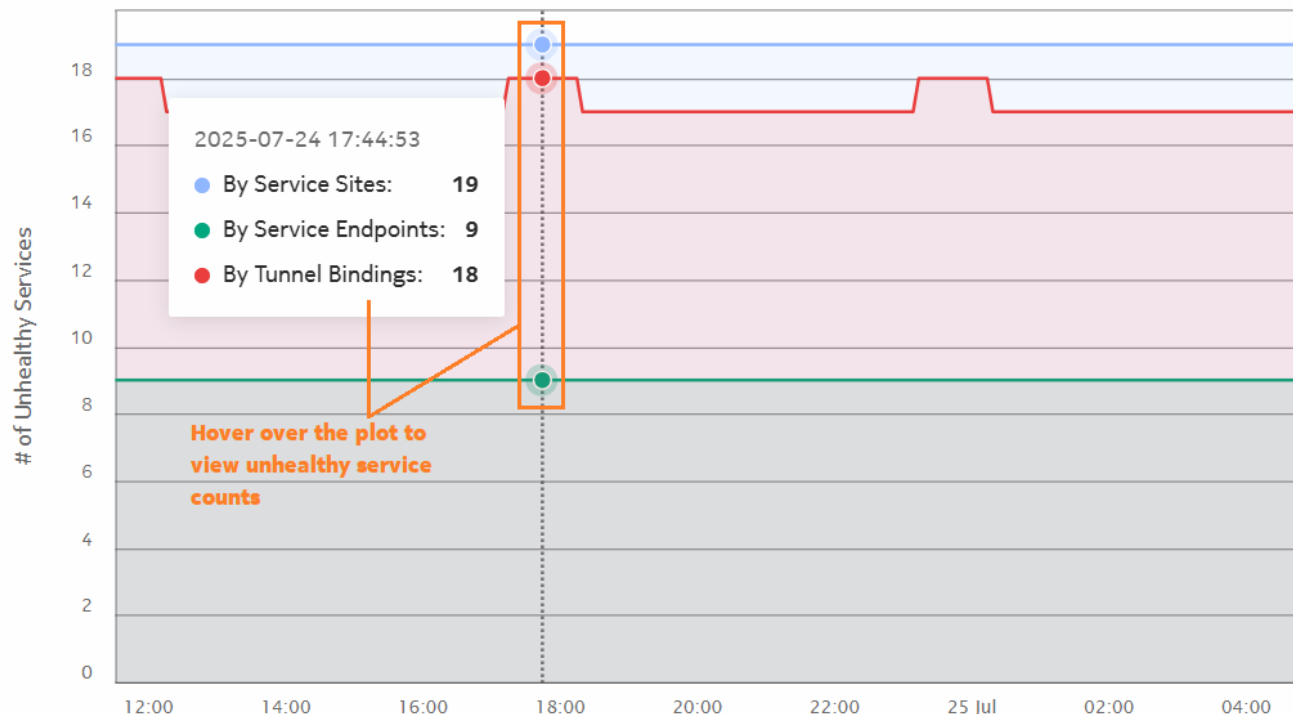
Unhealthy Services

See which network objects are affecting services

Last 24 Hours ▾

Click on these KPIs to display or hide their plots

● By Service Sites ● By Service Endpoints ● By Tunnel Bindings



1.10.2 To access the Unhealthy Services dashlet

1

Open Network Map and Health, Network Health View.

The Unhealthy Services dashlet appears at the bottom of the Network Health view.

END OF STEPS

1.10.3 Network monitoring pathway

Use the following dashlet features to expand on your network health investigation:

- **Scan unhealthy service counts:** Hover over the plot to view the unhealthy service count by network object at a given time point. The unhealthyservice counts update as you move the cursor to the left or right along the plot.

Because of the scale of the unhealthy service plots, small fluctuations in unhealthy service counts may not be visible in the plots.

- **List service-affecting objects:** view network objects that are affecting your services; see [1.11 "How do I list service-affecting network objects?" \(p. 15\)](#)

1.11 How do I list service-affecting network objects?

1.11.1 Steps

1

Open Network Map and Health, Network Health View.

2

In the Unhealthy Services dashlet, click **More, Show [Service Sites | Service Endpoints | Tunnel Bindings] Affecting Services**.

A list of the selected object type opens in Network Search and Inventory.

3

Return to the Network Health view by selecting Overview from the View menu at the top of the page.

END OF STEPS

1.12 How do I set KPI preferences?

1.12.1 Purpose

Use this procedure to configure the behaviour of KPI indicators in the Network Health view.



Note: If you have upgraded from NSP release 23.8 or earlier, your user preferences, such as list column selections, widths, ordering, matrix view settings, and KPI preferences will have been reset to defaults and you will need to reconfigure them. This action is only required once.

1.12.2 Steps

- 1 _____
Open Network Map and Health.
- 2 _____
Click **More, Settings**.
- 3 _____
In the Fault Management and Assurance Settings form, click **KPI Preferences**.
- 4 _____
Select a **Trend Time** from the drop-down menu.
The Trend Time is the time span over which KPIs are monitored to determine if they are trending upwards, trending downwards, or have remained unchanged.
- 5 _____
Under KPI Threshold (%), set the **Healthy**, **Low**, and **Medium** threshold percentages.
The threshold percentages specify the point at which a KPI indicator changes colour as a result of KPI change.
- 6 _____
Click **Save** to apply your changes.

END OF STEPS _____

News Feed

1.13 How do I view alarms in the News Feed?

1.13.1 Recent network events

The News Feed provides a live feed of unacknowledged root cause network alarms, as they occur in real time. Alarm severity levels of Warning, Minor, Major and Critical are displayed.

1.13.2 To access the News Feed

1

Open Network Map and Health, Network Health View.

The News Feed appears on the right-hand side of the Network Health view.

END OF STEPS

1.13.3 Troubleshooting pathway

You can cross-launch from an alarm object in the News Feed to an alternate NSP view to see more information about the alarm or the network object it originated from. Click **More** on an alarm and select a cross-launch option. Cross-launch view availability varies depending on originating object for the alarm.

1.14 How do I stop/start automatic updates in the News Feed?

1.14.1 Purpose

The News Feed is automatically updated every 30 seconds. You can switch to manual updates.

1.14.2 Steps

1

Open Network Map and Health, Network Health View.

2

In the News Feed dashlet, click **More, Set To Manual Refresh** to stop automatic News Feed updates. A manual update button is added to the News Feed that you can click as needed.

When in manual update mode, you can click **More, Set To Auto Refresh** to return to automatic updates.

END OF STEPS

1.15 How do I change the News Feed sort order?

1.15.1 Steps

- 1

Open Network Map and Health, Network Health View.
- 2

In the News Feed dashlet, click on the **Sort** filter and select a sorting rule.
The News Feed is reordered according to the new sorting rule.

END OF STEPS

Watchlist

1.16 How do I monitor network objects in the Watchlist?

1.16.1 Purpose

The Watchlist opens from the Network Map and Health dashboard or Object Troubleshooting dashboard to a separate browser tab. It allows you to monitor a shortlist of NEs and services that you want to keep under scrutiny and access easily. Objects in the Watchlist display basic KPIs, and the Trending indicator tells you if an object's condition is changing.

You add network objects to the Watchlist from the Topology map view and from object lists in Network Search and Inventory.

You can also add NEs to the Watchlist from:

- Current Alarms, Unhealthy NEs view
- Top Problems view, alarm NE Matrix tile
- Unhealthy NE tile

Information in the Watchlist is refreshed every 30 seconds.



Note: If you have upgraded from NSP release 23.8 or earlier, NEs and services in user watch lists are not carried over into the Watchlist and you will need to re-add them. This action is only required once.

1.16.2 Object status trending

An object in the Watchlist displays a Trending indicator that tells you if the object is stable or affected by problems:

- An Upward trending arrow means the object is impacted by new issues.
- A Downward trending arrow means the number of issues impacting the object is diminishing.
- A Flat trending indicator means the object's condition has remained unchanged.
- A Change indicator means the current value is the same as the initial value, but it has fluctuated during the trending period (configured in the Network Map and Health view settings).

1.16.3 To open the Watchlist

1

Open Network Map and Health, Network Health View.

2

On the toolbar, click **Watchlist**.

3

The Watchlist opens in a separate browser tab, displaying network objects set for monitoring.

END OF STEPS

1.16.4 Troubleshooting pathway

- From an unhealthy object in the Watchlist click **More, View in Object Troubleshooting**.
The Object Troubleshooting dashboard to displays complete status information for the object.

1.17 How do I set the trending KPI for NEs in the Watchlist?

1.17.1 Purpose

You can specify which type of KPI is indicated by the Trending indicator for NE objects in the Watchlist.

When the Watchlist is in manual update mode, the NE trending indicator value is not shown after changing trending the KPI for an NE. You must click the manual update button to see the NE trending indicator value.

1.17.2 Steps

1

Open Network Map and Health, Network Health View.

2

On the toolbar, click  **Watchlist**.

3

On the toolbar, click  **Watchlist Settings**.

4

In the Watchlist Settings form, select an NE KPI from the drop-down list.

5

Click **Save**.

END OF STEPS

1.18 How do I stop/start automatic updates in the Watchlist?

1.18.1 Purpose

The Watchlist is refreshed every 30 seconds. You can switch to manual updates.

When the Watchlist is in manual update mode, the NE trending indicator value is not shown after changing trending the KPI for an NE. You must click the manual update button to see the NE trending indicator value.

1.18.2 Steps

- 1 _____
Open Network Map and Health, Network Health View.
- 2 _____
On the toolbar, click  **Watchlist**.
- 3 _____
In the Watchlist, click **List Actions** , **Set To Manual Refresh** to stop automatic Watchlist updates. A manual update button is added to the Watchlist that you can click as needed.
When in manual update mode, you can click **List Actions** , **Set To Auto Refresh** to return to automatic updates.

END OF STEPS

1.19 How do I adjust the Watchlist contents?

1.19.1 Purpose

You can filter the Watchlist to display only certain object types, and you can change the sort order of the list.

Different sort options are available, depending on the filter selection; for example:

- services and NEs can be sorted by object type or name
- NEs can be sorted by chassis type, name, NE KPI, or system address
- services can be sorted by service type, name, or Unhealthy service state

1.19.2 Steps

- 1 _____
Open Network Map and Health, Network Health View.
- 2 _____
On the toolbar, click  **Watchlist**.
- 3 _____
Do either of the following:
 - Click the **Filter** chip and select **Service**, **NE**, or **Service and NE** from the drop-down list.

The list contents adjust to show only the object type(s) selected.

- Click the **Sort** chip and select **Object Type** or **Name** from the drop-down list.
The list sort order changes to follow the selected criterion.

END OF STEPS

Network Map

1.20 How do I check network health in the Network Map?

1.20.1 Map view

The Network Map is a graphical display of your network equipment and its interconnections. If the view is configured with a background map layer, equipment can be positioned on the map, based on its actual physical location. NEs are grouped into geographical regions and zones. The map can be zoomed out to view equipment from a broad perspective, or zoomed in to view only a handful of NEs.



Note: The Network Map does not support LAG link members, links to unmanaged endpoints, or LLDP links with endpoints with the destination MAC address set to Nearest Customer or Nearest Non TPMR.

The background map layer is configured by an NSP administrator in the NSP system settings. Administrative users can view subnets and links to subnets as objects on the Network Map. Non-administrative users whose access rights are defined through UAC cannot view subnets and links to subnets because the Network Map is intended to display networking equipment. Subnets are not actual equipment.

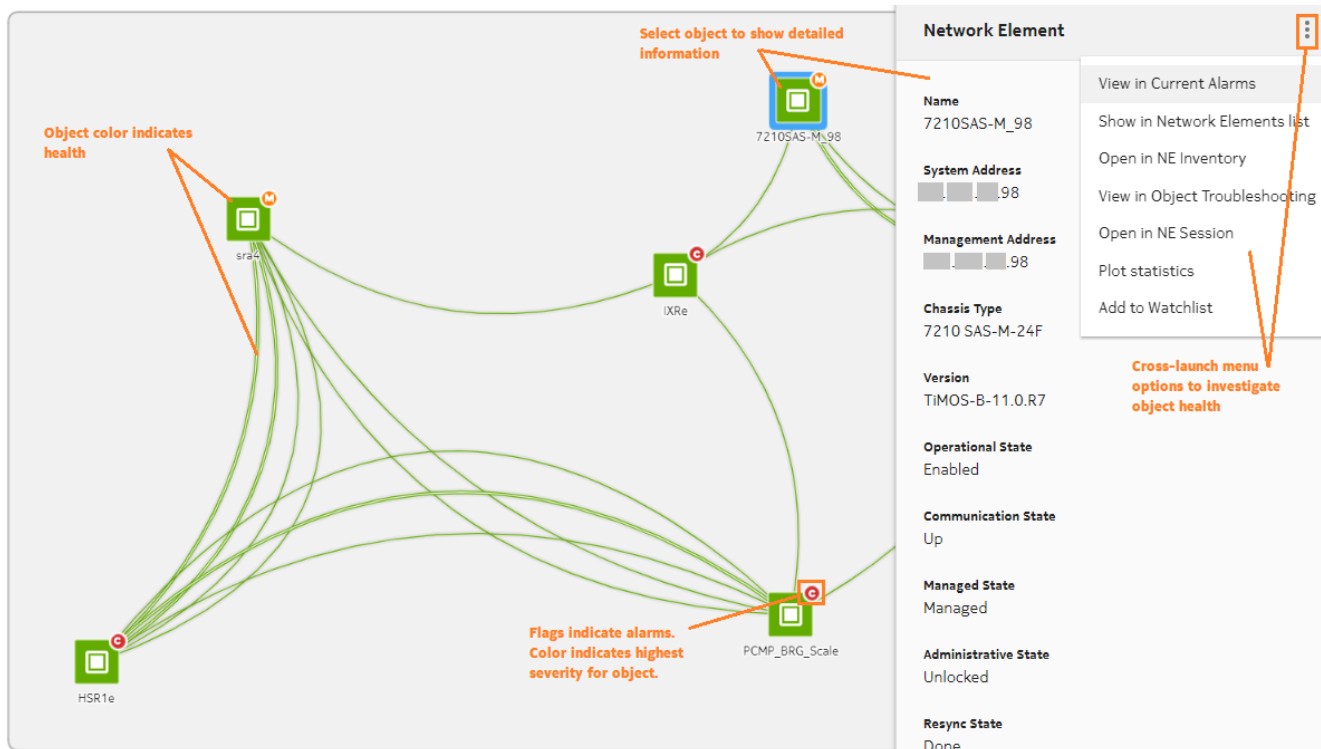
The Network Map does not support LLDP links with endpoints with the destination MAC address set to Nearest Customer.

Alarm and status information in the Network Map is automatically refreshed from the network every 30 seconds. You can update the map display manually by clicking  **Refresh**. The Refresh command can take significant time to complete on large networks. Use it only if there have been changes to the NSP common map layout (for example, recently-added NEs, links) and you want them to appear in the Network Map view, or if the map data is stale (in which case you are prompted to refresh the map).



Note: At initial startup, KPIs are calculated while objects are being loaded. It may take a few hours before final counts are available.

If a region titled NEs Without a Region appears on the map, it is an auto-created region containing NEs that exist in the Network but have not been grouped under a specific region. Your administrator must place the NEs in a region.



1.20.2 Map object clustering for small networks

In small NSP deployments of fewer than 2000 NEs and 3000 links, where no common map layout is configured, map objects can be clustered automatically based on their proximity, if the Cluster by Proximity option is enabled in the Map Palette. If the object limits are exceeded, The Network Map displays an error message, indicating that the administrator must configure a common map layout, dividing NEs into regions.

1.20.3 Common map layout

The Network Map layout can be derived from the NSP common map layout, as configured by an NSP administrator under Map Layouts and Groups. With this arrangement, map objects are clustered into regions and zones, as configured by the administrator.

1.20.4 Network monitoring pathway

Use the following dashlet features to expand on your network health investigation:

- Search for an object in the map; see [1.22 “How do I search for an object in the map?”](#) (p. 26).
- Examine an NE in an external GUI; see [1.24 “How do I view information about an object in the map?”](#) (p. 27).
- View network layers; see [1.21 “How do I view multiple network layers in the Network Map?”](#) (p. 26).

1.21 How do I view multiple network layers in the Network Map?

1.21.1 Multi-layer map

The Multi-layer map shows the relationships between equipment objects in various layers of the network; for example, the physical layer and the IGP layer. The IGP layer is displayed for physical NEs contained in a single resource group where the corresponding IP links and routers can span over multiple administrative domains. You can see how problems in one layer may be affecting, or affected by, other layers.

The Multi-layer map shares similar functionality (object information, view options) with the Topology map.

To access the Multi-layer map:

1. Open Network Map and Health, Network Map View.
2. Select **Multi-layer** from the **Type** drop-down list.

1.22 How do I search for an object in the map?

1.22.1 Steps

- 1 _____
Open Network Map and Health, Network Map View.
- 2 _____
Click **Find in Map** . Select a search option from the drop-down list and type a search string. You can click on an entry in the results list to re-center the map on the object's location.

END OF STEPS _____

1.23 How do I view the NEs in a map region or zone?

1.23.1 Steps

- 1 _____
Open Network Map and Health, Network Map View.
- 2 _____
Double-click on a region or zone to expand it and view its NEs.
To return to the top-level map, click on the left-most item in the map breadcrumb.

END OF STEPS _____

1.24 How do I view information about an object in the map?

1.24.1 Purpose

You can view information about NEs or links in the map, either in short-form in a tooltip, in the info panel, or in detail in separate NSP GUI.

1.24.2 Steps

1

Open Network Map and Health, Network Map View.

2

To display basic identification and status information for a map object, hover over the object. The information appears in a tooltip.

3

To view detailed information about an object in an external NSP view, right-click on the object and select one of the following options (options vary, according to object type):

Cross-launch options for NEs

- **View In Current Alarms** opens the Current Alarms list for the NE.
- **Show In Network Elements list** displays the NE in Network Search and Inventory, Network Element list.
- **Open In NE Inventory** displays an NE in the Device Management, NE Inventory on a separate browser tab, along with all configured objects.
- **View in Object Troubleshooting** displays the NE in the Object Troubleshooting dashboard with detailed performance and alarm information.
- **Open in NE Session** opens a Telnet session with the NE.
- **Plot Statistics** displays KPI plots for the NE in the Data Collection and Analysis view.
- **Add to Watchlist** adds the NE to the Watchlist view for monitoring.

Cross-launch options for links

- **View In Current Alarms** opens the Current Alarms list for the link.
- **Show in Link List** displays the link in the Network Search and Inventory, Link list.

END OF STEPS

1.25 How do I filter the map?

1.25.1 Purpose

You can configure filters in the map to control the display of information and reduce clutter. The Topology map allows filtering, with or without the Utilization option. You can add up to five filters. If

filters of the same type are added, they are linked by boolean OR operator. Dissimilar links are joined by a boolean AND operator.

1.25.2 Steps

1

In the map, click  **Add Filter** and select one of the filter types. A chip filter  is added to the map.

In the Topology map, the filter types are Alarm Severity, Chassis Type, Network Type, Product Type, and Status. You can add up to three filters. If the Utilization option is enabled, the filter types are Utilization and Capacity.

2

Click **Filter** and select a criterion related to the filter type.

The chip filter is applied to the map.

You can repeat this step to add up to five filters.

3

Click  **Close** on a chip filter to remove it from the map.

END OF STEPS

1.26 How do I customize my map view?

1.26.1 Purpose

You can set your own zoom level and adjust the behaviour and appearance of the map and objects using the Map Palette controls. You can also save your personal map layout settings so that they are retained in your future NSP sessions, or restore the map to the default common layout.



Note: If you have upgraded from NSP release 23.8 or earlier, your Map Palette settings will have been reset to defaults and you will need to reconfigure them. This action is only required once.

1.26.2 Steps

Use the Map Palette controls on the map palette to adjust the map layout. When objects share the same physical location, the map shows a multi-layered icon shaded in blue. To see the co-located objects individually, drag them off of the multi-layered icon.

1

Open Network Map and Health, Network Map View.

2

Adjust the Map Palette controls, as described in the table below.

Table 1-1 Map Palette controls

 Fit to Screen	Click this option to zoom the map to fit the selected region to available screen area.
 Clustering controls	Map cluster display options for region-based map (available when a common map layout is configured in NSP): - Options to display NE cluster health as a pie chart or a solid circle on the cluster. Object color indicates health. - Display or hide region and zone boundaries. Click Show More to access the following options: - Option to move all contained objects when moving a region or zone. - Group NEs external to a region or zone with their immediate parent zone or region; the map displays all connectors to zones or subzones that contain the external NEs. This option shows greater detail. - Group external NEs with their top-level region; the map displays a single connector to the region icon. This option shows less detail. Map cluster display options for cluster-based map (available in network of fewer than 2000 NEs and 3000 links, where no common map layout is configured in NSP): - Option to arrange NE into clusters, based on proximity - Options to display NE cluster health as a pie chart or a solid circle on the cluster. Object color indicates health. Click Show More to access the following options: - Option to adjust cluster inclusion range. - Option to adjust the cluster creation threshold for the entire network. - Option to adjust the cluster creation threshold for what is visible on the screen.
 Adjust vertices	Adjust vertices as follows: - Show/hide text labels for map objects. - Adjust icon size for NEs, zones, and regions.

Table 1-1 Map Palette controls (continued)

 Adjust Links	Adjust link display as follows: • Show or hide links between NEs, zones, and regions. • Show or hide links when the objects they connect to are outside the map view. • Adjust link curvature (i.e., how deep of an arc) between objects. • Adjust link grouping threshold. Click Show More to access the following options: • Options to display link group health as a pie chart or a solid circle on the group. Object color indicates health. • Show or hide the number of links in a group.
 Map View	Turn on Bird's-eye View (shows the entire map in a small inset in the corner). Adjust the opacity of the background map.
   Zoom	Zoom into and out from the map.

END OF STEPS

1.26.3 Steps

Save your personalized map layout or restore it to the default layout.

1

Click **More** , **Save As My Layout** to save the map display settings as they are currently configured.

If you want to return to the default map display settings, click **More** , **Restore To Default Layout**.

END OF STEPS

1.27 How do I check link utilization in the map?

1.27.1 Link utilization

The Network Map has a Utilization view option that shows how much available capacity is being utilized on network links. The Utilization view lets you quickly assess how efficiently your network is managing traffic, and identify links that are over- or under-utilized. Utilization is displayed as colored arrows on link objects. Network utilization can be analyzed in the Network Map for all links in a link group.

The Utilization view option supports a limit of five active users at any one time. If more than five users are accessing the Utilization view, You will see a notification.

Information in the Utilization view is based on port egress statistics collected for IP links on Ethernet physical ports. Statistics must be supported and available for utilization to be displayed; see [1.32 “How are utilization statistics collected?” \(p. 35\)](#).

The Utilization view option displays usage data for the following link types:

- Point-to-Point (IP / IGP / CUPS)
- LAG
- Cross-domain
- Radio microwave
- Optical

The Utilization view option supports physical map layout and region-based clustering, but region and zone icons cannot be opened in the Utilization view.

The Utilization Map shows NEs connected by link lines that represent physical connections between endpoints. When you hover over a link, its utilization level is displayed in the object tooltip. The links have the following features:

- **Thickness.** The relative capacity on the link is indicated by a thin, medium, or thick line. Thinner lines indicate lower capacity, thicker lines indicate higher capacity. Link capacity is based on the operational port speed configured for the port.
- **Color.** Physical links between endpoints are shown in grey. Utilization is shown as a green, orange, or red arrow along the grey line. Each color indicates a range of utilization: low, medium, or high. The colors change as utilization (in percent) crosses preset thresholds.
- **Arrow length.** The length of the colored arrow shows the relative utilization of the capacity on the link. Arrows grow from minimal utilization at an endpoint, to 100% utilization at the mid-point crossbar (for bidirectional links). The crossbar represents 100% utilization from either direction. Utilization rates near zero will show a disproportionately long arrow (it may look like about 5%) to provide a visual cue that there is utilization on the link.

A grey line with no colored arrow means either zero utilization, or there is no data available for that link.

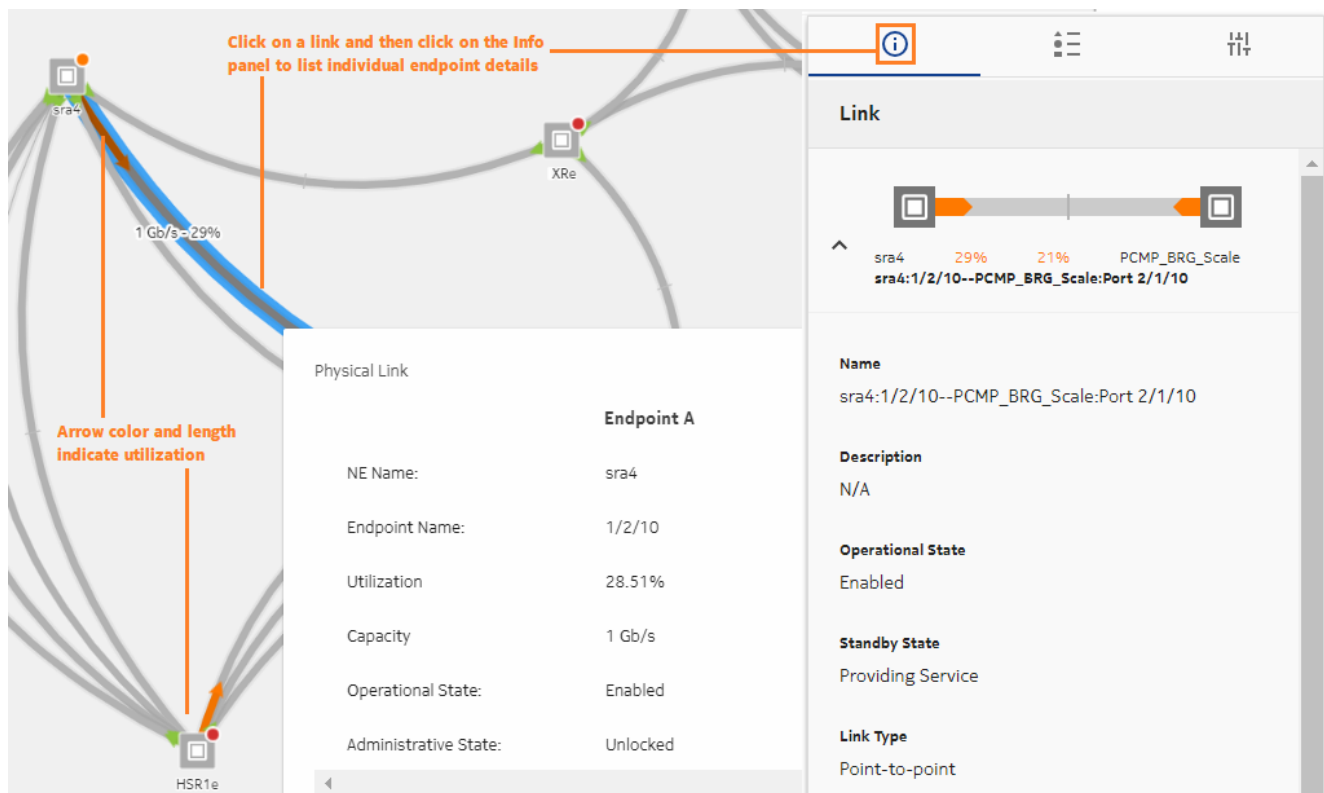
If utilization statistics are not supported on an NE, traffic may be present, but no utilization arrow is displayed.

1.27.2 To display link utilization

1. Open Network Map and Health, Network Map View.
2. Double-click a region or zone object to display the links you want to monitor.
3. Select **Utilization**, from the **View** drop-down menu.
Utilization data is displayed on the links.
4. Hover over a link to display its utilization information in brief, as a tooltip.

1.27.3 Detailed link information

You can click on a link and display detailed information relating to links and individual endpoints on the Info panel.



1.27.4 Managing link utilization map performance

The size of the resource group may affect performance. Consider the following:

- If the number of links in the resource group is large, there may be a delay before the **Utilization** view option if fully enabled.

To maintain system performance and to avoid exhausting available statistics counters, consider creating groups of no more than 50 NEs for the purpose of link utilization monitoring. Your system administrator can create and modify resource groups.

- Statistics are collected by subscription from qualified ports. If there are too many qualified ports in the resource group, performance may be affected. For information about utilization map scale limits, see the *NSP Planning Guide*.
- Be aware that utilization map performance can also vary based on the number of subscriptions and on other telemetry gathering in the NSP system.

1.28 How do I highlight links by type in the map?

1.28.1 Purpose

The Network Map provides an option to highlight selected link types, using colors to identify the type of link. View options are available in the Topology view of the map. You can highlight the following link types:

- Copper: Ethernet link using coaxial copper cable
- Fiber: Ethernet SFP link using optical fiber cable
- LAG N+0: LAG link without protected ports
- LAG N+N: LAG link with member ports protected (supported for Wavence NEs only)
- Protected: protected radio link (supported for Wavence NEs only)
- Unprotected: unprotected radio link (supported for Wavence NEs only)

1.28.2 Steps

- 1 _____
Open Network Map and Health, Network Map View.
- 2 _____
Click  **Highlight** to open the Link Highlight Options panel.
- 3 _____
Click on the link types in the list to  **Enable** or  **Disable** highlighting for each. The links on the map show colors indicating the link type.

END OF STEPS _____

1.29 How do I access cross-domain links in the Network Map?

1.29.1 Purpose

Cross-domain links between IP and optical equipment are shown on topology maps as dashed lines, and on multi-layer maps as solid lines. They are also included in the Network Search and Inventory Links list.

You can access a list of optical services that terminate on the optical endpoint of a cross-domain link. The list shows information about those optical services, and provides additional options.

1.29.2 Steps

1

To open a list of optical services in the map:

1. Open Network Map and Health, Network Map View.
2. Select a cross-domain link on the map.
3. Open the  Info panel.
4. In the Info panel, click  **More, Show Optical Services**. A list of optical services that terminate on the optical endpoint of the cross-domain link is displayed.

2

To open a list of optical services from the Links dashlet:

1. Open Network Map and Health, Network Search and Inventory.
2. Expand the Links list and select a cross-domain link in the list.
3. Click  **Table Row Actions , Show Optical Services**. A list of optical services that terminate on the optical endpoint of the cross-domain link is displayed.

END OF STEPS

1.30 How do I optimize map performance?

1.30.1 Map object limits

Consider the following performance information when working in maps.

Nokia recommends a maximum of 2000 NEs per region for the Topology map. The Multi-layer map is limited to a maximum of 4000 objects for the entire network.

Users should expect the following Multi-layer map loading times with different numbers of NEs:

- for 250 NEs (125 physical links); approximately six seconds for the initial page loading and four seconds to reload
- for 500 NEs (250 physical links); approximately nine seconds for the initial page loading and six seconds to reload
- for 2000 NEs (1000 physical links); approximately 50 seconds for the initial page loading and 28 seconds to reload

1.31 How do I set Utilization map preferences?

1.31.1 Purpose

Use this procedure to configure the automatic refresh rate of the Utilization map, and to set threshold values for KPI indicator color changes.

1.31.2 Steps

- 1 _____
Open Network Map and Health.
- 2 _____
Click  **More, Settings**.
- 3 _____
In the Fault Management and Assurance Settings form, click **Utilization Map Preferences**.
- 4 _____
Configure the **Refresh Rate** parameter, which specifies the interval of automatic refreshes of the map.
- 5 _____
Under Port Speed Settings, set the **Low** and **Medium** port speed thresholds in Mbps.
The threshold percentages specify the point at which a KPI indicator changes colour as a result of KPI change.
- 6 _____
Under Color Settings for Utilization Ranges, set the **Low** and **Medium** utilization threshold percentages.
The threshold percentages specify the point at which a KPI indicator changes colour as a result of KPI change.
- 7 _____
Click **Save** to apply your changes.

END OF STEPS

1.32 How are utilization statistics collected?

1.32.1 Map statistics collection

Information in the Utilization map is based on port egress statistics collected for IP links on Ethernet physical ports. Statistics must be supported and available for utilization to be displayed. The Operational State of NEs and ports must be Up.

Utilization statistics are collected from NEs using SNMP for NFMP NEs and gRPC for MDM NEs.

When a user switches from the Operational Map to the Utilization Map, the first two telemetry subscriptions listed below are created. If LAG links are present on the map, the third subscription listed below is created.

1. Subscription for SNMP and MDM managed NEs and ports on the current map.

SNMP-managed 7750 SR family NEs respond to the filter.

The following MDM-managed NEs respond to the filter: 7750 SR, 7250 IXR / VSR(I), 7450 ESS, and 7950 XRS

This subscription determines utilization directly from the output-utilization statistic counter.

2. Subscription for SNMP managed 7705 SAR and 7210 SAS NEs and ports on the current map.

This subscription calculates utilization from the following statistics: operational-speed, transmitted-broadcast-packets-periodic, transmitted-multicast-packets-periodic, transmitted-octets-periodic, and transmitted-unicast-packets-periodic.

SNMP utilization statistics are calculated from NFM-P counters using this formula:

```
output-utilization(%) = (transmittedTotalOctetsPeriodic +  
( (transmitted-unicast-packets-periodic +  
transmitted-multicast-packets-periodic +  
transmitted-broadcast-packets-periodic) * 20)) *  
8/t/operational-speed/1000 * 100
```

Where *t* is the collection interval in seconds, as specified in the Utilization Map preferences.

3. Subscription for SNMP-managed NEs and LAG links on the current map.

This subscription calculates utilization from the following statistics: speed, transmitted-broadcast-packets-periodic, transmitted-multicast-packets-periodic, transmitted-octets-periodic, and transmitted-unicast-packets-periodic.

SNMP utilization statistics are calculated from NFM-P counters using this formula:

```
output-utilization = (transmitted-octets-periodic +  
( (transmitted-unicast-packets-periodic +  
transmitted-multicast-packets-periodic +  
transmitted-broadcast-packets-periodic) * 20)) *  
8/t/operational-speed/1000 * 100
```

Where *t* is the collection interval in seconds, as specified in the Utilization Map preferences.

For SNMP, NEs must be managed by the NFM-P and reachable using SNMP. The following are used for utilization data:

- MIB name: TIMETRA-PORT-MIB
- MIB entry name: tmnxPortEtherEntry
- MIB counter name: tmnxPortEtherUtilStatsOutput
- Statistics group: Additional Ethernet Stats
- Counter: utilStatsOutput (in centi-percent)

The following NEs support the required SNMP statistics for the Utilization map:

- 7250 IXR
- 7450 ESS
- 7750 SR and VSR

-
- 7950 XRS

For MDM, the following gRPC schema path is used for utilization data:

- /state/port[port-id]/ethernet/statistics/out-utilization

SR OS NEs support the required MDM statistics for the Utilization map. The supporting chassis types are:

- 7250 IXR
- 7450 ESS
- 7750 SR
- 7950 XRS

1.32.2 Utilization statistics for the 7705 SAR and 7210 SAS

The following (periodic) counters are used for utilization calculation for the 7705 SAR and 7210 SAS:

- MIB name: IF-MIB
- MIB entry name: ifXEntry
- MIB counter names: ifHCOutOctets, ifHCOutUcastPkts, ifHCOutMulticastPkts, ifHCOutBroadcastPkts
- NFMP statistics group: Interface Additional Statistics
- NFMP counters: transmittedBroadcastPackets, transmittedMulticastPackets, transmittedTotalOctets, transmittedUnicastPackets

The following information is used to establish a reference speed:

- MIB name: TIMETRA-PORT-MIB
- MIB entry name: tmnxPortEtherEntry
- MIB counter name: tmnxPortEtherOperSpeed (mbps)
- NFMP class: equipment.PhysicalPort
- NFMP counter: actualSpeed (kbps)

1.33 What is a Simplified Microwave Router?

1.33.1 Simplified microwave router

In networks where multiple Wavence UBT-SA devices are linked to a single 7250 IXR or 7705 SAR NE, the NSP provides a Simplified Microwave Router (SMR). To facilitate network monitoring, the SMR shows the router and its linked UBTs, including CA (Carrier Aggregated) UBTs, as a single logical site with the following display features:

- The Network Map shows the NE and its linked UBT-SAs as a single router NE; the UBT-SAs are not displayed.
- KPIs, and current and historical alarms on UBT-SAs are propagated to the linked router.

- The NE Inventory shows the UBT-SAs as child objects of the router, under Radio Equipment.
- You can cross-launch to the external EMS for UBT-SA devices by right-clicking on their object in the NE Inventory.
- You can search for a UBT-SA object using the object name or IP address.

1.34 How do I open a CLI session with an NE?

1.34.1 Purpose

You can open an NE session from menus in the following Network Map and Health dashboard locations:

- the information panel for a selected NE on the Network Map
- NEs in Network Search and Inventory
- NEs and links in the Network Map

There is typically a brief delay before the **Open in NE Session** menu item becomes active.

 **Note:** Opening an NE session requires that your access privileges include execute permission for the selected NE. See your network administrator for more information.

1.34.2 Steps

- 1 _____
Open Network Map and Health, Network Health View.
- 2 _____
Perform one of the following:
 - a. To open a session from the Network Map: Click on the NE in the map, then click  **Info** to open the Information panel. Click  **More, Open in NE Session**.
 - b. To open a session from the Network Search and Inventory, Network Elements list: On a list item, click  **More, Open NE Session**.An NE Session form opens in a new tab.
- 3 _____
Click **CONNECT**. NEs that are managed using MDM only use the session type configured in the CLI mediation policy. For SSH sessions with NEs managed using NFM-P, a Login window appears.
- 4 _____
Enter the username and password for the NE in the Login window for an SSH session, or in the terminal window for a Telnet session.

5

Click **DISCONNECT** when your session is finished to log out and close the session.

END OF STEPS

Network Search and Inventory

1.35 What does Network Search and Inventory show me?

1.35.1 Access network objects

Network Search and Inventory provide a GUI for network operators to examine equipment across the network. It is a repository of network objects, categorized into the Equipment Inventory, Service Inventory, and Network Topology. These inventories are further categorized into objects types, and presented in a series of object lists.

Network Search and Inventory lets you browse lists of various equipment, including NEs, cards, ports, links, LSPs, services, and service components. These object lists can be filtered to display smaller subsets of objects, and exported to CSV files.

An NSP operator could use Network Search and Inventory to:

- access a list of cards in the network. Filter the list for cards of a certain type or version.
- access a list of cards in the network. Filter the list for all ports that have optical components (SFP).

1.35.2 Object lists

The object lists display detailed data displayed for each list item in columnar format. The total number of objects in the list is displayed in the **Row Count** parameter at the bottom of the list.

The object lists have a variety of tools available to help you control what you see, including sorting and filtering options:

- **Show the Inventory categories:** Expand  the navigation panel on the left-hand side of the Inventory. The Equipment Inventory, Service Inventory, and Network Topology categories appear. You can expand the inventory categories to show object lists, categorized by object type.
- **Search the inventories:** Type an object type in the  **Search** field. Results matching the search string appear in the navigation panel.
- **Refresh the list:** Click  **Refresh**. The last refresh date and time appear at the bottom of the list.
- **View detailed information about a list item:** On a list item, click  **Table Row Actions** and select the NSP GUI in which to view the object; see [1.38 “How do I open an inventory object in a different NSP GUI?” \(p. 44\)](#).
- **Filter the object list under a specific column:** Type a text string in the text field or select a filter option at the top of a column.

The object list automatically updates with filter results as you type your filter string. The **Row Count** parameter adjusts as filters are applied to the list.

Where applicable, click  **Filter Menu** next to the text field, select an operator from the drop-down list, and type a filter string.

Next to the column headers, click  **Table Settings and Actions**, **Clear Filters** to clear column filters.

Lists with a filter applied do not update with new data after a refresh. You must clear and reapply the filter(s) to refresh the list.

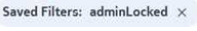
- **Sort the object list under a specific column:** Click on a column header to sort the list under that column. Click on the header again to toggle between ascending and descending sorting. Next to the column headers, click  **Table Settings and Actions**, **Clear Sorting** to clear column sorting.
-  **Advanced Filters** allow you reduce the contents displayed in Network Search and Inventory view using custom filter expressions; see [1.36 “How do I manage advanced filters in Network Search and Inventory?” \(p. 40\)](#).
- **Export objects:** Next to the column headers, click , **Table Settings and Actions**, **Export All** to export all objects in the list, **Export filtered** to export a reduced portion of the list, or **Export Selected** to export selected objects. The exported objects are copied to a CSV file.

1.36 How do I manage advanced filters in Network Search and Inventory?

1.36.1 Purpose

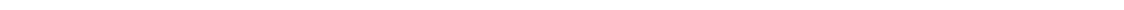
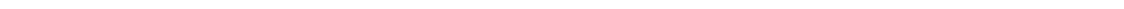
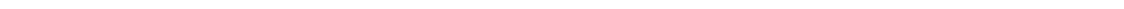
You can configure advanced filters to reduce the number of objects listed in Network Search and Inventory. You configure advanced filters by defining customized filter expressions, and combining the expressions using boolean operators. By combining filter expressions in this way, you can focus the inventory view to a very specific group of objects. After you configure an advanced filter, you can save it to the filter menu for reuse in a future NSP session.

If you are working in the Network Elements or Services inventory, you can also select the pre-configured Unhealthy NEs or Unhealthy Services filters.

When a filter is applied to Network Search and Inventory, a filter chip  appears at the top of the objects list. You can apply only one filter to Network Search and Inventory at a time.

1.36.2 Steps

Configure an advanced filter

- 1 
Open Network Map and Health, Network Search and Inventory View.
- 2 
On the left-hand panel of Network Search and Inventory, select the object inventory you want to filter.
- 3 
Click  **Filter, Advanced Filter**.
The Advanced Filter form opens with a selection of filter attributes for the object type you selected.

4

Do one of the following:

- a. In the Expression Builder panel, select a **Filter Attribute**, **Filter Operator**, and **Filter Value**, and then click **+Add**.

The filter expression is added to the Filter Expression panel. To add more expressions to the filter, select a combining operator (**and**, **or**, **not**, **()**) and repeat this step.

- b. If you are familiar with filter syntax, you can type a filter expression directly in the Filter Expression panel, or edit existing filter expressions.

Type an attribute, followed by a Boolean operator and an attribute value. The system suggests possible attributes, operators, and attribute values as you type, and displays error messages in red when an expression is invalid. You can combine attribute-value expressions using AND and OR operators.

5

Specify a **Filter Name**.

6

If you want your filter to be available to other NSP users, enable the **Public** option.

7

If you want to save the filter for future use, click **Save New**.

To apply your filter to the inventory, click **Apply Filter**.

The Network Search and Inventory list is reduced to display only objects that match the filter expression(s).



Note: If the filter code contains errors (signified by red text in the Filter Expression panel), the Apply Filter button is not accessible.

END OF STEPS

1.36.3 Steps

Remove a filter

1

Open Network Map and Health, Network Search and Inventory View.

2

Remove an applied filter by clicking **×** **Close** on the filter chip.

END OF STEPS

1.36.4 Steps

Apply or edit a saved filter

- 1

Open Network Map and Health, Network Search and Inventory View.
- 2

On the left-hand panel of Network Search and Inventory, select the object inventory you want to access.
- 3

Click  **Filter, Saved Filter**.
If you are working in the Network Elements or Services inventory, you can also select the pre-configured **Unhealthy NEs** or **Unhealthy Services** filter.
- 4

In the **Saved Filter** chip, click the **Search** menu and select a saved filter.
- 5

The Network Search and Inventory list is reduced to display only objects that match the applied filter.
If you need to alter the filter expressions, click  **More, Edit**. In the Advanced Filter form, edit the filter expressions as described in [“Configure an advanced filter” \(p. 41\)](#).

END OF STEPS

1.37 How do I view statistics plots for inventory objects?

1.37.1 Purpose

You can view statistics plots directly from objects in Network Search and Inventory. The types of statistics available depend on the object you are viewing and your statistics collection configuration in Data Collection and Analysis.

You can open statistics plots from the following objects:

- NEs
- cards
- links
- ports
- service endpoints

1.37.2 Steps

- 1 _____
Open Network Map and Health, Network Search and Inventory View.
- 2 _____
Expand **Service Inventory** or **Equipment Inventory** and select the inventory list from which you want to view statistics.
- 3 _____
On a list item, click  **Table Row Actions** and select **Plot Statistics** from the menu.
The statistics plot opens in a chart view in Data Collection and Analysis Visualizations.

END OF STEPS _____

1.38 How do I open an inventory object in a different NSP GUI?

1.38.1 Purpose

You can cross-launch from objects in Network Search and Inventory object lists to other NSP views. Cross-launch availability varies depending on the object type. The following table lists the cross-launch commands available for various objects, and the NSP view that opens for each command.

1.38.2 Steps

- 1 _____
Open Network Map and Health, Network Search and Inventory View.
- 2 _____
On a list item, click  **Table Row Actions** and select the NSP GUI in which to view the object.
- 3 _____
The following cross-launch options are available:

Table 1-2 Cross-launch options from Network Search and Inventory objects

Cross-launch command	Opens NSP view	Available for objects
View In Current Alarms	Current Alarms List	NEs, Links, Ports, Service Sites, L2 and L3 Service Endpoints, Tunnel Bindings, Shelves, Fans, Power Supplies, Cards, Modules, LAGs

Table 1-2 Cross-launch options from Network Search and Inventory objects (continued)

Cross-launch command	Opens NSP view	Available for objects
Open In NE Inventory	Device Management, Inventory view	NEs, Ports, Shelves, Fans, Power Supplies, Slots, Cards, Modules, LAGs, Pluggables, Port Adaptors, L3 Service Endpoints (some)
Show In Network Map ¹	Network Map	NEs and Links
View in Object Troubleshooting	Object Troubleshooting dashboard	NEs, Links, Ports, Service Sites, Service Endpoints, Tunnel Bindings
Open in NE Session	CLI	NEs
Plot Statistics	Data Collection and Analysis Visualizations	NEs, Cards, Service Endpoints
Plot Utilization Statistics	Data Collection and Analysis Visualizations	Links and Ports
Plot Error Statistics	Data Collection and Analysis Visualizations	Links and Ports
View In Service Management	Service Management	Services
View In Configuration Deployments	Device Management	NEs

Notes:

1. The Network Map does not support cross-launch for LLDP links with endpoints with the destination MAC address set to Nearest Customer.

END OF STEPS

1.39 How do I configure manual links for the Topology Map?

1.39.1 Purpose

Use this procedure to configure manual links between NEs, ports, or LAGs in the Network Map, Topology view.

 **Note:** You must be logged in as an Administrative user to configure manual links.

1.39.2 Steps

- 1 _____
Open Network Search and Inventory.
- 2 _____
On the left-hand panel of Network Search and Inventory, expand the **Network Topology** category and select **Links**.
- 3 _____
On the NSP banner, click  **Create Manual Link**.
- 4 _____
In the Create Manual Link form, specify the **Name**, **Description**, **Latency**, and **Link Type** for the link.
- 5 _____
Click **+ ADD** to specify endpoints for the link.
- 6 _____
Specify the endpoint type (NE, port, or LAG) for the link.
- 7 _____
In the Add Two Network Elements|Ports|LAGs form, click on two items to select them in the left-hand list. You can search the list by specifying name or address strings in the fields at the top of the list.
As you select items in the list, they appear in the right-hand list.
- 8 _____
When you have selected two endpoints, click **Add**.
- 9 _____
In the Create Manual Link form, click **Create**.

END OF STEPS

1.40 How do I delete a link from Network Search and Inventory?

1.40.1 Purpose

You can delete a link object from Network Search and Inventory if it was created manually in NSP, or if it is an MDM link and is operationally Down.

1.40.2 Steps

- 1 _____
Open Network Map and Health, Network Search and Inventory View.
- 2 _____
On a link item and click  **Table Row Actions, Delete Link**.

END OF STEPS _____

1.41 How do I modify the Network Search and Inventory view?

1.41.1 Purpose

You can change Network Search and Inventory to hide unused object categories from the navigation panel. You can also change the displayed order of the categories.

1.41.2 Steps

- 1 _____
Open Network Map and Health, Network Search and Inventory View.
- 2 _____
In Network Search and Inventory, click  **Edit**.
Network Search and Inventory switches to Edit mode, with all object categories displayed in the navigation panel.
- 3 _____
Click the  **Displayed** or  **Hidden** icon next to any of the categories to toggle the category to the opposite state.
- 4 _____
To move an object category up or down in the navigation list, click the  **Drag** icon and move the category to the desired position.
- 5 _____
Click **Save** to save your inventory appearance changes.
You can also click **Discard Changes** followed by  **Close Edit Mode** if you choose not to make the changes.

END OF STEPS _____

1.42 How do I manage LSPs in Network Search and Inventory?

1.42.1 Purpose

You can monitor and create LSPs in Network Search and Inventory. You must modify the Network Map and Health dashboard to display LSPs in Network Search and Inventory; see “How do I customize a dashboard?” in the *NSP User Guide*.

Network Search and Inventory can display up to 50 000 LSPs.

1.42.2 Prerequisites

Before you can perform this procedure, the following prerequisites must be completed.

1. Obtain the latest Predefined IETF Intents artifact bundle and service management artifact bundle for the current release from the [Nokia NSP software download site](#).
2. Import the IETF intent types into NSP; see “How do I import an intent type from my computer?” in the *NSP Network Automation Guide*.
3. Install the service fulfillment artifact bundle; see “How do I install an artifact bundle?” in the *NSP Network Automation Guide*.
4. If you will be creating LSPs through the IETF-TE-TUNNEL model, run the IETF_Yang_Intent_Mapping request by selecting the IntentMapping.json from Postman.
5. Create a configuration template using the `icm-te-tunnel` intent type. The name of the template must be DefaultLspTemplate. See “How do I create a configuration template?” in the *NSP Device Management Guide*.

1.42.3 To create an LSP

1

Open Network Map and Health, Network Search and Inventory View.

2

In the LSPs inventory, click  **Size Settings and Actions, Create LSP**.

The Deploy Logical Configuration form opens in Device Management, with the DefaultLspTemplate selected.

3

Configure your LSP; see “How do I create a logical configuration deployment?” in the *NSP Device Management Guide*.

END OF STEPS

Subscriber Monitoring

1.43 What is the Subscriber Monitoring view?

1.43.1 Subscriber monitoring

The Subscriber Monitoring View provides you with KPIs and statistics for BNG/FWA CUPS NEs with MD interfaces. The Subscriber Monitoring View lets you see NE information in the control and user planes, and provides subscriber session statistics. Sites are listed by subscriber count, and by ODSA pools usage.

The Subscriber Monitoring View displays the most recent subscriber count in the database. If subscriber streaming is broken between an NE and NSP, the view displays the value in the database and not what is present on the NE.



Note: The Subscriber Monitoring view is hidden by default. You must modify the Network Map and Health dashboard to display the Subscriber Monitoring dashlets; see “How do I customize a dashboard?” in the *NSP User Guide*.

The view displays subscriber information through the following dashlets

- BNG/FWA CUPS Summary - the number of control plane (active and standby) and user plane NEs in the network.
Click the KPI indicators to list NEs in an expanded data page.
- User Plane Peer Connectivity - the number of user plane NE peer connections at CP level that are operationally Up versus Down.
Click the KPI indicators to list the peers in an expanded data page.
- Subscriber Performance Metrics - total number of control plane active subscribers.
- Sites by Subscriber Count - list of active control plane and user plane sites by subscriber count.
Expand the dashlet to see the entire list.
- BNG/FWA CUPS Network Elements - list of BNG/FWA CUPS NEs.
Expand the dashlet to see the entire list.
- PFCP Peer View - list of PFCP peers.
Expand the dashlet to see the entire list.
- Sites by ODSA pools usage - list of active control plane sites by ODSA pool usage.
Expand the dashlet to see the entire list.
- RADIUS Group Peer Metrics - list of control plane active sites by RADIUS group peer metrics.
Expand the dashlet to see the entire list.

2 Troubleshooting network objects

2.1 What is the Object Troubleshooting dashboard?

2.1.1 Object troubleshooting

The Object Troubleshooting dashboard allows you to check the performance of a selected service object or piece of network equipment. The view allows you to view summarized performance information, and to drill down into specific objects and view performance details, opening objects in external views where necessary.

The data in the Object Troubleshooting dashboard is updated every 30 seconds.

You can search for objects to troubleshoot under the following contexts:

- Network Elements
- Services
- Composite Services
- Ports
- Links
- Subscribers



Note: If you have upgraded from NSP release 23.8 or earlier, your user preferences, such as list column selections, widths, ordering, and KPI preferences will have been reset to defaults and you will need to reconfigure them. This action is only required once.

User access control for Object Troubleshooting

To view equipment and services in the Object Troubleshooting dashboard, user accounts must have an access level above "none" to the associated resource groups to see unhealthy NEs, ports, and services. See the *NSP System Administrator Guide* for more information about configuring user access control.

2.2 How do I troubleshoot a network object?

2.2.1 Purpose

When you open the Object Troubleshooting view, you must specify what type of object you want to troubleshoot and then select a specific object. Once you have selected an object, you are taken to a troubleshooting dashboard with performance details for the object.

If you are already in an object troubleshooting summary dashboard and want to examine a different object, click **Change Target** and select a new object as described from [Step 2](#) below. If you want to return to an object you had previously opened in the Object Troubleshooting dashboard, click **History** and select an object from the drop-down list.

2.2.2 Steps

1

Open Object Troubleshooting.

The Select a Troubleshooting Target form opens in the Troubleshooting Summary dashboard.

2

Select the type of object you want to troubleshoot from the **Target Type** drop-down list.

3

To filter the Troubleshooting Target list contents, select a search criterion from the drop-down list , type a search string in the field and click  **Add Filter**.

The list is reduced to the filtered items. You can configure up to two more filters if needed.

4

Select the object you want to troubleshoot from the Troubleshooting Target list.

5

Click **Choose**.

A Troubleshooting Summary dashboard opens. Depending on the type of object you selected, the dashboard contains different selections of dashlets and views.

END OF STEPS

2.3 What can an NE show me in the Troubleshooting Summary dashboard?

2.3.1 NE troubleshooting

The NE Troubleshooting Summary dashboard consists of a selection of dashlets intended to show an overall picture of a selected network element, providing the necessary information to troubleshoot it. On alarm dashlets, you can click on an alarm KPI to open the related alarms in a list view.

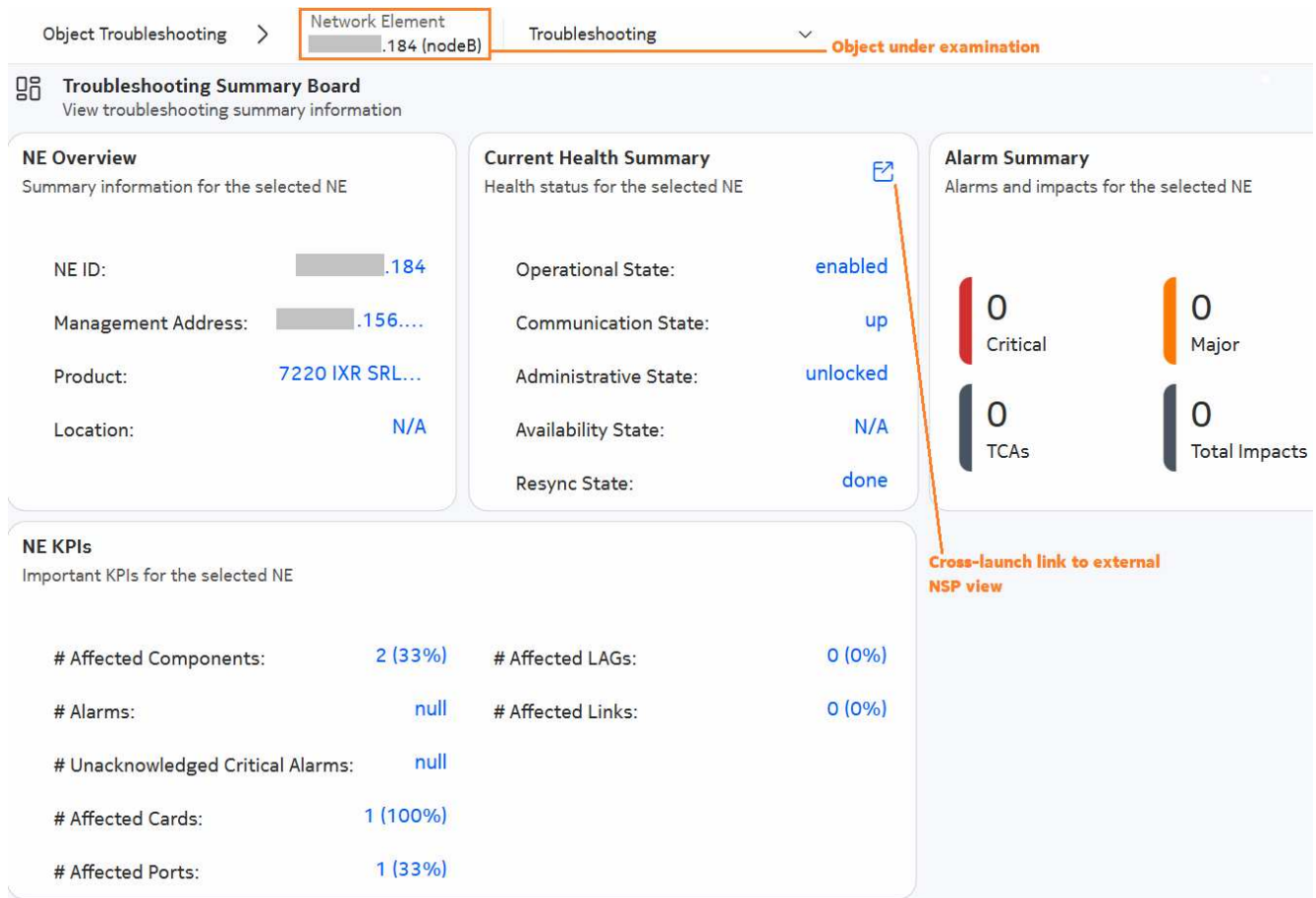
The Troubleshooting Summary dashboard provides information in a series of common dashlets that appear for all object types:

- NE Overview - IP address, model, and location information for the selected NE
- Current Health Summary - operational status of the selected NE
- Alarm Summary - alarm counts for the selected NE
Click on an alarm KPI to view the alarms in the Current Alarms list, filtered to the KPI.
- Analytics reports - a list of Analytics reports that can be run on the selected NE
- NE KPIs - performance data for the selected NE
- Event Timeline Summary

- Troubleshooting Map

If you need to refer back to an object you had previously opened in the Object Troubleshooting dashboard, click  **Target History** and select an object from the drop-down list.

Figure 2-1 Sample Troubleshooting Summary for an NE



2.3.2 Troubleshooting pathway

Use the following dashboard features to troubleshoot an NE:

- View all alarms for an NE KPI in the Alarms list; see 2.9 “How do I check alarms for a KPI?” (p. 58).
- View all alarms for the NE; in the Alarm Summary dashlet; click  **View In Current Alarms**.
- Run Analytics reports for the NE.
- Plot statistics for the NE; see 2.24 “How do I plot performance statistics for an object?” (p. 78)

- View equipment on the NE; in the Current Health Summary dashlet, click **Open In NE Inventory** to display the NE in the NE Inventory, in the context of its related equipment (shelves, cards, ports).
- Execute an NSP workflow on the NE; see [2.21 “How do I execute a workflow on an object?” \(p. 75\)](#).

2.4 Subscriber Troubleshooting view

2.4.1 Subscriber troubleshooting

The Subscriber Troubleshooting view is an optional component of the NE Troubleshooting Summary dashboard. It provides KPIs and statistics information on a BNG/FWA CUPS NE through a series of user-enabled dashlets.



Note: The Subscriber Troubleshooting View is hidden by default. You must modify the Object Troubleshooting dashboard to display the Subscriber Troubleshooting View dashlets; see “How do I customize a dashboard?” in the *NSP User Guide*.
The following dashlets are available

- **Subscriber Performance Metrics:** total number of subscribers and subscriber session counts by protocol.
Control Plane NEs: IPoE Session Count, PPPoE Session Count, FWA Session Count, IPv4 Session Count, IPv6 Session Count, Subscriber Count.
User Plane NEs: IPoE Session Count, PPPoE Session Count and Subscriber Count.
- **NE Performance Metrics:** NE system performance KPIs.
Control Plane NEs: CPU Usage (%), Memory Usage (%) and ODSA Pool Usage (%)
User Plane NEs: CPU Usage (%), Memory Usage (%)³
- **User Plane Peer Connectivity:** the number of user plane NE peer connections at control plane level that are operationally Up versus Down.
Click the KPIs to list the peers in an expanded data page.
- **BNG/FWA CUPS Health Summary:** NE control plane and user plane details.
- **PFCP Peer View:** details of packet frame control protocol peers on the NE.
Expand the dashlet to see the entire list with added details.
- **System VM Metrics:** system VM statics for the NE.
Expand the dashlet to see the entire list with added details.

If you need to refer back to an object you had previously opened in the Object Troubleshooting dashboard, click **Target History** and select an object from the drop-down list.

2.5 What can a service show me in the Troubleshooting Summary dashboard?

2.5.1 Service troubleshooting

The service Troubleshooting Summary dashboard consists of a selection of dashlets intended to show an overall picture of a selected service, providing the necessary information to troubleshoot it.

On alarm dashlets, you can click on an alarm KPI to open the related alarms in a list view. Cross-launch to alternate views is not possible if the related object is not part of a resource group.

The dashboard provides information in a series of dashlets:

- Service Overview - customer ID, service type, and site count for the selected service, plus membership count for composite services (if any)
- Current Health Summary - overall operational status of the selected service
- Sites Health Summary - operational status of service sites
- Endpoints Health Summary - operational status of service tunnel endpoints
- Tunnel Bindings Health Summary - operational status of service tunnel bindings
- Alarm Summary - alarm counts for the selected service
- Event Timeline Summary
- Service Connectivity Map
- Analytics reports - a list of Analytics reports that can be run on the selected service

If you need to refer back to an object you had previously opened in the Object Troubleshooting dashboard, click  **Target History** and select an object from the drop-down list.

2.5.2 Service Inventory

The Service Inventory displays operational information relating to service objects:

- Service Sites - list all sites for the selected service
- Service Endpoints - list all endpoints for the selected service
- Tunnel Bindings - list all tunnel bindings for the selected service, or for a selected site

2.5.3 Troubleshooting pathway

Use the following dashboard features to troubleshoot a service:

- View all alarms for a service KPI in the Alarms list; see [2.9 “How do I check alarms for a KPI?” \(p. 58\)](#).
- View all alarms for the service; in the Alarm Summary dashlet; click  **View In Current Alarms**.
- Run Analytics reports for the service.
- Execute an NSP workflow on the service; see [2.21 “How do I execute a workflow on an object?” \(p. 75\)](#).
- View OAM test results for a service; see [2.22 “How do I run an OAM test on a service?” \(p. 76\)](#).
- If the service is a member of one or more composite services, you can click on the Number of Composite Services count in the Service Overview dashlet to list the composite services.

2.6 What can a composite service show me in the Troubleshooting Summary dashboard?

2.6.1 Composite service troubleshooting

The composite service Troubleshooting Summary dashboard consists of a selection of dashlets intended to show an overall operational picture of a selected composite service, providing the necessary information to troubleshoot it. On the Alarm Summary dashlet, you can click on an alarm KPI to open the related alarms in a list view.

The dashboard provides information in a series of dashlets:

- Composite Service Overview - identification and operational information for the selected composite service
- Composite Service Health Summary - total service count plus number of unhealthy and misaligned services for the selected composite service
- Alarm Summary - alarm counts for the selected service

If you need to refer back to an object you had previously opened in the Object Troubleshooting dashboard, click  **Target History** and select an object from the drop-down list.

2.6.2 Troubleshooting pathway

Use the following dashboard features to troubleshoot a composite service:

- View services included in the composite service; see [2.10 “How do I view the services contained in a composite service?” \(p. 58\)](#).
- View all alarms for a composite service KPI in the Alarms list; see [2.9 “How do I check alarms for a KPI?” \(p. 58\)](#).
- View all alarms for the composite service; in the Alarm Summary dashlet; click  **View In Current Alarms**.
- Execute an NSP workflow on the composite service; see [2.21 “How do I execute a workflow on an object?” \(p. 75\)](#).

2.7 What can a port show me in the Troubleshooting Summary dashboard?

2.7.1 Port troubleshooting

The Port Troubleshooting Summary dashboard consists of a selection of dashlets intended to show an overall picture of a selected port, providing the necessary information to troubleshoot it. On alarm dashlets, you can click on an alarm KPI to open the related alarms in a list view. Cross-launch to an alternate view via links or alarm circles is not possible if the related object is not part of a resource group.

The Port Summary dashboard provides NE information in a series of dashlets:

- Port Overview - type and address information for the selected port
- Current Health Summary - overall operational status of the selected port

- Alarm Summary - alarm counts for the selected port
- Analytics reports - a list of Analytics reports that can be run on the selected port
- Equipment Overview - equipment model and version information for the selected port

If you need to refer back to an object you had previously opened in the Object Troubleshooting dashboard, click  **Target History** and select an object from the drop-down list.

2.7.2 Troubleshooting pathway

Use the following dashboard features to troubleshoot a port:

- View all alarms for a port KPI in the Alarms list; see [2.9 “How do I check alarms for a KPI?” \(p. 58\)](#).
- View all alarms for the port; in the Alarm Summary dashlet; click  **View In Current Alarms**.
- Run Analytics reports for the port
- Plot utilization or error statistics for the port; see [2.24 “How do I plot performance statistics for an object?” \(p. 78\)](#)
- In the Current Health Summary dashlet, click **Open In NE Inventory** to display the port in the NE Inventory on a separate browser tab, in the context of its related equipment (card, shelf, NE).
- Execute an NSP workflow on the port; see [2.21 “How do I execute a workflow on an object?” \(p. 75\)](#).

2.8 What can a link show me in the Troubleshooting Summary dashboard?

2.8.1 Link troubleshooting

The link Troubleshooting Summary dashboard consists of a selection of dashlets intended to show an overall picture of a link, providing the necessary information to troubleshoot it. On alarm dashlets, you can click on an alarm KPI to open the related alarms in a list view. Cross-launch to an alternate view via links or alarm circles is not possible if the related object is not part of a resource group.

The Link Summary dashboard provides information in a series of dashlets:

- Link Endpoints Overview - port and address information for the selected link
- Current Health Summary - overall operational status of the selected link
- Alarm Summary - Link - alarm counts for the selected link
- Alarm Summary - Endpoints - alarm counts for the selected link endpoints

If you need to refer back to an object you had previously opened in the Object Troubleshooting dashboard, click  **Target History** and select an object from the drop-down list.

2.8.2 Troubleshooting pathway

Use the following dashboard features to troubleshoot a link:

- View all alarms for a link KPI in the Alarms list; see [2.9 “How do I check alarms for a KPI?” \(p. 57\)](#).
- View all alarms for the link; in the Alarm Summary dashlet; click  **View In Current Alarms**.
- Open the link in IP Optical Coordination; see [2.11 “How do I open a link in IP Optical Coordination?” \(p. 59\)](#)
- Plot utilization or error statistics for the link; see [2.24 “How do I plot performance statistics for an object?” \(p. 78\)](#)
- Execute an NSP workflow on the link; see [2.21 “How do I execute a workflow on an object?” \(p. 75\)](#).

2.9 How do I check alarms for a KPI?

2.9.1 Purpose

You can cross-launch from a KPI in an Alarm Summary dashlet to the Current Alarms list. The alarm list is filtered to display only alarms related to the KPI you clicked.

2.9.2 Steps

You can list network root cause alarms, filtered by severity.

- 1 _____
Open Object Troubleshooting.
- 2 _____
Select an object to troubleshoot as described in [2.2 “How do I troubleshoot a network object?” \(p. 51\)](#)
- 3 _____
In the Alarm Summary dashlet, click an alarm KPI icon.
Current Alarms opens with the list filtered by the KPI you clicked.

END OF STEPS _____

2.10 How do I view the services contained in a composite service?

2.10.1 Purpose

Use this procedure to view a list of services that are included in a composite service. From the list, you can cross-launch member services in other NSP GUIs.

2.10.2 Steps

- 1 _____
Open Object Troubleshooting.
- 2 _____
Select a composite service as described in [2.2 “How do I troubleshoot a network object?” \(p. 51\)](#).
- 3 _____
In the Composite Service Troubleshooting Summary dashboard, select **Contained Services** from the view menu.

The Composite Service Inventory > Contained Services page opens with a list of member services. On each service item in the list, you can click  **More** and cross-launch the service in the following NSP GUIs:
 - Current Alarms
 - Object Troubleshooting
 - Service Management
 - associated composite services
 - add to Watchlist

END OF STEPS

2.11 How do I open a link in IP Optical Coordination?



Note: In an IP/optical shared-mode NSP deployment, equipment manufacture details are not available for optical network elements.

2.11.1 Steps

- 1 _____
Open Object Troubleshooting.
- 2 _____
Select a link as described in [2.2 “How do I troubleshoot a network object?” \(p. 51\)](#).
Scroll down to the map.
- 3 _____
Click  **More, Open in IP Optical Coordination.**

END OF STEPS

2.12 What is the Troubleshooting map?

2.12.1 View object information

The Troubleshooting map displays an NE or link in a graphical context, in relation to associated objects. You can view information about NEs and links, either in short-form in a pop-up window, or in detail in separate NSP GUI.



Note: The Troubleshooting Map does not support LAG link members, links to unmanaged endpoints, or LLDP links with endpoints with the destination MAC address set to Nearest Customer or Nearest Non TPMR.

2.12.2 Troubleshooting pathway

1

Open Object Troubleshooting.

2

select an NE as described in [2.2 “How do I troubleshoot a network object?” \(p. 51\)](#).
Scroll down to the map.

3

To display basic identification and status information for an NE, hover over the object.
The information appears in a pop-up window.

4

To display further details, click on the object and click **Details**.
The Information panel opens, with expanded object information.

5

To view further information about an object, right-click on the object and select one of the following options (options vary, according to object type):

Troubleshooting options for NEs

- **Explore** displays all links terminating at the NE
- **View In Current Alarms** opens the Current Alarms list for the NE.
- **Open In NE Inventory** displays an NE in the Device Management, NE Inventory on a separate browser tab, along with all configured objects.
- **View in Object Troubleshooting** for a map object that is not currently displayed in the Object Troubleshooting dashboard, this option opens the object for troubleshooting.
- **Open in NE Session** opens a Telnet session with the NE.

- **Add to Watchlist** adds the NE to the Watchlist view for monitoring.

Cross-launch options for links

- **View In Current Alarms** opens the Current Alarms list for the link.
- **View in Object Troubleshooting** for a link object that is not currently displayed in the Object Troubleshooting dashboard, this option opens the link for troubleshooting.

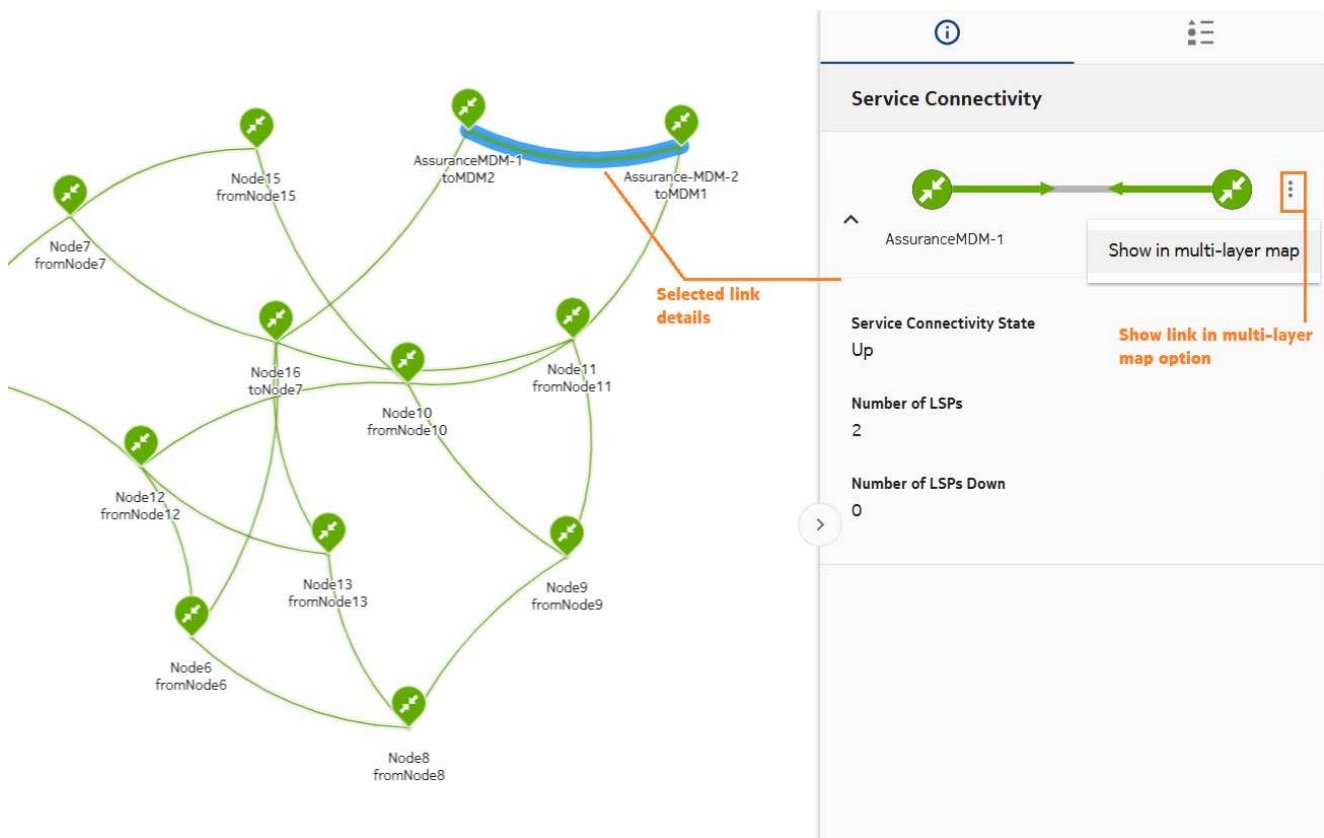
END OF STEPS

2.13 What is the Service Troubleshooting map?

2.13.1 Service troubleshooting

The Service Troubleshooting map displays a service object in a graphical context, in relation to associated objects. The map can be useful in identifying a service segment that is experiencing problems. The map lets you trace services in the network through various layers, such as service tunnels, MPLS, and IGP.

Figure 2-2 Service troubleshooting map



2.13.2 Troubleshooting pathway

Use the following Service Troubleshooting Map features to troubleshoot service objects:

- View map object details; see [2.15 “How do I view information about an object in the Service Troubleshooting map?”](#) (p. 66).
- View the map in separate network layers; see [2.14 “What is the multi-layer map?”](#) (p. 61)

2.14 What is the multi-layer map?

2.14.1 Multi-layer map

The multi-layer map is a display option on the Service Troubleshooting map. It allows you to view services through a series of separate network layers.

The map displays the service path (MPLS and IGP layers) for services using autobind. NSP supports the provisioning of various autobind tunnel configurations on services and the ability to determine the likely active tunnel path between service sites. Users can visualize the active service tunnel path as an overlay across the IGP and physical network topology. This support applies only to SROS NEs.

The selected LSP for the tunnel is determined by the SROS NE, tracking the state until the map data is refreshed.

The service map displays autobind tunnels for EVPN-EPIPE, EVPN-VPLS, and VPRN services.

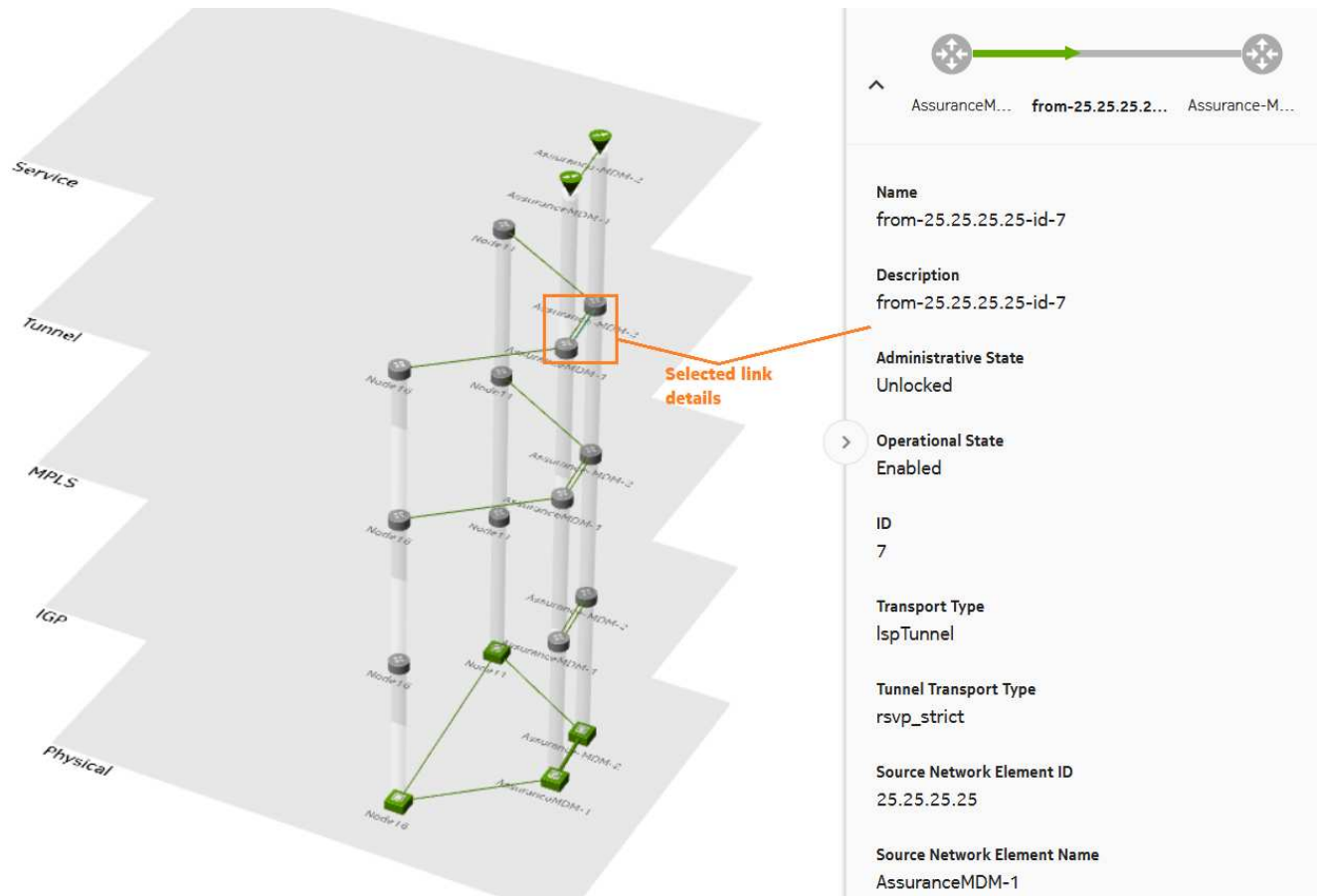
i Note: The MPLS, IGP, and Physical layers are sourced through either CPAM/CPAA or VSR/NRC (NRC-P), which must be configured in NSP in order for the map to display services on managed NEs. The IGP topology layer displays a persisted and stateful combination of both the ISIS and OSPF topologies.

The feature requires the Service Oper Model to be deployed (included in the feature packs `serviceAssurance`, `networkInfrastructureManagement-basicManagement`, or `serviceActivationAndConfiguration-intentBasedServiceFulfillment`). The VSR-NRC+IP MPLS optimization visualizer feature pack is required to view IGP and Physical layers. Autobind must be enabled in the Service Oper Model.

The multi-layer map uses the SPF algorithm to calculate IGP paths. If Flex algorithms are being used to calculate IGP paths elsewhere in NSP, the service map may display differing results.

The MPLS layer does not display if the tunnels configured for a service do not have any associated LSPs.

Figure 2-3 Multi-layer map



2.14.2 Troubleshooting pathway

Use the following Multi-layer Map features to troubleshoot map objects:

- View map object details; see [2.15 “How do I view information about an object in the Service Troubleshooting map?”](#) (p. 66).

2.14.3 Protocol support

The multi-layer map supports the following protocols and protocol-adjacent technologies: PCEP (if NSP PCE is deployed), RSVP, SR-TE, MPLS, IGP (OSPF/ISIS), and BGP (BGP-LS). See “How does NSP enable path control?” and (if NSP PCE is deployed) “How does NSP function as a PCE?” in the *NSP Path Control and Simulation Guide* for more information.

2.14.4 Tunnel support

The tables below list the different supported tunnel types in which an LSP path can be determined based on the IGP topology source (CPAM or VSR-NRC) to populate the IGP layer in the multi-layer map.

Table 2-1 Multi-layer map tunnel support for SDP

Tunnel type	Supported in map?	Support notes
GRE	Yes	CPAM/CPAA runs SPF protocol to get actual path of LSP recorded in NFM-P. Both endpoints need to be in the same admin domain. VSR-NRC runs NRC-P path calculation. Endpoints do not need to be in the same admin domain.
MPLS – LDP	Yes	CPAM/CPAA runs SPF protocol to get actual path of LSP recorded in NFM-P. Both endpoints need to be in the same admin domain. VSR-NRC runs NRC-P path calculation. Endpoints do not need to be in the same admin domain.
MPLS – LSP	Yes	CPAM/CPAA runs SPF protocol to get actual path of LSP recorded in NFM-P. Both endpoints need to be in the same admin domain. VSR-NRC returns PCEP related path discovered by NRC-P. If no PCEP related path is available, NRC-P runs path calculation to determine path. Endpoints do not need to be in the same admin domain.
MPLS – SR-TE-LSP	Yes	VSR-NRC returns PCEP related path discovered by NRC-P. If no PCEP related path is available, NRC-P runs path calculation to determine path. Endpoints do not need to be in the same admin domain.
MPLS – SR-ISIS	Yes	VSR-NRC runs NRC-P path calculation. Endpoints do not need to be in the same admin domain.
MPLS – SR-OSPF	Yes	VSR-NRC runs NRC-P path calculation. Endpoints do not need to be in the same admin domain.
RSVP/TE (PCE)	Yes	CPAM/CPAA runs SPF protocol to get actual path of LSP recorded in NFM-P. Both endpoints need to be in the same admin domain. VSR-NRC returns PCEP related path discovered by NRC-P. If no PCEP related path is available, NRC-P runs path calculation to determine path. Endpoints do not need to be in the same admin domain.

Table 2-1 Multi-layer map tunnel support for SDP (continued)

L2TPv3 MPLS – BGP Tunnel MPLS – Class forward- ing MPLS – Mixed LSP Mode Eth- GRE- Bridged Static	No	Not supported.
--	----	----------------

Table 2-2 Multi-layer map tunnel support for autobind

Tunnel type	Supported in map?	Support notes
LDP	Yes	CPAM/CPAA runs SPF protocol to get actual path of LSP recorded in NFM-P. Both endpoints need to be in the same admin domain. VSR-NRC runs NRC-P path calculation. Endpoints do not need to be in the same admin domain.
SR – TE	Yes	—
RSVP – TE	Yes	—

Table 2-2 Multi-layer map tunnel support for autobind (continued)

UDP BGP RIB – API SR – ISIS SR – OSPF SR – Policy MPLS For- warding Policy	No	Not supported.
---	----	----------------

2.15 How do I view information about an object in the Service Troubleshooting map?

2.15.1 Purpose

You can view information about objects in the Service Troubleshooting map, either in short-form in a pop-up window, or in detail in separate NSP GUI.

2.15.2 Steps

- 1 _____
Open Object Troubleshooting.
- 2 _____
select a service as described in [2.2 “How do I troubleshoot a network object?”](#) (p. 51).
Scroll down to the map.
- 3 _____
To display basic identification and status information for a service, hover over the object.
The information appears in a pop-up window.
- 4 _____
To display further details, click on the object and click ⓘ **Details**.
The Information panel opens, with expanded object information.

5

To display an object in the Multi-layer Map, right-click on the object and select **Show in Multi-layer Map**.

END OF STEPS

2.16 How do I view past events on an object?

2.16.1 Event Timeline summary view

The Event Timeline displays events related to alarms, configuration and state change notifications, and OAM test failure notifications, to help determine the root cause of a problem with an NE or service. The Event Timeline is displayed as a summary in the Troubleshooting dashboard. The default timeline displays a plot along the bottom of the view and event categories, such as alarms or updates, are listed with total event counts.

View events that occurred prior to a hardware problem, or an alarm being raised to determine a possible cause (for example, an object configuration change).

When you are troubleshooting an NE, the timeline shows events on equipment, physical links originating or terminating on the NE, service site, service endpoint, tunnel bindings, tunnels, and LSPs originating on the NE.

When you are troubleshooting a service, the timeline shows events on services sites, endpoints, tunnel bindings, supporting LSPs, physical links, LAGs and supporting ports (for endpoints).

To open a more detailed view, click **View in Event Timeline** for event details by category in shorter time frames, with associated events listed.



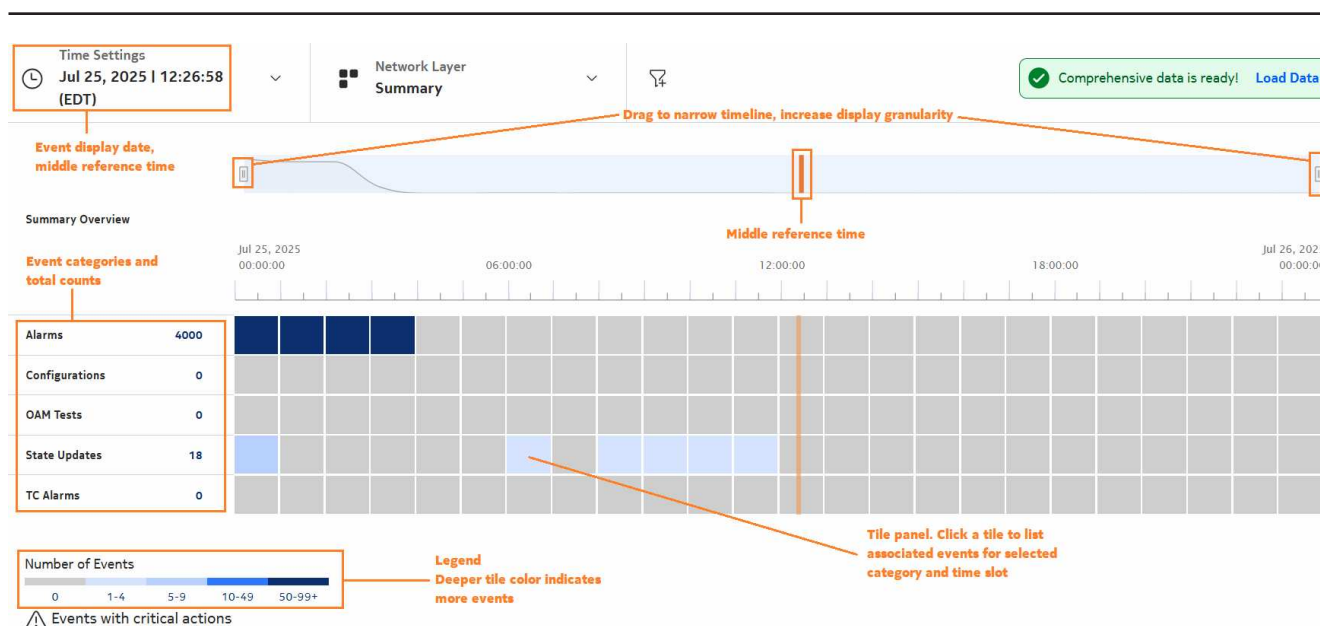
Note: The Event Timeline does not auto-refresh. You must refresh the page manually OR change the time span to update the timeline, either in the summary view or expanded view.

2.16.2 Expanded view

The timeline is displayed across the top of the view, and event categories, such as alarms or updates, are listed on the left. Event counts are displayed as colored tiles. A deeper color indicates a high number of events within the specified time span.

By default, the expanded Event Timeline view displays the same basic event information as in the summary view. After you open the expanded view, comprehensive event data begins loading for the monitored object. The comprehensive data can be displayed when it has finished loading; see [2.18 "How do I display comprehensive event data?"](#) (p. 69).

A **Critical** icon is displayed if the time span has at least one critical event, such as a critical alarm or a failed OAM test.



2.16.3 Troubleshooting pathway

- **Show event details:** click on a tile in the tile panel to list events for the selected time slot and category.

2.17 How do I set the event time frame?

2.17.1 Purpose

You can specify the date for which the Event Timeline displays events, and set the mid-point time of the displayed time frame. By default, the Event Timeline displays events for 12 hours before and 12 hours after the specified date and time. This time frame can be narrowed to view the tile panel in greater detail.

If you set the time frame in the summary view and then click **View in Event Timeline**, you are taken to the expanded view with the same time frame setting. You can also reset the time frame from the expanded view.

2.17.2 Steps

1

Open an object in the Object Troubleshooting dashboard as described in [2.2 "How do I troubleshoot a network object?"](#) (p. 51).

2

In the Event Timeline, click  **Time Settings** and choose an option **Select Date**.

- Choose **Today** to set the timeline to display the current day, with the current time as the mid-point of the collection time frame.
- Choose **Select Date** to set the timeline to display a specific date and mid-point time. You can set the mid-point time by hour, minute, and second.

3

Click **Save**.

END OF STEPS

2.18 How do I display comprehensive event data?

2.18.1 Purpose

In the Event Timeline, Expanded view, you can display event data, depending on the object you are monitoring. Comprehensive event data begins to load for the object when you open the Expanded view. If there is a lot of comprehensive data, the loading can take a few minutes. A flag in the upper-right corner of the view indicates when comprehensive event data is ready to display. The default timeline view displays only event data for the selected object on its network layer. The comprehensive timeline view displays event data for the selected object on the Physical, Service, LSP, Tunnel, and IGP layers.

For NEs on the Event Timeline

- The default view only displays events on the Physical network layer (NE cards, ports, lags).
- The comprehensive view displays events on all five layers (Physical, Service, LSP, Tunnel, and IGP).

For services on the Event Timeline

- The default view only displays events on the Service network layer (service, endpoints, tunnel bindings, sites).
- The comprehensive view displays events on all five layers (Service, Physical, LSP, Tunnel, and IGP).

2.18.2 Steps

1

Open an object in the Object Troubleshooting dashboard as described in [2.2 “How do I troubleshoot a network object?” \(p. 51\)](#).

2

In the Event Timeline, summary view, click **View in Event Timeline**.

3

In the expanded view, wait for the Comprehensive Data is Ready flag to appear and then click **Load Data**.

The additional information appears in the Summary Overview, and in the event list when you click on a tile in the Summary.

END OF STEPS

2.19 How are NSP assurance events retrieved and recorded?

2.19.1 Assurance events

Assurance events are used as a troubleshooting tool to give the user insight into the events that led to a certain state. For example, an operator could examine events that occurred on an NE or service prior to a critical alarm and see if there was a configuration change that resulted in an alarm.

Assurance events are sequential historical events recorded on NE or service objects and their hierarchical sub-components. The type of events that can generate an assurance event are:

- object creation, including alarm creation (AlarmRaised)
- attribute change, which can include configuration change (ConfigurationChange) and state change (StateChange), depending on which attributes have changed
- object deletion, including alarm deletion (AlarmCleared)
- alarm update

Event recording is handled by NFM-P for NFM-P-managed objects, while NSP-managed object recording is handled in NSPOS by the timedevent-recording-app pod. An NSP administrator enables event logging globally for NSP and NFM-P in the Network Assurance Settings; see [2.20 “How do I manage assurance event recording?” \(p. 74\)](#) Any changes to the NFM-P event recording policy made directly in the NFM-P GUI will have no effect on the NSP event recording settings.

2.19.2 Assurance event recording in NSPOS

For objects managed by MDM or WS-NOC, event logging is implemented using time series model recording. .

NSPOS records events for the following objects:

Network Element	Shelf	Slot	Card
Port	Lag	Physical Link	Physical Link Site
Physical Link Endpoint	Service	Service Site	Service Endpoint
Service Tunnel Binding	Tunnel	LSP	LSP Path

2.19.3 Assurance event format

The following is a sample NSPOS event:

```
{
  "nsp-model-notification:object-creation": {
    "schema-nodeid": "/nsp-equipment:
network/network-element/hardware-component/container",
    "instance-id": "/nsp-equipment:network/network-element[ne-id='xx.xx.
xx.xx']/hardware-component/container[component-id='shelf=1/sfmSlot=2']",
    "context": "nsp-db-synchronizer",
    "tree": {
      "/nsp-equipment:
network/network-element/hardware-component/container": {
        "@": {
          "nsp-model:schema-nodeid": "/nsp-equipment:
network/network-element/hardware-component/container",
          "nsp-model:identifier": "/nsp-equipment:network/network-element
[ne-id='xx.xx.xx.xx']/hardware-component/
container[component-id='shelf=1/sfmSlot=2']",
          "nsp-model:creation-time": "20xx-04-17T13:08:05.968Z",
          "nsp-model:last-modified-time": "20xx-04-17T13:08:05.968Z",
          "nsp-model:sources": [
            "fdn:app:mdm-ami-cmodel:16.16.16.16:equipment:
EquipmentHolder:/sfm[sfm-slot='2']"
          ]
        },
        "mac-address": null,
        "mfg-deviations": null,
        "mfg-name": null,
        "availability-state": [],
        "oper-state": "disabled",
        "description": null,
        "mfg-date": "",
        "mfg-assembly-number": "",
        "hw-mac-address": null,
        "admin-state": "unlocked",
        "actual-type": "unassigned",
        "parent-rel-pos": 2,
        "ne-id": "xx.xx.xx.xx"
      }
    },
    "serial-num": "",
    "clei-code": null,
    "holder-state": "not-expected-not-installed",
    "component-id": "shelf=1/sfmSlot=2",
    "ne-name": "Assurance-MDM-2",
    "provisioned-type": "unassigned",
    "part-number": "",
    "standby-state": "providing-service",
    "name": "SFM Slot-2 (unassigned)"
  }
}
```

```
        "state-reason": [],  
        "class": "container"  
    }  
},  
    "event-time": "20xx-17T13:08:06.695604Z"  
}  
}
```

The following is a sample NFM-P event:

```
{  
  "eventId": "294a9268-b926-40d9-a08f-c71d6c73a0d2",  
  "samUser": "admin",  
  "timeIssued": 1713384849182,  
  "objectDisplayName": "Card - 1",  
  "objectClass": "equipment.BaseCard",  
  "changes": [  
    {  
      "newValue": "outOfService",  
      "propertyName": "administrativeState",  
      "oldValue": "inService"  
    }  
  ],  
  "objectName": "network:xx.xx.xx.xx:shelf-1:cardSlot-1:card",  
  "eventType": "ConfigurationEvent",  
  "version": "1.0"  
}
```

Each NFM-P or NSPOS event contains the following information:

2.19.4 Assurance event recording in NFM-P

Assurance events for NFM-P-managed objects are recorded using NFM-P event generation (JMS). The assurance event recorder subscribes to the JMS events. If an event is on an object of interest, it is translated into an assurance event record and logged using the event logging framework.

By default, assurance events are recorded in an Oracle database, but if the customer has configured an auxiliary database, the recording is automatically routed to the auxiliary database.

An NFM-P Event policy allows you to enable/disable event recording (Admin State Up/Down) and the log retention time. To configure an Event policy, open NFM-P, click **Tools** menu, **Events, Event Policies**. The framework also provides tools to purge events.

Event log retention time defaults and minimum/maximum values depend on the type of database you are using.

Table 2-3 Event log retention time defaults and minimum/maximum values

Database type	Default retention time	Minimum retention time	Maximum retention time
---------------	------------------------	------------------------	------------------------

Table 2-3 Event log retention time defaults and minimum/maximum values (continued)

Oracle	168 hours (one week)	One hour	720 hours (1one month)
Auxiliary	720 hours (one month)	One day	8760 hours (one year)

In addition to NFM-P Event Policy configuration, you must enable event recording in the Timeline Settings form. Click [More, Timeline Settings](#). You can enable event recording for specific objects:

Table 2-4 Objects with event recording

Object type	NFM-P class
Network Element	netw.NetworkElement
Link	netw.AbstractPhysicalLink
Card	equipment.Card
Port	equipment.Port
Site	rtr.ProtocolSite
Network Interface	rtr.NetworkInterface
LAG Interface	lag.Interface
Site Sync	sonet.SiteSync
MPLS	rsvp.Interface
IGP	isis.Interface
IGP	ospf.Interface
VNF Instance	nfv.VNFInstance
VNF Component	nfv.VNFComponent

2.19.5 Assurance events retrieval

NSP retrieves events from the appropriate database (NFM-P or NSPOS), based on the original source of the object.

Assurance events are retrieved using API commands from the web component library (assurance-share-md), which provides web components to display events on a timeline. Assurance events are recorded and stored on a server that is accessible from NSP using the web component library.

The web component library:

- Retries events from the NFM-P or NSPOS database, based on the source of the referenced object.

- Displays events in a timeline view. Events are grouped into categories in the display, and multiple categories are presented on the same event timeline.
- Has the ability to filter assurance events based on the source object type or category.
- Provides the option to select individual events and examine event details.
- Provides the option to select a set of events to form a pattern, and then search the Event Timeline for a similar pattern of events.

The Event Timeline displays events for multiple objects at the same time if all of the selected objects are part of the same object hierarchy, and all of the objects are at the same level in the hierarchy (for example, multiple service sites for the same service). As the number of related/child objects to a troubleshooting object increases, response times for retrieved events will also increase.

2.19.6 Assurance event purging

An NSP administrator can purge event records in the Network Assurance Settings; see [2.20 “How do I manage assurance event recording?” \(p. 73\)](#) Event purging only removes events stored in the NSP PostgreSQL database; not events recorded and stored in NFM-P. NFM-P event purging is performed from the assurance event recording policy in the NFM-P GUI.

2.20 How do I manage assurance event recording?

2.20.1 Purpose

Use this procedure to enable event recording policies globally for NSP and NFM-P, for the Event Timeline feature, and to purge previously-recorded messages.

 **Note:** This procedure requires NSP administrator access permission.

2.20.2 Steps

- 1 _____
Open Network Map and Health.
- 2 _____
Click  **More, Settings**.
- 3 _____
In the Fault Management and Assurance Settings form, click **Timeline**.
- 4 _____
Enable the **Enable Event Recording Feature** check box.
- 5 _____
Enable the check box for all object types for which you want to record events.

6

To purge previously-recorded events, click **Delete Stored Events** and then click **Ok** to confirm the deletion.

7

Click **Save**.

END OF STEPS

2.21 How do I execute a workflow on an object?

2.21.1 Purpose

Use this procedure to execute a workflow on the following objects in the Object Troubleshooting dashboard:

- NE
- service
- port
- link
- subscriber

Your workflows must be configured in NSP Workflows before you can execute them in the Object Troubleshooting dashboard; see Using workflows in the *NSP Network Automation Guide*.

You must configure your workflows with an Input parameter called `targetObject` and then Autogenerate the Input form. The workflows must also include a tag for the target object type, as shown in the following table:

Target object type	Tag values
NE	<Object-Troubleshooting> <NE>
Service	<Object-Troubleshooting> <SERVICE>
Port	<Object-Troubleshooting> <PORT>
Link	<Object-Troubleshooting> <LINK>
Subscriber	<Object-Troubleshooting> <SUBSCRIBER>

2.21.2 Steps

1

Open Object Troubleshooting and select the object you want to execute a workflow on, as described in [2.2 “How do I troubleshoot a network object?” \(p. 51\)](#).

The target object opens in the Object Troubleshooting dashboard.

2

Click  **Execute Workflow** in the upper right-hand corner of the Troubleshooting Summary board.

The Workflow Execution form opens with the target object name displayed.

3

Click in the **Workflow Name** field and select the workflow you want to execute.

4

Specify any required workflow inputs.

5

Click **Execute**.

When the workflow completes, the Workflow Execution form displays its status in the Input and Output tabs.

After you execute the workflow, you can access it in the  **Workflow Basket** on the NSP banner. The Workflow Basket displays the 10 most-recently executed workflows.

END OF STEPS

2.22 How do I run an OAM test on a service?

2.22.1 Purpose

Use this procedure to run an OAM test on an MDM-managed or classic-managed service. For information about OAM tests on services, see the *NSP Data Collection and Analysis Guide*.

Some tests on classic-managed services are only provided through the dashboard and are not supported through Data Collection and Analysis. These tests can only display results immediately in the NSP GUI, and the results cannot be retrieved later in NSP. These one-time validation tests are:

- ICMP Ping
- ICMP Trace
- LSP Ping
- LSP Trace
- VCCV Ping
- VCCV Trace
- Tunnel Ping
- MTU Ping



Note: In the case of MTU Ping tests, if you select a tunnel endpoint with a large MTU, the test may time out before it completes and display a failed result.

2.22.2 Steps

1

Open Object Troubleshooting and select a service to test, as described in [2.2 “How do I troubleshoot a network object?”](#) (p. 51).

2

Click **Run OAM Test**.

The Run OAM Test form opens with the default Test Type for the service pre-selected.

3

Click **Select** in the Entity Type list.

The type of service entities listed depends on the type of service you are testing.

4

In the Select [service entity] form, select the service entities you want to test and click **Select**.

5

Click **Run**.

The test is run and the results are displayed in NSP.

END OF STEPS

2.23 How do I view historical OAM test results for a service?

2.23.1 Purpose

Use this procedure to examine past OAM test results for a service. This functionality applies to MDM services on MDM-managed NEs only. For information about OAM tests, see the *NSP Data Collection and Analysis Guide*.

2.23.2 Steps

1

Open Object Troubleshooting and select a service component to test, as described in [2.2 “How do I troubleshoot a network object?”](#) (p. 51).

2

Click **More, View OAM Test Results**.

A list of tests opens in Data Collection and Analysis Management.

-
- 3 _____
- Select a test item in the list and click  **Table Row Actions, View Results** on the selected item.
- A list of test results opens.

- 4 _____
- Select a test result and click **View Results**.

END OF STEPS _____

2.24 How do I plot performance statistics for an object?

2.24.1 Purpose

Use this procedure to test an object in the Object Troubleshooting dashboard for errors or performance. For information about OAM tests, see the *NSP Data Collection and Analysis Guide*.

2.24.2 Steps

- 1 _____
- Open Object Troubleshooting and select an object to test, as described in [2.2 “How do I troubleshoot a network object?” \(p. 51\)](#).
- 2 _____
- Click  **More, Plot Statistics**.
- The component is opened as a statistics plot in Data Collection and Analysis Visualizations.

END OF STEPS _____

3 Network health alarm views

3.1 How does the NSP manage alarms?

3.1.1 NSP fault management

The NSP provides alarm monitoring, correlation, and troubleshooting for the most unhealthy NEs in the network. You can use the NSP to filter alarm lists, identify root causes, and determine alarm impacts.

The NSP receives alarms from managed equipment through a variety of possible means, depending on your network. The NSP collects those alarms and then applies root cause analysis and any configured alarm policies. The resulting alarm information is displayed in the Network Health dashboard in several views and lists, through which you can interact with and manage the alarms.

3.1.2 What alarm severity levels are supported?

The NSP supports the following alarm severity levels:

- Cleared
- Indeterminate
- Info
- Condition
- Warning
- Minor
- Major
- Critical

Alarm severity levels are color-coded. An administrator can change the colors assigned to each severity level; see the *NSP System Administrator Guide* for information about modifying alarm severity colors. You can manually change the severity of an alarm, or configure the NSP to change the severity of an alarm when it is received; see [4.25 "How do I automate alarm management using a policy?"](#) (p. 106).

3.1.3 How can I browse alarm definitions?

There are two tools you can use to view descriptive information about alarms.

The on-product NSP Alarm Search Tool contains a searchable list of alarms that are relevant to the NSP installation; the list expands as required when new artifacts and adaptors are installed. To view the NSP Alarm Search tool, navigate to the NSP Help Center and select Alarm Search from the Reference drop-down menu.

The offline NSP Alarm Search Tool contains a comprehensive searchable list of all NSP and NFM-P alarms for all supported NEs. To access the offline NSP Alarm Search Tool, navigate to the NSP

documentation collection on the [Nokia Documentation Center](#) and select NSP Alarm Search Tool from the Additional Resources drop-down menu.

3.1.4 How does user access control affect what alarms are displayed?

Alarm-related navigation actions require write or execute access to the object affected by the alarm. Opening an NE Session requires execute access to the corresponding NE.

User access control is not supported on the historical and merged alarms lists, or on historical alarm REST and RESTCONF requests, which may show alarms outside of a user's assigned role. User access control for optical trail alarms is only supported in deployments that include IP/Optical Coordination.

3.1.5 What happens when alarms are reloaded from MDM and WS-NOC sources?

When alarm messages from MDM and WS-NOC sources are modified or deleted in NSP, the change is recorded in the NSP database, but not at the alarm source. If alarms are bulk-reloaded from an MDM or WS-NOC source to the NSP, for example during a node resync after a loss of communication, previously modified or deleted alarms from that source are handled in the following manner:

- For alarms with modified fields, any data already in the NSP database is not overwritten by the reloaded alarm.
- Alarms in the NSP database that are tagged as Transient (i.e., not standing alarms) are not deleted by the reload, even if they are no longer present on the source system.

3.1.6 How do I use advanced filters from previous NSP versions?

Advanced filters that were created in NSP Release 23.8 and earlier are not compatible with NSP Release 23.11 or later. Upgrading to this release of the NSP automatically updates any advanced filters to the current version. Filters that require your attention because the migration failed to produce a usable filter have * before the filter name and are displayed in red; unusable filters appear in the filter list but cannot be added to functions such as watched filters and e-mail policies. Filters with duplicate names (for example, when a filter from 23.8 is migrated but a new filter already exists with the same name) have "(migrated)" appended to the filter name.

Filters that were used with an e-mail policy are removed from the policy, and the policy becomes disabled. Associate the filter with the e-mail policy and re-enable it to restore the policy.

 **Note:** Check all migrated filters before use to ensure the automatic migration produced a filter that provides the results you need. The automatic migration is a tool to assist you in migrating filters, but the results are not guaranteed to filter exactly the same as the original filter.

If you have a filter you need to manually upgrade, perform the following modifications to the filter strings:

- ensure there are double quotation marks around keys and values
- ensure that all operators (and, or, not) are lower-case
- ensure timestamp comparisons are valid; timestamps can only be compared up to the second, not millisecond
- separate all 'between' operators into two conditions, one $\leq$ and one $\geq$

- remove the following unsupported operators:
 - is null
 - In the Past
 - equals (match case)
 - not equal (match case)
- update operators to React format as described in the following table:

Previous format (Smartclient)	Current format (React)	Notes
equals	=	This operator is case-sensitive
notEqual	<>	This operator is case-sensitive
less than	<	
greater than	>	
less than or equal to	<=	
greater than or	>=	
contains	contains	
does not contain	not contains	
between (inclusive)	between	This operator is internally translated to >= and <= operators
is null	= 'null'	
wildcard	like, not like, starts with, ends with	
AND	and	
OR	or	
NOT	not	
In the Past	>	SmartClient uses this as a relative offset, but a similar function does not exist in React. The React operator only compares time in absolute reference time.

3.2 How do I view a list of alarms in my network?

3.2.1 Alarm list views

The three alarm list views display alarms in your network as a table. You can configure which columns appear in the table, and filter the results to refine your view. There are three categories of alarm lists, available from the drop-down list:

- The **Current Alarms** view displays all active alarms in the network, or for a specific NEs.
- The **Merged Alarms** view displays all active and previously-active alarms in the network (or for specific NEs) over a specified time period.
- The **Historical Alarms** view displays all previously-active alarms in the network (or for specific NEs) over a specified time period.

For information about configuring and filtering alarm lists, see [“Displaying alarms”](#) (p. 87).

Selecting multiple alarms

You can select up to 200 alarms from the alarm list, from among currently displayed alarms. You can select alarms in batches, but cannot make a selection that includes alarms that have not been loaded into the NSP alarm list (for example, by starting a selection, then scrolling past the currently loaded alarms). Selecting large numbers of alarms from a long list is not recommended; see [4.11 “How do I create an advanced alarms filter?”](#) (p. 94) for information about filtering alarms.

Note: If a selected alarm is deleted during the selection process, the deleted alarm is not included in the action performed on the other selected alarms.

3.3 How do I view root cause distribution in the network?

3.3.1 Alarm Distribution diagram

The Alarm Distribution diagram shows all root cause trees for the most impacting alarms for the network. Root cause alarms with no impacts are not shown.

3.3.2 How do I use the Alarm Distribution diagram?

Select Alarm Distribution from the drop-down list in the Network Map and Health dashboard. The inner circle is each root cause alarm. The outer circles are objects impacted by the alarm, with the width of the blocks representing the impact magnitude. Click on an alarm to see its information in the panel on the right.

Note: If the managed network yields only alarms with no impact, the Alarm Distribution diagram is blank.

In cases where the number of impacted objects is very high, the Alarm Distribution diagram data content is scaled down to maintain diagram readability. A message appears at the bottom of the diagram, indicating that it has been scaled due to high impact counts.

On the Info panel:

- Click Show Impacts  to open the Impacts diagram.
- Click Alarm List  to open the alarm list for the selected alarm, filtered by the alarm object full name.

Manage the order and content of your alarm distribution diagram using the following tasks:

- **Control what's visible in the alarm distribution diagram.** Click on the Filter button  and select one of the options (date range, name, site ID, product, topology group, or saved filter). Depending on what you select, additional filters options appear. The filters appear as chip filters  at the top of the diagram. Click the Close button on a chip filter to remove it from the diagram.
- **Hide specific alarm types.** Click More  and select the appropriate menu option, then click Refresh to hide acknowledged alarms or maintenance (admin state) alarms.

- **Hide alarms of specific severity.** Click More  and de-select the appropriate severity options, then click Refresh.

3.4 How do I view network alarms as a chart?

3.4.1 Using the Alarm Statistics chart

The Alarm Statistics chart displays network alarm counts by alarm severity. Select Alarm Statistics from the Network Map and Health dashboard to see the chart. Click on a bar in the graph to navigate to the Current Alarms view.

To download the chart data as a CSV file, click on the More button and select Export Data to CSV.

3.5 How do I view which NEs have the most alarms?

3.5.1 Unhealthy NEs view

The Unhealthy NEs view displays the NEs in your network with the highest number of alarms in a matrix. NEs are represented as tiles, with alarm count information and links to alarm lists for the selected NE.

3.5.2 What can I do from the Top Unhealthy NEs view?

Manage the order and content of your Top Unhealthy NEs view using the following tasks:

- **Control what's visible in the matrix.** Click on the Filter button  and select one of the options to filter the matrix.
The filters appear as chip filters  at the top of the matrix. Click the Close button on a chip filter to remove it from the matrix.
- **Sort the matrix NE tiles.** Click the sort menu in the top right-hand corner of the matrix and select one of these options:
 - # of Alarms - NE tiles are sorted by the number of alarms against them.
 - # of Unacknowledged Alarms - NE tiles are sorted by the number of unacknowledged alarms against them.
 - # of Impacts - NE tiles are sorted by the number of network objects impacted by alarms on each NE.
- **View current alarms on an NE.** Hover over the NE tile and click on the More button then select View in Current Alarms .
- **View current and historical alarms on an NE.** Hover over the NE tile and click on the More button then select View in Merged Alarms .
- **View historical alarms on an NE.** Hover over the NE tile and click on the More button then select View in Historical Alarms .
- **View additional details and KPIs.** Hover over the NE tile and click on the More button then select View Details
- **Open an NE session.** Hover over the NE tile and click on the More button then select Open in NE Session.

- **Download the chart data as a CSV file.** Click on the More button and select Export Data to CSV.

3.6 How do I view which alarms are occurring the most?

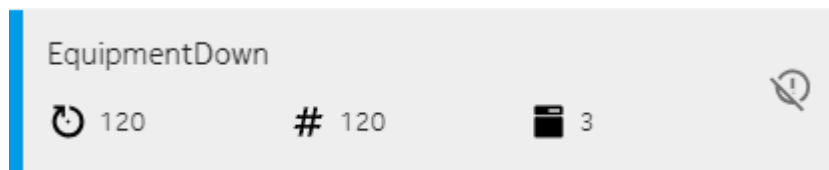
3.6.1 Top Problems view

The Top Problems view helps identify the largest problems in your network by displaying issues in the form of a bar chart. By default, the view displays the alarm types with the most occurrences in the network. Each bar represents a specific alarm type, and its size represents the number of occurrences. The top problems are polled according to the time interval your administrator set in the system preferences or that you set in your user preferences, from the Settings menu on the NSP settings page.

You can configure the graph to instead display alarms grouped by probable cause or specific cause, and count by total number of alarms, total alarm occurrences, or total NEs affected.

Top Problems Alarm Type List

The Alarm Type list displays name and alarm count information for the top 50 alarm types displayed in the chart. The first icon in an alarm type list item shows the number of occurrences of the alarm type, the second icon shows the total alarm count, and the third icon shows the number of NEs affected by the alarm type. When you hover over an alarm type in the list, the corresponding bar in the chart is highlighted.



Hide Alarm Types. Hide an alarm type from the list and the chart by clicking the Hide button  that appears when you hover on the right-hand side of an alarm type item. If any alarms are hidden, the Hidden Alarm Type button is enabled on the main toolbar. Click this button to remove the alarm type from the list of hidden alarms (the alarm type is re-displayed).

Export as a CSV file. You can export the alarm list as a CSV file by clicking on the More button and selecting Export Data to CSV.

3.6.2 How do I configure the Top Problems view?

When you hover over a bar in the chart, the corresponding alarm type in the list is highlighted.

- **Configure the graph.** Use the drop-down menus at the top of the graph to configure what bars are displayed, and what criteria are counted for the bars
- **Filter alarms by severity.** Click on the Filter button  (appears only when the alarms are displayed by alarm name) and select a severity level from the drop-down list. Only alarms of the selected severity are displayed in the chart. Click the Close button on the chip filter to remove it from the chart.

3.7 Fault management API and tools support

3.7.1 Fault management API developer portal documentation

NSP fault management functions are available for OSS using programmable APIs. For general information about developer support, visit the [Network Developer Portal \(https://network.developer.nokia.com/\)](https://network.developer.nokia.com/). For API documentation, see the [NSP API documentation portal \(https://network.developer.nokia.com/api-documentation/\)](https://network.developer.nokia.com/api-documentation/).

For specific documentation about REST APIs for fault management, see:

<https://nsp-server-ip/FaultManagement/api-docs>.

3.7.2 Alarm correlation rules

To see the available correlation rules or rules that have been evicted for NSP, navigate to:

<https://nsp-server-ip/FaultManagement/api-docs/rules.html>

Speak with your Nokia customer relations representative for information about creating and applying custom correlation rules, or enabling and disabling existing rules.

4 Managing network alarms

Displaying alarms

4.1 How do I configure current alarm list settings?

4.1.1 Purpose

Configure aging and overflow settings for the current alarm list. These settings apply to WS-NOC, NSP, and MDM alarms. NFM-P alarm settings are configured in NFM-P; see the *NFM-P Administrator Guide*.

You must have administrator privileges to configure alarm settings.

i **Note:** Time display settings (for example, whether to use 24-hour time) are based on your browser settings and are not configured in the NSP. See your browser documentation for information about configuring how time is displayed.

4.1.2 Steps

- 1

In the **Network Map and Health, Current Alarms** view, click the More button  and select Settings. The Alarm Settings form opens.
- 2

Click Current Alarms on the left-hand panel.
- 3

Enable the Aging Settings option and specify the number of days after which alarms are deleted.
- 4

Under Overflow Settings, specify the percentage of the maximum alarm count at which an alarm overflow warning is issued.

i **Note:** The maximum count for alarms from sources other than the NFM-P is 250 000 if the WS-NOC and NSP are integrated, or 300 000 if the WS-NOC is deployed as an independent system.
- 5

Specify an overflow action. If you choose Halt, all new alarms are dropped. If you choose Wrap, alarms are purged from the database, based on the following settings:
 - Purge Amount - the percentage of the current alarm count to be deleted.

- Purge Policy - either the lowest severity alarms are deleted first or the oldest alarms are deleted first.

6

Save your changes.

END OF STEPS

4.2 How do I configure historical alarm list settings?

4.2.1 Purpose

Configure logging and overflow settings for the historical alarm list. These settings apply to NSP, WS-NOC and MDM alarms. NFM-P alarm settings are configured in NFM-P; see the *NFM-P Administrator Guide*.

You must have administrator privileges to configure alarm settings.



Note: Time display settings (for example, whether to use 24-hour time) are based on your browser settings and are not configured in the NSP. See your browser documentation for information about configuring how time is displayed.

4.2.2 Steps

1

In the **Network Map And Health, Historical Alarms** view, click the More button  and select Settings. The Alarm Settings form opens, with the Historical Alarms panel displayed.

2

Enable the Archive Settings option and enable either or both of the Log on Change and Log on Deletion options.

3

Under Overflow Settings, specify the maximum alarm count at which an alarm overflow warning is issued.

4

Specify a warning threshold as a percentage of the maximum alarm count, and specify the percentage of the alarm count to be purged at the time of the warning message.

5

Specify a critical threshold as a percentage of the maximum alarm count, and specify the percentage of the alarm count to be purged at the time of the critical message.

6

Save your changes.

END OF STEPS

4.3 How do I configure which columns are displayed in an alarm list?

4.3.1 Steps

1

In any alarm list, click on the More  button at the end of the column headers and select Manage columns.

2

Click on the names of the columns that you want to display.

3

Click Apply to save your changes and close the form.

END OF STEPS

4.4 How do I show alarm severity using row color?

4.4.1 Purpose

By default, alarm severity colors only affect the Severity column in an alarm list. You can color the entire background of the row to match the alarm severity using the Row color with severity toggle.

4.4.2 Steps

1

In any view, click the drop-down menu in the title bar and select **Settings**.

2

Perform one of the following:

- a. To enable row color for the current user only: in the User Preferences panel, enable the Row color with severity toggle.
- b. To enable row color for all users: in the System Settings panel, enable the Row color with severity toggle. Only an administrator can enable this toggle.

3

Navigate to any alarm list, or refresh an open alarm list, to view the row colors.

END OF STEPS

4.5 How do I pause the current alarm list?

4.5.1 Purpose

The current alarm list is updated in real-time. You can pause the updates using the Live Data toggle. While updates are paused, the time elapsed since the last update is displayed next to the toggle.



Note: When you scroll down the current alarm list far enough to load alarms that have not yet been displayed, information about those alarms is loaded from the NSP alarm database. The information is current to the time it was loaded, and not to the time when alarm updating was paused. The last refresh time is updated when this occurs.

4.5.2 Steps

1

In the **Network Map And Health, Current Alarms** view click on the Live Data toggle in the lower left. The indicator changes to display the date and time of the last refresh.

2

To resume alarm updates, click on the toggle again.

END OF STEPS

4.6 How do I apply a quick alarms filter?

4.6.1 Purpose

You can use the search fields in a column header to filter the current alarms list. If you have an advanced filter applied, you can combine the quick filter and the advanced filter to create a new advanced filter.

4.6.2 Steps

1

In the **Network Map And Health, Current Alarms** view click on the filter symbol in the header and select an operator. The available filters vary by column.

2

In the search field in the column header, enter a value or choose one from the drop-down list, and press Enter.

3

To save the quick filter as an advanced filter, click on the Filter  button and select Advanced Filter. Enter a name for the filter and click Save Filter.



Note: Do not use a filter name that starts with a * character. The NSP uses this character to identify unusable filters.

END OF STEPS

4.7 How do I view current alarms of a specific severity?

4.7.1 Purpose

In the Current Alarms view you can restrict the view to display only alarms of a selected severity.

4.7.2 Steps

1

In the **Network Map And Health, Current Alarms** view, click on an alarm severity level icon in the Severity filter selector  to display only alarms of that particular severity level.

2

Click on the Clear Filter button  to clear filters.

END OF STEPS

4.8 How do I trigger a sound alert when a new alarm is raised in my current view?

4.8.1 Purpose

You can configure the NSP to play a sound when a new alarm is created in the Current Alarms view. The sound is played when all of the following criteria are satisfied:

- the alarm is a new alarm that was detected in the last 2 seconds, based on the First Time Detected parameter.
- the alarm has not been acknowledged
- there is not already an alarm sound playing
- the alarm appears in the current view based on applied filters (alarms that are filtered out do not trigger a sound)
- the alarm list is not paused
- network time is synchronized across all devices, including the browser displaying the NSP
- sound is enabled and working normally on the device displaying the NSP



Note: When using UAC, a user's role must have Read access to the NSP File Server in order to hear custom sounds. After a user's role is modified, they must refresh their browser and re-login in order to hear the alarm sound.

4.8.2 Limitations

Alarm sounds may not function under the following circumstances:

- systems with high volumes of alarms
- systems where network time is not synchronized between elements (including the browser displaying the NSP)
- Current Alarms views which have a Root Cause filter applied

4.8.3 Steps

1

Ensure that your Internet browser has support for autoplay sounds enabled. See your browser documentation for information about configuring sound preferences.

2

In the **Network Map And Health, Current Alarms** view (or any alarms view) click the More button  and select Settings. The Alarm Settings form opens.

3

Select User Preferences in the left panel. The User Preferences panel appears.

4

Enable the Audible Alarms toggle and click Save. Alarm sounds are enabled for the current user.

5

Refresh any open Current Alarms views by navigating to any other view and then returning to the **Network Map And Health, Current Alarms** view.

END OF STEPS

4.9 How do I configure a custom sound for audible alerts?

4.9.1 Purpose

An administrator can upload a custom sound file to use in audible alarm alerts; for information about audible alarms, see [4.8 "How do I trigger a sound alert when a new alarm is raised in my current view?" \(p. 91\)](#) . One sound file is configured for all users, and the file must conform to the following requirements:

- a .wav format file

-
- named default_alarm_sound.wav
 - 100Kb or less in size
 - two seconds or less in duration

After an administrator changes the default alarm sound, active users must refresh their browser window or re-login in order to hear the newly-configured sound.

4.9.2 Steps

1

In the **File Server** view, navigate to the following directory:

`/nokia/nsp/faultManagement/sounds/`

If the directory does not exist, you must first create it. See the *NSP System Administrator Guide* for information about using the File Server.

2

Upload the custom default_alarm_sound.wav file.

END OF STEPS

4.10 How do I view current alarms based on their root cause analysis status?

4.10.1 Purpose

You can filter the Current Alarms view to display alarms based on whether or not it has a root cause determined by the NSP.

4.10.2 Steps

1

Click on the Filter button  and select Root Causes. A chip filter appears at the top of the list.

2

Choose an option from the drop-down list to configure the filter. Select True to show only root cause alarms, False to exclude root cause alarms, or Unknown to show alarms where the root cause status is being determined.

3

Click on the close button  to clear the Root Causes filter.

END OF STEPS

4.11 How do I create an advanced alarms filter?

4.11.1 Purpose

You can create and save an advanced search filter. Advanced search filters are more detailed than simple search filters. After applying an advanced filter, you can further refine the results using quick filters and save the refinements to a new filter.

4.11.2 Steps

1

In the **Network Map And Health, Current Alarms** view click on the Filter  button to open the advanced filter configuration form.

2

Enter a filter name and description, and specify whether you want the filter to be public or private. Multiple filter properties in an advanced search filter are combined using Boolean operators.



Note: Do not use a filter name that starts with a * character. The NSP uses this character to identify unusable filters.

3

Choose a Boolean operation from the drop-down list:

- AND - When you combine properties using the AND operator, the search returns objects that meet both or all of the specified criteria
- OR - When you combine properties using the OR operator, the search returns objects that meet at least one of the specified criteria
- NOT - When a filter property is preceded by the NOT operator, items that meet the criteria in that filter property are excluded from the results

4

Choose an alarm attribute from the drop-down list, choose a search function, such as equals, and then choose or enter a value. You can click on the + or - button to add or remove clauses. Click on the +{} button to add sub-clauses.

5

To share the filter with other users, enable the Public option.

6

Click Apply or Save Filter. Advanced search filters that you save appear in the Saved Filters list.

END OF STEPS

4.12 How do I delete a saved alarms filter?

4.12.1 Purpose

You can delete a saved filter you have created, including public filters. An administrator can delete public filters created by other users. Before deleting a public filter, ensure other users are not using the filter (for example, in a Watched Filters list or e-mail policy). When you delete a filter you are using, a warning appears; a warning is not produced when you delete a filter being used by other users.

4.12.2 Steps

1

In the **Network Map And Health, Current Alarms** view click on the Filter  button and click Saved Filter.

2

Select the filter you need to delete from the drop-down list and click on the Delete button. A confirmation dialog appears listing any users or objects affected by deleting the filter.



Note: Deleting a public filter that is currently in use by a watched filters list or an e-mail policy disables them when the filter is deleted. You must have permission to edit e-mail policies to delete a filter being used by an e-mail policy.

END OF STEPS

4.13 How do I combine an advanced filter and a quick filter?

4.13.1 Purpose

You can apply a quick filter to an advanced filter to create a new filter.

4.13.2 Steps

1

In the **Network Map And Health, Current Alarms** view click on the Filter  button and click Saved Filter to load an existing filter, or Advanced Filter to create a new filter. See [4.11 “How do I create an advanced alarms filter?” \(p. 94\)](#) for more information about using advanced filters.

2

Apply a quick filter to the results of the advanced filter. See [4.6 “How do I apply a quick alarms filter?” \(p. 90\)](#) for more information about using quick filters.

3

Click on the advanced filter chip filter. The Advanced Filter form opens with the quick filters you applied included in the filter expression.

4 _____
Enter a new name for the filter.

5 _____
Click Save Filter to save the new filter, then click on Apply to return to the alarm list.

END OF STEPS _____

4.14 How do I add filters to the watched filters list?

4.14.1 Purpose

You can use the Watched Filters list to display a summary of up to ten saved filters and apply those filters to the Alarm List with a click. Perform the following to add saved filters to the Watched Filters list.

4.14.2 Steps

1 _____
In the **Network Map And Health, Current Alarms** view click the Watched Filters  button in the details panel to open the Watched Filters list.

2 _____
Click Add in the Watched Filters panel and select one or more saved filters from the displayed list.

3 _____
Click Add in the list of filters to add the selected filters to the Watched Filters list.

4 _____
Click on a filter in the Watched Filters list to apply the selected filter to the Alarm List.

5 _____
Click and drag a filter row in the Watched Filters list to move the filter in the list.

END OF STEPS _____

4.15 How do I export alarms to a file?

4.15.1 Exporting alarms to a CSV file

You can export the alarms in an alarm view to a CSV file. You can export all alarms in a view, for example all current alarms or all historical alarms, export currently selected alarms, or export alarms displayed with the current filter.

When alarms are exported to a CSV file, the following rules are applied to the data:

- Empty fields become empty cells
- Null fields become a - character
- ENUM fields become the ENUM's display value; boolean ENUMs are exported as TRUE/FALSE.
- When performing an export using the Export selected action dates are displayed according to the NSP's timezone settings and the locale of the browser that triggered the export; when using the Export filtered or Export all actions, only the NSP timezone settings are considered.
- Localization is supported for the column name, data, and the file name

The exported data is in plain text; if the Custom Text or Additional Text fields contain sensitive data that should not be exported, ensure that these columns are not displayed before performing an export.

Before performing an export, ensure that your browser is configured to accept downloads.

 **Note:** To avoid losing data, do not navigate away from the alarm view, close the tab, or close your browser while the download is in progress. Depending on the number of alarms in the database and the applied UAC or filters, it can take several minutes to generate the file.

4.15.2 Limitations

The following limitations apply when exporting alarms.

- Role-based access does not apply to the historical and merged alarm lists
- You cannot export more than 500 000 alarms. Use a filter and the Export filtered action to reduce the number of alarms to export.
- To export all alarms on the IAP, object impact list, and email policies views, remove any filters and use the Export filtered action. The Export All action is not available on these views.
- For alarm lists, alarm data exported using Export all or Export filtered may differ from the data displayed in the UI, as the alarm data is collected from the database and will reflect changes that have occurred since the data in the UI was last refreshed.
- When an alarm export is already in progress when you trigger a new Export all or Export filtered export in an alarm list, the export file generated only contains the following message: Export file request terminated. An export operation is already in progress.

4.15.3 Steps

1

Navigate to any alarms list (including the Email Policies and Squelch configuration panels), or an alarm matrix or diagram such as Unhealthy NEs, Alarm Statistics, or the Root Causes diagram.

2

To export data in an alarm list, at the end of the alarm table columns click the Table settings & actions button.

- a. To export the alarms you have currently selected, select Export selected.
- b. To export the alarms currently displayed in the alarm view based on any applied filters, select Export filtered.
- c. To export all alarms in the current view, select Export current page.

3

To export data in a matrix or diagram, click the More button and select Export Data to CSV.

4

Depending on your browser configuration, you may receive a prompt to save or accept the download. Save the file as appropriate to begin downloading the file.

END OF STEPS

Investigating alarms

4.16 How do I find the object affected by an alarm?

4.16.1 Steps

1

Click More  on the right side of a row and choose Open Affected Object to open the affected object of the alarm for configuration in a separate GUI.



Note: When you choose Open Affected Object on an alarm on a classically-managed or MDM-managed NE, or an optical alarm on a WS-NOC-managed NE, the NSP attempts to cross-launch to the NE in the appropriate application (the NFM-P, MDC, and WS-NOC, respectively). For classically-managed NEs, the NFM-P GUI must already be open for the cross-launch to succeed.

END OF STEPS

4.17 How do I list other objects impacted by an alarm?

4.17.1 Steps

1

In any **Network Map and Health** alarm view, click on the More button and select View Object Impacts to open the Object Impacts list for the selected alarm.



Note: This function is not available for alarm messages that are not involved in correlation.

2

To view information about an impacted object, click **(Table row actions)** , **Open impacted object**. When you perform this action on a classically-managed NE, the NFM-P client opens.

END OF STEPS

4.18 How do I display the root cause of an alarm?

4.18.1 Steps

1

In the **Network Map And Health, Current Alarms** view click on the More button and select View Root Causes to open the Root Causes diagram for the selected alarm.



Note: This function is not available for alarm messages that are not involved in correlation.

END OF STEPS

4.19 How do I configure e-mail notifications for alarms?

4.19.1 E-mail policies

An e-mail policy allows administrative users to configure e-mail notification policies for specific alarm messages in NFM-P, WS-NOC, MDM, and NSP. An e-mail policy is configured with a filter. When an alarm message matches the filter criteria, the policy sends alarm notifications to a specified list of up to 20 user e-mail addresses. In order to manage bursts of alarms, e-mail notifications are pooled for up to one minute before sending. If there are more than ten alarms within one minute for a specific policy, a single e-mail notification is sent with a list of ten alarms. Each e-mail policy specifies the maximum number of e-mail notifications (up to 10) that can be sent over a one-hour period. The recipients are notified when the maximum has been reached. You can configure a maximum of 20 alarm e-mail notification policies.



Note: LI and mirror service alarms are not sent in e-mail notifications.

E-mails are not sent for alarm attribute change events, only for alarm creation. For example, if an alarm is created with a severity of major, and the severity is subsequently changed to critical, alarm e-mail policy filters for critical alarms will not include this alarm.

In order for NSP to send e-mail notifications, you must configure connection information for an e-mail server through the NSP settings page. See the *NSP System Administrator Guide* for information about configuring an e-mail server.

Example e-mail notification

The e-mail sent by the NSP consists of a link to the affected alarm, and a set of information about the alarm. The following is an example e-mail sent for a single NSP system alarm:

Subject: NSP Notification - New NSP Health-Alarm Received

Select the link(s) below to launch the filtered alarm list for each alarm. To view the alarm you will be required to have access to an NSP system and to use your existing credentials for log in.

<https://233.252.0.0:443/FaultManagement/?view=alarmListImpacts&alarmId=fdn%3Amodel%3Afm%3AAlarm%3A213673>

Severity: major

Alarm Name: NspApplicationPodDown

Alarm Type: communicationsAlarm

Alarm ID: fdn:model:fm:Alarm:213673

Probable Cause: systemFailed

Alarmed Object ID: fdn:app:server:cam-im-deployer-app-1676634176763-job-rr2sr:233.252.0.0

Alarmed Object Type: NmsSystem

Alarmed Object Name: cam-im-deployer-app-1676634176763-job-rr2sr:233.252.0.0

Last Time Detected: 2023/02/17 11:46:04 159 UTC

Is Service Affecting: false

Site ID: cam-im-deployer-app-1676634176763-job-rr2sr:233.252.0.0

Site Name: cam-im-deployer-app-1676634176763-job-rr2sr:233.252.0.0

Implicitly Cleared: false

Administrative State: unknown

Source Type: nsp

Source System: fdn:app:server

Additional Text: null

Custom Text: N/A

When the NSP has reached the configured maximum number of e-mails permitted in one hour, the following statement is added:

The configured maximum number of e-mail notifications has been reached for this e-mail notification filter. NSP will not send further notifications for up to one hour. Open the FM App to verify current alarm information.

When there are more than 10 new alarms to be included in the e-mail, the following statement is added:

A set of 10 or more alarms exist for this e-mail notification policy. This e-mail notification only displays 10 of the reported set. Open the FM App to verify all the current alarm information.

4.19.2 Steps

You must have administrator privileges to configure alarm e-mail policies.

- 1

In the **Network Map And Health, Current Alarms** view click the More button  and select Settings. The Alarm Settings form opens.
- 2

Click E-mail Policies on the left-hand panel.
- 3

Click on the **+ Email Policy** button .
- 4

Type a name for the policy.
- 5

Select Enabled or Disabled in the Policy Status drop-down menu.
- 6

Select an alarm filter. For information about creating and saving alarm filters, see [4.11 “How do I create an advanced alarms filter?”](#) (p. 94).
- 7

Adjust the Max E-mails Per Hour slider to set the maximum number of alarm notifications that can be sent per hour.
- 8

In the Recipient List field, specify the e-mail addresses of the intended recipients for the alarm notification. You can specify up to 20 recipients.
- 9

Click Save to save the e-mail policy.

END OF STEPS

4.20 How do I stop e-mail notifications for alarms?

4.20.1 Purpose

You can temporarily disable or permanently delete an e-mail policy

4.20.2 Steps

- 1

In the **Network Map And Health, Current Alarms** view click the More button  and select Settings. The Alarm Settings form opens.

2

Click E-mail Policies on the left-hand panel.

3

Perform one of the following:

- **Disable an e-mail policy.** Select an e-mail policy and disable the Enable parameter in the details panel. The policy is disabled until you enable the parameter again.
- **Delete an e-mail policy.** Select an e-mail policy and click the Delete button at the end of the row.

END OF STEPS

Managing alarms

4.21 How do I configure global alarm settings?

4.21.1 Purpose

Use this procedure to configure alarm handling options for all alarms. These settings apply to NSP, WS-NOC, and MDM alarms.

You must have administrator privileges to configure alarm settings.

4.21.2 Steps

1

In the **Network Map And Health, Current Alarms** view click the More button  and select Settings. The Alarm Settings form opens.

2

Click System Settings on the left-hand panel to configure system-wide alarm settings.

3

Perform any of the following configurations, as required:

- **Enable manual changes to alarms** Enable the Manual Settings parameter, then enable options to allow users to promote, demote, or clear alarms.
- **Configure when users can delete alarms** Enable the Alarm Deletion Settings and Manual Alarm Deletion Settings parameters, then choose an option that specifies when a user can delete a system alarm (for example, only after it has been acknowledged, or anytime without restriction).
- **Enable notifications for deleting correlated alarms** Enable the Alarm Deletion Settings and Correlated Alarm Settings for Manually Deleted Alarms parameters, then choose an option that specifies whether an alert is displayed when correlated alarms would be deleted by manually deleting a system alarm.
- **Configure automatic deletion of alarms** Enable the Alarm Deletion Settings and Automatic Alarm Deletion Settings parameters, then choose an option that specifies when to automatically delete alarms.
- **Configure automatic acknowledgement of correlated alarms.** Enable the Alarm Acknowledgement Policy and Correlated Alarm Settings for Manually Acknowledged Alarms parameters to automatically acknowledge any correlated alarms when a system alarm is manually acknowledged, and specify whether a GUI notification occurs when the correlated alarms are acknowledged.
- **Configure automatic acknowledgement of cleared alarms.** Enable the Alarm Acknowledgement Policy and Acknowledge alarms when cleared parameters to automatically acknowledge any alarms when they are cleared.

END OF STEPS

4.22 How do I acknowledge an alarm?

4.22.1 Steps

1

In the **Network Map And Health, Current Alarms** view click More  on the right side of a row and choose Edit Alarm(s).

2

Set the Acknowledge parameter to Acknowledge.

3

You can acknowledge or unacknowledge multiple alarms by using the Ctrl key and selecting the rows before using Edit Alarms(s). When you select multiple alarms, the following limitations apply:

- When you select multiple NFM-P alarms, you can configure the Severity, and Acknowledgement Note.
- When you select multiple NSP, WS-NOC, or MDM alarms, you can configure the Acknowledgement Note.
- When you select a mix of NFM-P and NSP, WS-NOC, or MDM alarms, you can configure the Acknowledgement Note.

4

As required, configure the Severity and Acknowledgement Note and click Save. For NSP, WS-NOC, or MDM alarms, configure the Acknowledgement Note and click Save.

END OF STEPS

4.23 How do I delete or clear an alarm?

4.23.1 Steps

1

In the **Network Map And Health, Current Alarms** view click More  on the right side of a row and choose Delete Alarm(s) to remove or Clear Alarm(s) to clear the alarm or alarms. You can delete or clear multiple alarms by using the Ctrl key and selecting the alarms.



Note: If the Delete Alarm(s) or Clear Alarm(s) actions are not available, an administrator may need to enable manual alarm deletion or clearing.

2

Click OK on the dialog box to confirm the action, or Cancel to close the dialog box without deleting or clearing the alarms.

END OF STEPS

4.24 How do I edit alarm custom text?

4.24.1 Steps

1

In the **Network Map And Health, Current Alarms** view click More  on the right side of a row and choose Edit Alarm(s). You can edit custom alarm text for multiple alarms by using the Ctrl key and selecting the alarms.

2

Enter text in the Custom Text field and click SAVE. You can enter a URL, for example: `http://www.example.com`.

END OF STEPS

4.25 How do I automate alarm management using a policy?

4.25.1 Purpose

When the NSP receives an alarm for the first time, the NSP creates an alarm policy for that alarm. You can use an alarm policy to perform operations automatically on future instances of the alarm. Changes to alarm policies are not retroactive, and only apply to alarms received in the future; to modify existing alarms, use the Alarm List view

You can delete an alarm policy by clicking on the More  button, and selecting Delete. Stale alarm policy entries may occur due to drift in alarm dictionary keys; it is safe to delete these stale policies.

4.25.2 Steps

1

In the **Network Map And Health, Current Alarms** view, click on the More  button in the title banner of the current tab and select Settings.

2

Click on Individual Alarm Policies in the left panel. A list of alarm policies appears.

3

Select one or more alarm policies. You can use the column headers to filter or sort the list of alarm policies.

4

Click (Table row actions), Edit. The following table describes the alarm policy parameters.

Parameter	Effect
General Actions	
Squelch	Hides future instances of this alarm. Squelched alarms are not displayed in the Alarm List.
Initial Severity Assignment	Assigns the chosen severity to the alarm when it is received, overriding the severity assigned by the source.
Auto Acknowledge	Acknowledges the alarm when it is received.
History	Disables or enables historical alarm archiving for the alarm.
Custom Text	Applies the specified custom text to the alarm when it arrives, overwriting any custom text assigned by the source.
Escalation Policies	
Escalation Policy	Enable to configure an escalation policy for the alarm. An escalation policy changes the severity of the alarm when a specified threshold is crossed.
Severity	Specifies the severity to assign to the alarm when the threshold is crossed.
Threshold Type	Specifies the type of threshold to use; for example, the number of occurrences.
Threshold Value	The value that must be exceeded in order to trigger escalation; for example, 50 occurrences.
Escalation Policies	
De-escalation Policy	Enable to configure a de-escalation policy for the alarm. A de-escalation policy changes the severity of the alarm when the number of occurrences drops below a specified number.
Severity	Specifies the new severity to assign to the alarm when the policy is triggered.

Parameter	Effect
Threshold Value	Specifies the number of occurrences for the de-escalation threshold. When the number of occurrences of the alarm drops below this value, the de-escalation policy is triggered.
Alarm Debouncing	
Alarm Debouncing	Enables debouncing for the alarm. When alarm debouncing is enabled, alarm clear events are held until the specified hold period expires instead of being processed immediately. If an alarm raise event occurs before the hold period expires, the existing debounced alarm is updated with the new occurrence and the event is processed immediately. This prevents unnecessary historical alarm logging and lowers the frequency of alarm NBI notifications for highly active (flapping) alarms.

5

Click on the Save button to save your changes.

END OF STEPS

4.26 How do I suppress all alarms raised on a port, NE, or resource group?

4.26.1 Purpose

You can use the NSP to squelch all new alarms raised against an NE or a port, or all NEs and ports in a resource group, including service endpoints associated with the port. Squelched alarms are dropped when received, and do not appear as historical events. Squelched alarms received from an NFM-P do not appear in the current alarm list, but continue to appear in the historical and merged alarm lists. Alarms that are squelched in the NSP are not squelched at their originating data source (for example NFM-P or WS-NOC) and continue to appear in their respective clients



Note: You can individually squelch up to 1000 NEs and 1000 ports. You can use resource groups to squelch up to 250,000 objects (both ports and NEs combined).

Alarm squelching behavior

NSP discards alarms on squelched objects based on the Affected Object and Site ID parameters of the alarm. Squelching an NE discards alarms with a Site ID that matches the squelched NE. Squelching a port discards alarms with an Affected Object parameter that matches the squelched port. Squelching a port also discards alarms for service endpoints associated with the port.

Resource groups

The Resource Group panel on the Alarm Squelch page displays existing network supervision, network element, and equipment resource groups. For more information about using group directories and resource groups, see the *NSP System Administrator Guide*.

4.26.2 Steps

- 1

In the **Network Map And Health, Current Alarms** view, click on the More  button in the title banner of the current tab and select Settings.
- 2

Select Alarm Squelch in the left panel. The Alarm Squelch panel opens.
- 3

Click on the Port, NE, or Resource Group tab in the Alarm Squelch panel. A list of objects appears.
- 4

Select one or more objects and click on (Table row actions), Squelch to squelch the selected objects.

END OF STEPS

4.27 How do I open an SSH or Telnet session with an NE?

4.27.1 Purpose

You can use the NSP to open an SSH or Telnet session with an NE in a browser window. You can open a session from an alarm entry or NE tile in any view. You can only open a session with NEs that are managed using the NFM-P or MDM.

 **Note:** The NSP supports up to 100 concurrent NE sessions. Opening a session with an NE requires that your user privileges include execute permission for the selected NE. See your network administrator for more information.

4.27.2 Steps

- 1

Perform one of the following:
 - a. To open a session from the Unhealthy NEs view, click **Show More** , **Open in NE Session**.
 - b. To open a session from any alarm list, click **(Table row actions)** , **Open in NE Session**.An NE session form opens in a new tab.

2

Select the type of session in the drop-down list, if required, and click **CONNECT**. NEs that are managed using MDM only use the session type configured in the CLI mediation policy. For SSH sessions with nodes managed using the NFM-P, a Login window appears.

3

Enter the username and password for the NE in the Login window for a SSH session, or in the terminal window for a Telnet session.

4

Click the Theme toggle to switch the appearance of the session to black text on a white background.

5

Click **DISCONNECT** when your session is finished to log out and close the session.



Note: You can click **CONNECT** to open the session again, or enter an IP address in the IP field and click **CONNECT** to open another session with a different NE.

END OF STEPS

4.28 How do I manually escalate or de-escalate an alarm?

4.28.1 Purpose

You can promote or demote the severity of an alarm by editing the alarm directly. These settings apply to NSP, WS-NOC, and MDM alarms. NFM-P alarm settings are configured in the NFM-P GUI; see the *NSP System Administrator Guide*.

You must have administrator privileges to configure alarm settings.

4.28.2 Steps

1

In the **Network Map And Health, Current Alarms** view click More  on the right side of a row and choose Edit Alarm(s).

2

Click on the Severity drop-down menu and choose a new severity level for the alarm, then click Save to save your changes.

END OF STEPS

4.29 How do I automate escalating or de-escalating alarms?

4.29.1 Purpose

You can use an alarm policy to automatically escalate or de-escalate an alarm based on configurable triggers. These settings apply to NSP, WS-NOC, and MDM alarms. NFM-P alarm settings are configured in the NFM-P GUI; see the *NFM-P Administrator Guide*.

You must have administrator privileges to configure alarm settings.

4.29.2 Steps

- 1 _____
In the **Network Map and Health, Current Alarms** view, click on the More  button in the title banner of the current tab and select Settings.
- 2 _____
Click on Individual Alarm Policies in the left panel. A list of alarm policies appears.
- 3 _____
Select one or more alarm policies. You can use the column headers to filter or sort the list of alarm policies.
- 4 _____
Click on (Table row action), Edit. The Alarm Policy edit form appears.
- 5 _____
Navigate to the Escalation Policy panel and enable the Escalation Policy parameter.
- 6 _____
Configure the Severity parameter to specify a different severity to be applied to the alarm when the frequency threshold is reached.
- 7 _____
Select an Escalation Threshold Type. The following table describes the escalation threshold types. De-escalation policies only support the Frequency threshold type.

Threshold type	Description
Frequency	How many times the alarm has occurred in the last 24 hours.
Number of Occurrences	How many occurrences of the alarm have been reported.
Days without Ack/Clear	How many days have elapsed without the alarm being acknowledged or cleared.

8 _____
Configure the Escalation Threshold Value parameter for the selected escalation threshold type.

9 _____
Enable the De-Escalation Policy parameter.

10 _____
Configure the Severity parameter to a new severity to apply to the alarm when the alarm is de-escalated, and a De-Escalation Threshold Value.

11 _____
Click Save. The new policy is applied to the alarm.

END OF STEPS _____

A Assurance application evolution

A.1 How has assurance changed in NSP?

A.1.1 Overview

The NSP has significantly evolved its user interface in Release 23.11. The purpose of this appendix is to demonstrate graphically the evolution of the assurance applications in 23.11 and help customers who are upgrading to 23.11 to adapt quickly to the new layout and navigation.

Note: Starting with upgrades to NSP 23.11, any customizations to assurance applications that you may have saved as preferences will have to be reconfigured.

See the *NSP User Guide* for more information on the UI evolution and common behaviors of the new UI in 23.11.

A.2 Managing alarms

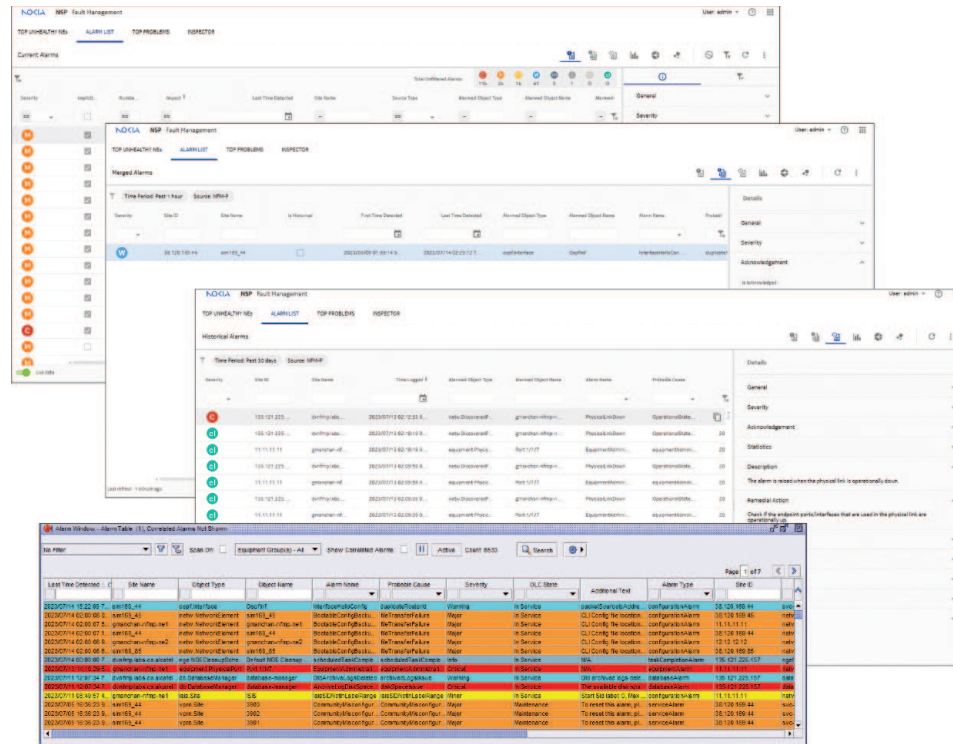
A.2.1 What do I use instead of the NFM-P Alarm Window or NSP Alarm Views?

All of the alarm management features available in pre-23.11 NSP releases remain available in 23.11. You can now use Current Alarms, or the Network Health or Troubleshooting dashboards to access alarm management features.

New navigation

Open **Network Map and Health**, or open **Current Alarms**

Fault Management Alarms Views + NFM-P Alarm window (pre 23.11)



Current Alarms (25.8)

Nokia Network Services Platform									
Network Map and Health									
Current Alarms									
Overview									
Alarm Statistics									
Severity	Alarm Name	Site Name	Alarm Type	Object Name	Alarm Time	Problem Cause	Severity	QoS Data	Alarm Type
M	NodeMemoryBelowThresh...	default:gmancn-node1	2025/07/23 04:28:44	NmsSystem	default:gmancn-node1				
M	NodeMemoryBelowThresh...	default:192.168.97.253	2025/07/23 04:28:44	NmsSystem	default:192.168.97.253				
M	NspBaseServiceDown	default:10.233.104.13...	2025/07/23 04:28:44	NmsSystem	default:10.233.104.13...				
M	NspBaseServiceDown	default:10.233.104.93...	2025/07/23 04:28:44	NmsSystem	default:10.233.104.93...				
M	NspBaseServiceDown	default:10.233.119.50...	2025/07/23 04:28:44	NmsSystem	default:10.233.119.50...				
M	NodeConfigurationPro...	Assurance-MDM-2	2025/07/23 04:28:00	necontrol.DiscoveredHe	16.16.16.16				
M	NodeConfigurationPro...	Assurance-MDM-1	2025/07/23 04:28:00	necontrol.DiscoveredHe	15.15.15.15				
M	EntitlementResourceAll...	-	2025/07/23 04:27:40	clm.NetworkPoolKey	-				
M	DiskSpaceBelowThresh...	default:gmancn-node1	2025/07/23 04:23:44	NmsSystem	default:gmancn-node1				
M	DiskSpaceBelowThresh...	default:192.168.97.253	2025/07/23 04:23:44	NmsSystem	default:192.168.97.253				
M	PeerConnectionDown	CALGABSM15	2025/07/23 03:53:00	bgp.Peer	92.168.96.28				
M	PeerConnectionDown	CALGABSM15	2025/07/23 03:53:00	bgp.Peer	92.168.96.28				
M	EquipmentDown	CALGABSM15	2025/07/23 03:52:53	equipment.PhysicalPort	92.168.96.28				
M	CommunityMisconfigur...	MINTLQCSM158	2025/07/23 03:52:53	vprn.Site	92.168.96.36				
M	CommunityMisconfigur...	MINTLQCSM158	2025/07/23 03:52:53	vprn.Site	92.168.96.36				
M	NodeRebooted	CALGABSM15	2025/07/23 03:52:51	netw.NetworkElement	92.168.96.28				
M	NodeRebooted	MINTLQCSM158	2025/07/23 03:52:51	netw.NetworkElement	92.168.96.36				
M	PeerConnectionDown	CALGABSM15	2025/07/23 02:52:02	bgp.Peer	92.168.96.28				
M	PeerConnectionDown	CALGABSM15	2025/07/23 02:52:02	bgp.Peer	92.168.96.28				
M	PeerConnectionDown	MINTLQCSM158	2025/07/23 02:52:02	bgp.Peer	92.168.96.36				
M	ClogMessageGenerated	cisco_srv_sup_labs...	2025/07/23 01:39:47	equipment.NetworkEle...	92.168.96.25				
M	ClogMessageGenerated	cisco_srv_sup_labs...	2025/07/23 01:39:47	equipment.NetworkEle...	92.168.96.25				
M	ConfigManagement	cisco_srv_sup_labs...	2025/07/23 01:39:47	equipment.NetworkEle...	92.168.96.25				

40437

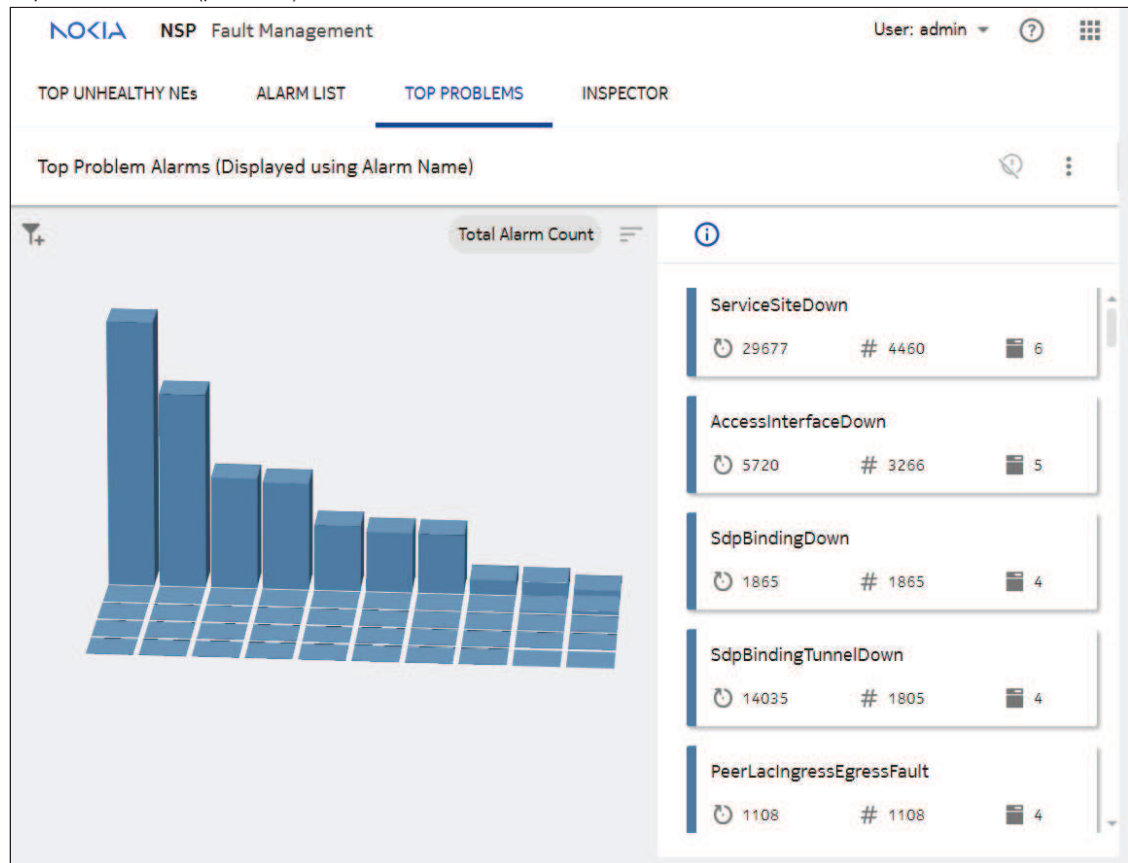
A.2.2 How have the fault management diagrams changed?

In NSP 23.11, the fault management diagrams have been simplified to improve readability, labelling, and zooming features. See examples: Top Problems, Impact Diagram.

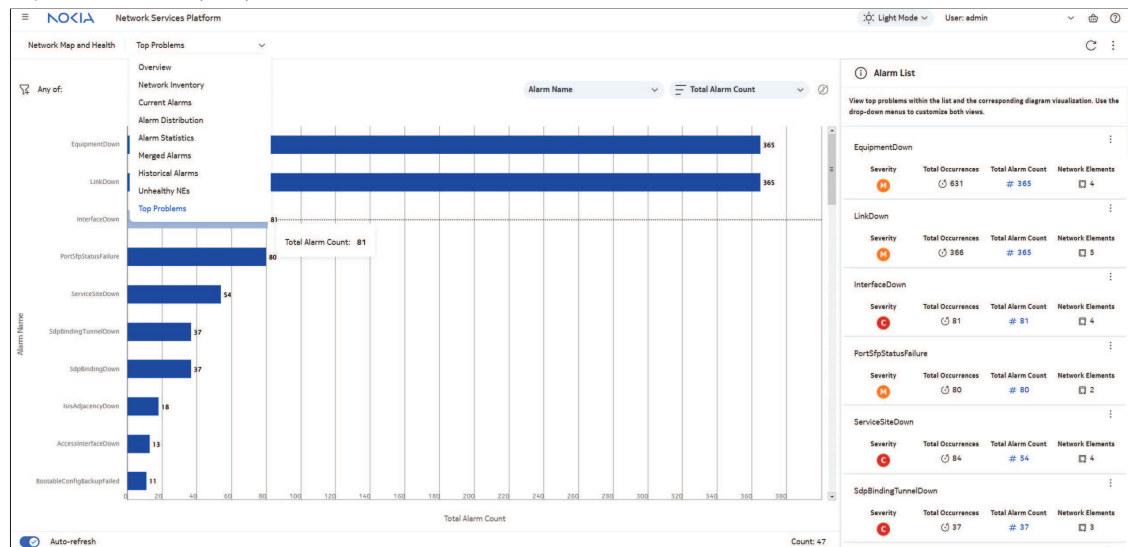
New navigation: Top Problems

Open **Network Map and Health, Top Problems**

Top Problems view (pre 23.11)



Top Problems view (25.8)



40438

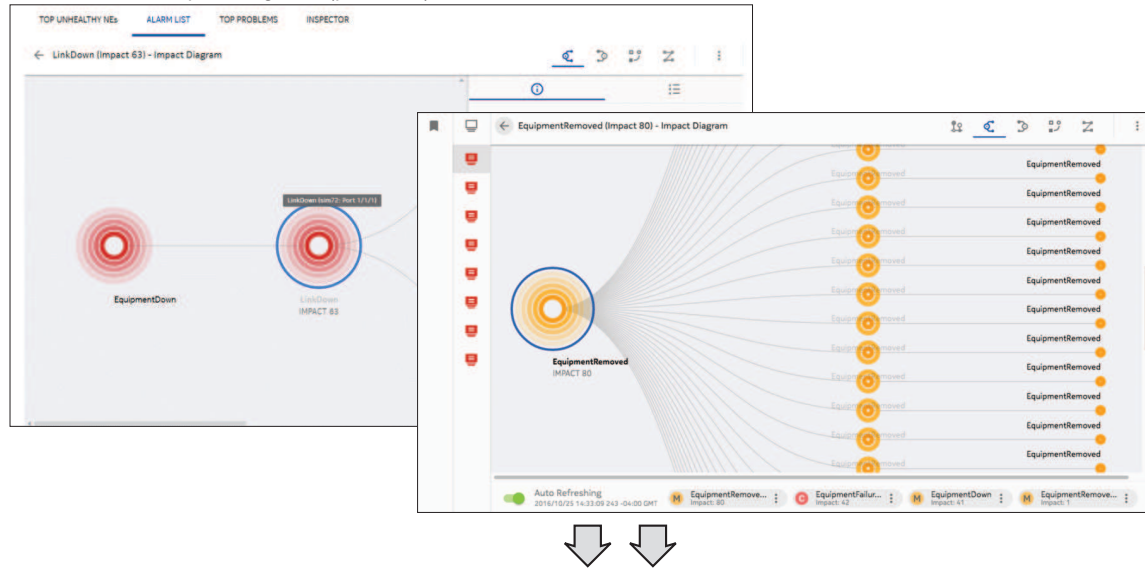
New navigation: Impact Diagram

Open **Network Map and Health**, select an alarm that has impacts, click  **Table Row Actions**,
View Impacts

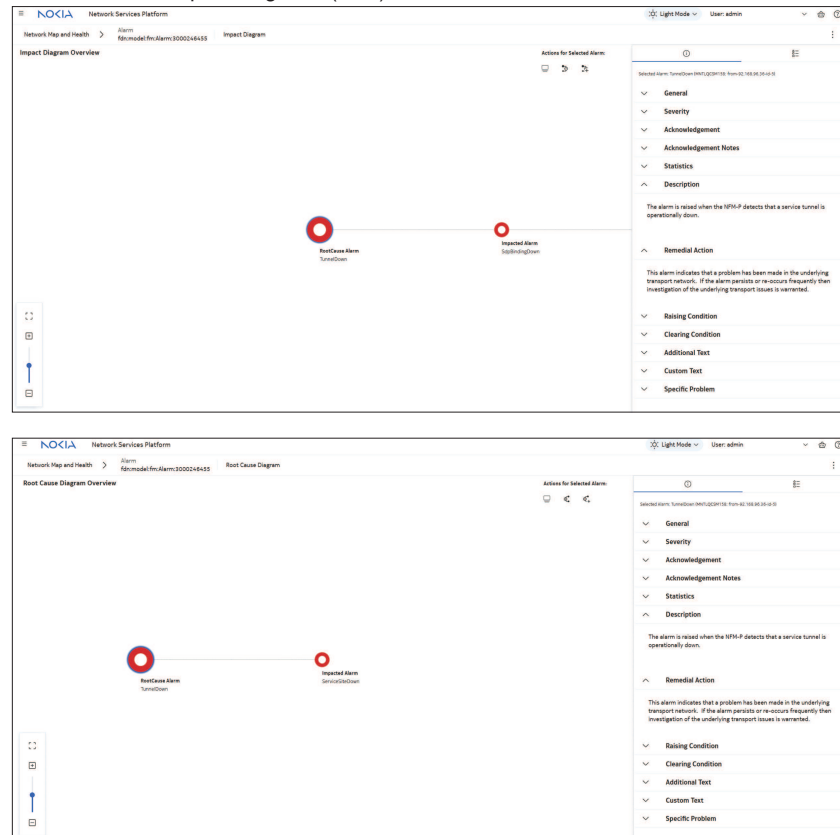
New navigation: Root Causes

Open **Network Map and Health**, select an alarm that has impacts, click  **Table Row Actions**,
View Root Causes

Root cause and impact diagrams (pre 23.11)



Root cause and impact diagrams (25.8)



40439

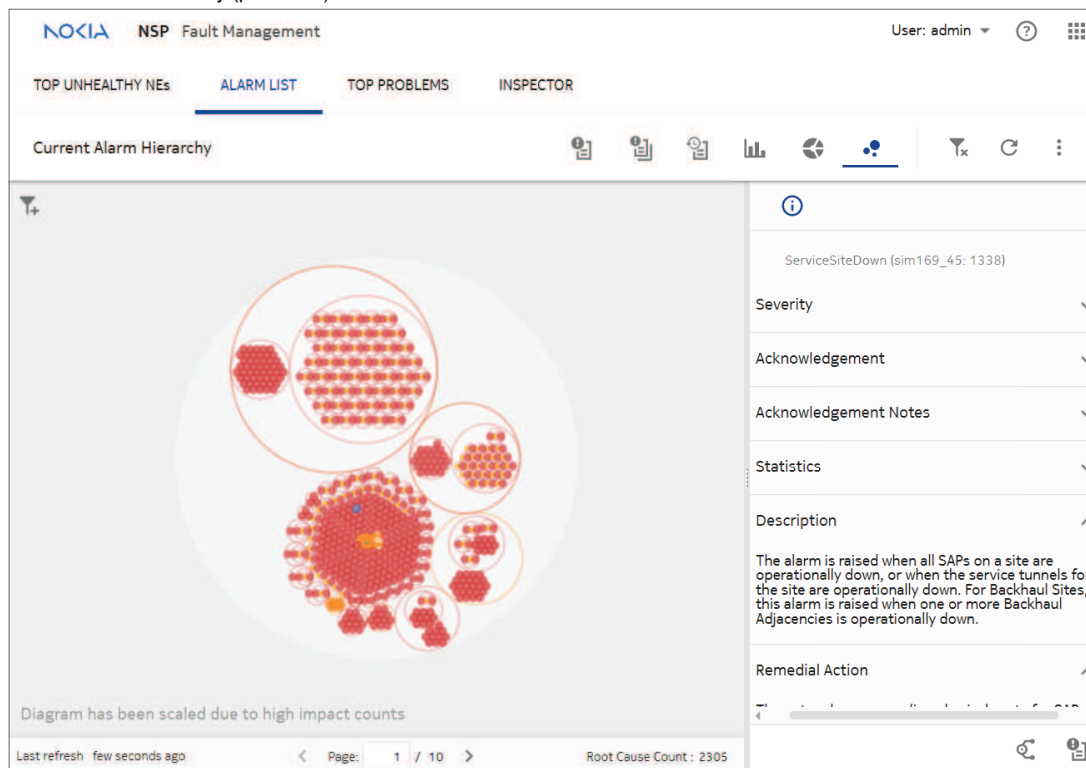
A.2.3 Where is the Current Alarm Hierarchy?

In NSP 23.11, Current Alarm Hierarchy has been removed and replaced by the Alarm Distribution diagram.

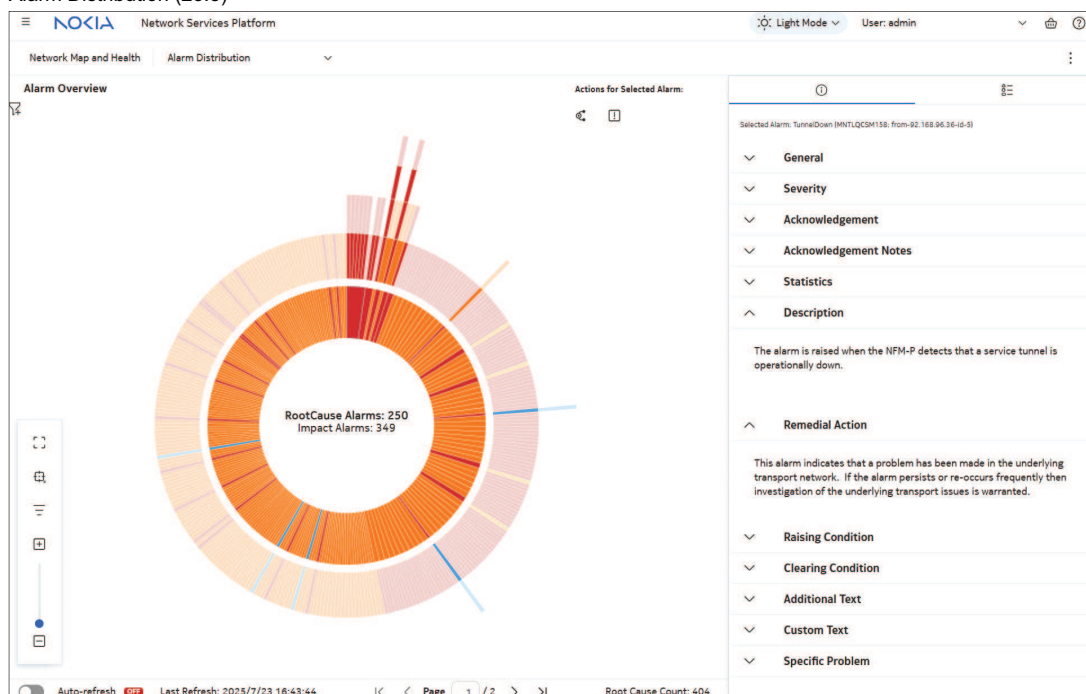
New navigation

Open **Network Map and Health, Alarm Distribution**

Current Alarm Hierarchy (pre 23.11)



Alarm Distribution (25.8)



40440

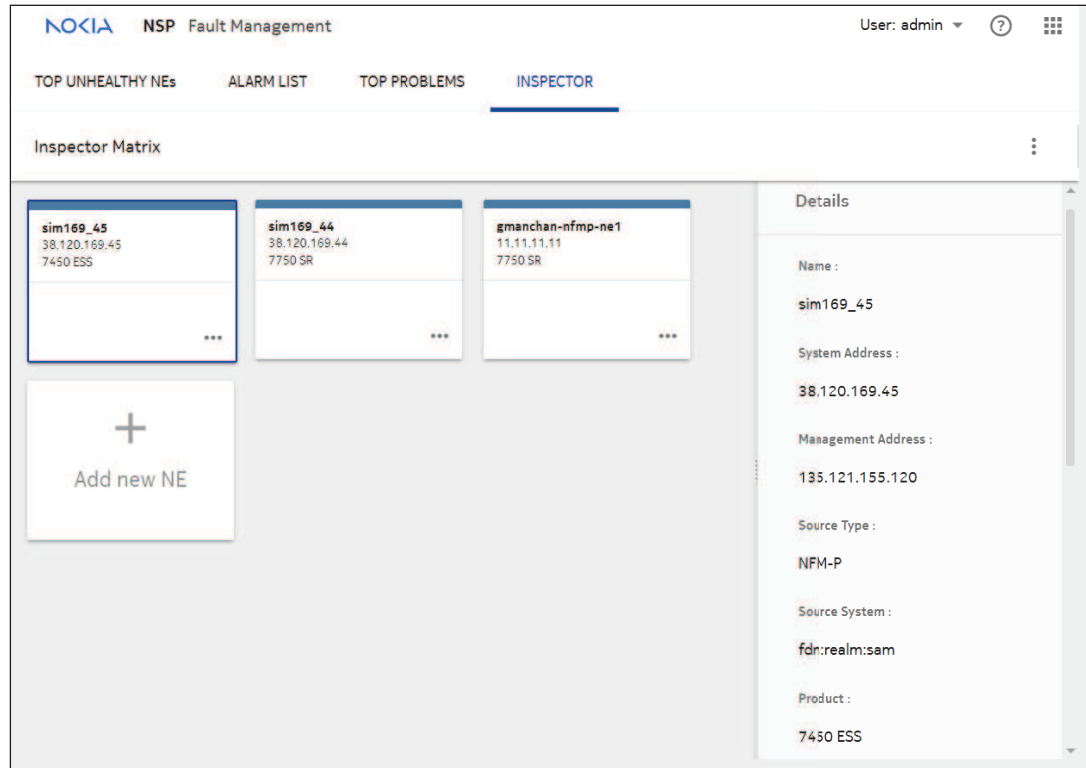
A.2.4 Where is the Inspector Matrix?

In NSP 23.11, Inspector Matrix has been removed and replaced by the NE Troubleshooting dashboard.

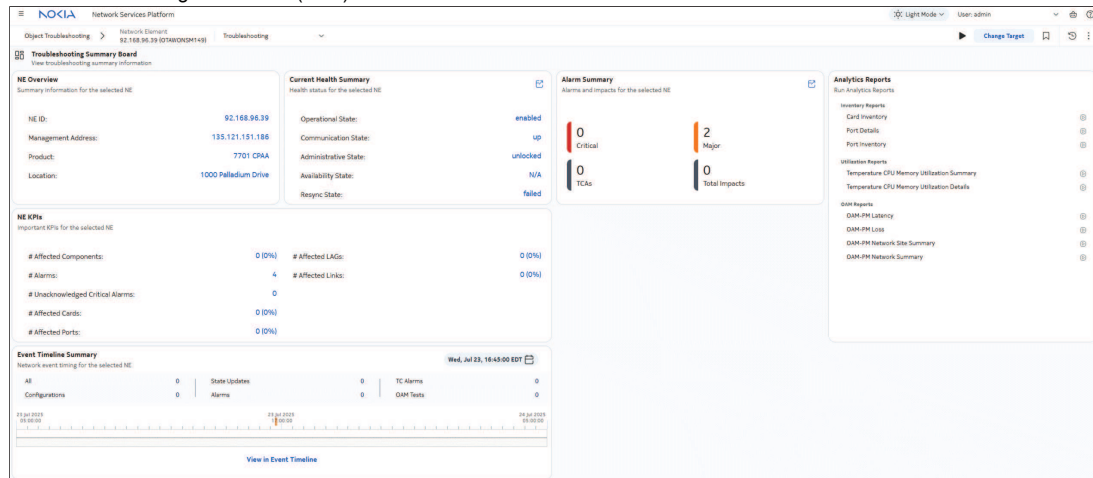
New navigation:

Open **Object Troubleshooting, Network Element**

Inspector Matrix (pre 23.11)



NE Troubleshooting dashboard (25.8)



40441

A.3 Monitoring the network and services

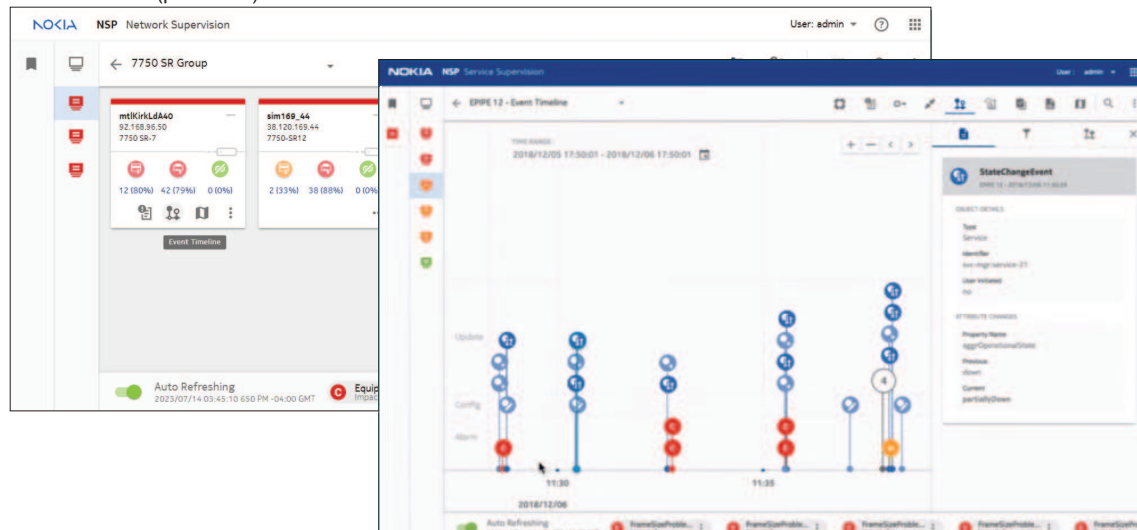
A.3.1 How has Event Timeline changed?

In NSP 23.11, this view has been updated to a simplified graphic to address scale issues and improve panning, zooming, and time scale labels. The Event Timeline has been moved to the NE and Service Troubleshooting dashboards.

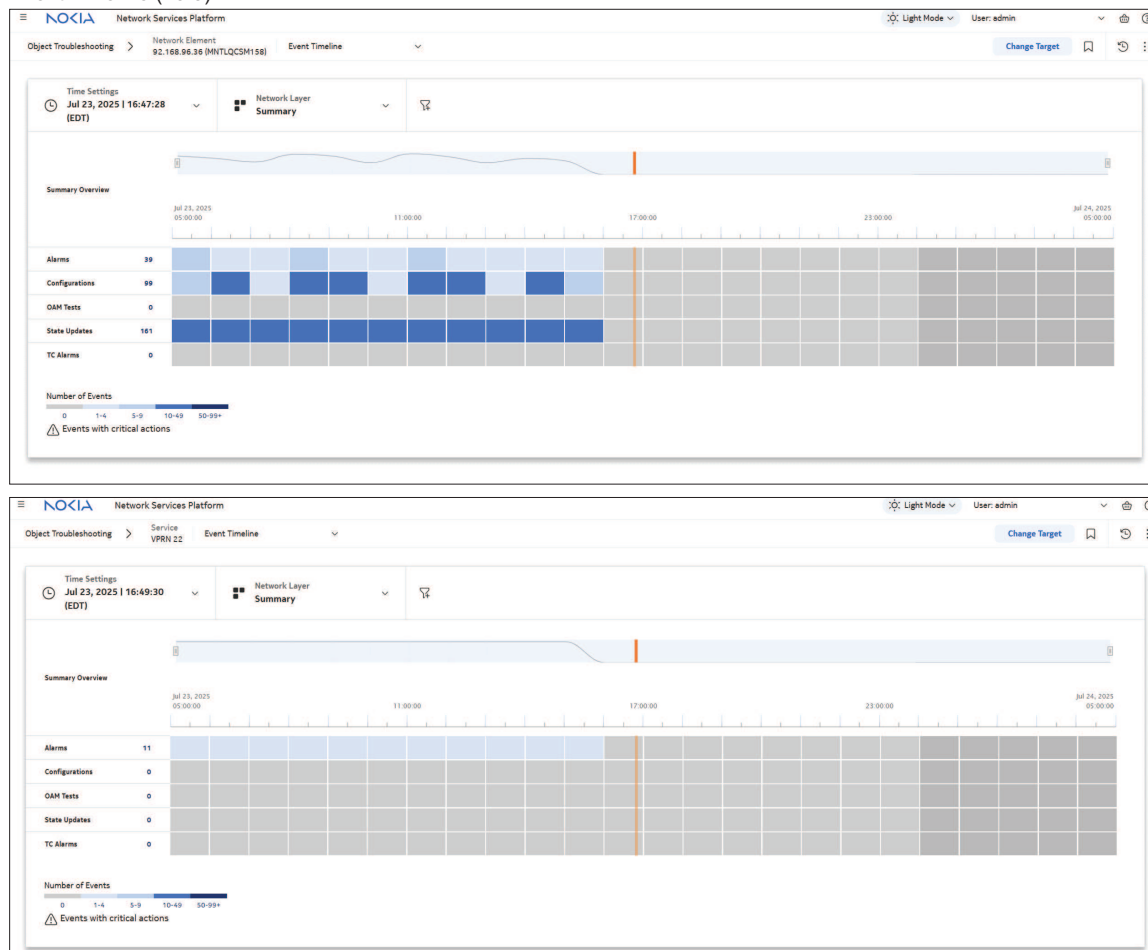
New navigation

Open **Object Troubleshooting, Network Element or Service, Event Timeline**

Event Timeline (pre 23.11)



Event Timeline (25.8)



40442

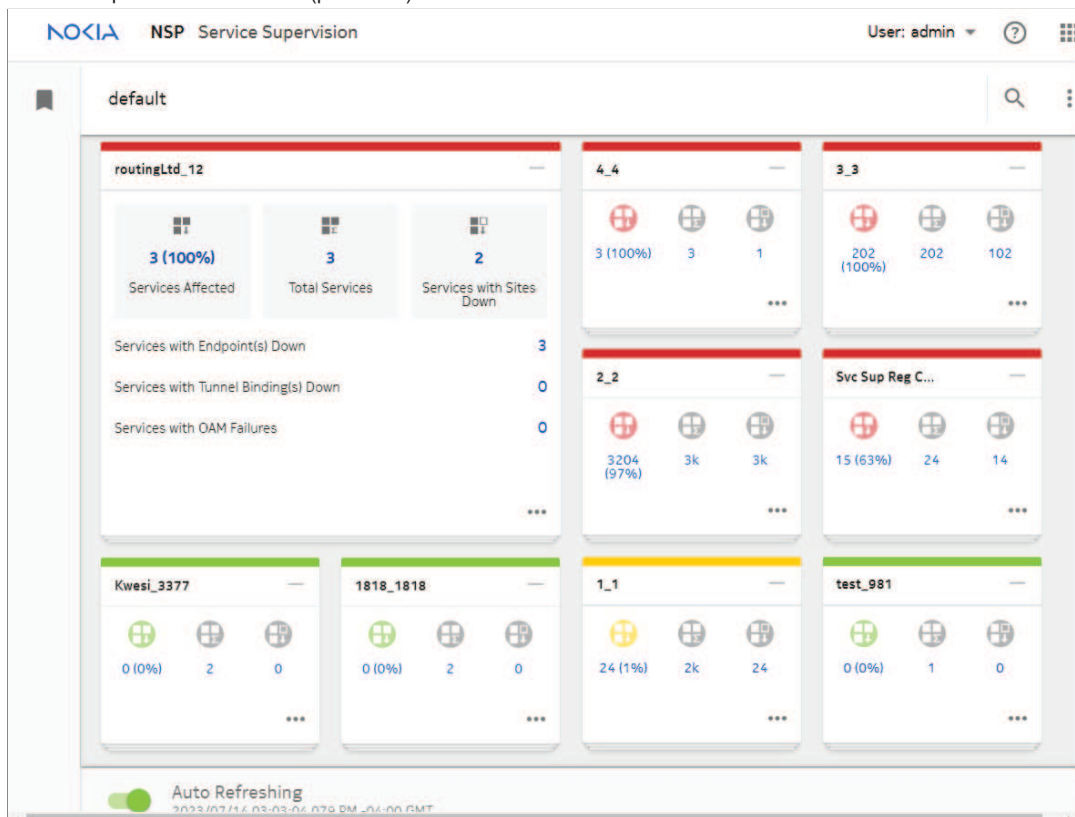
A.3.2 How has service supervision changed?

In NSP 23.11, the functionality previously delivered by the Service Supervision application (including Service health KPIs, Tunnel bindings, etc.) has been migrated to the Service Troubleshooting dashboard.

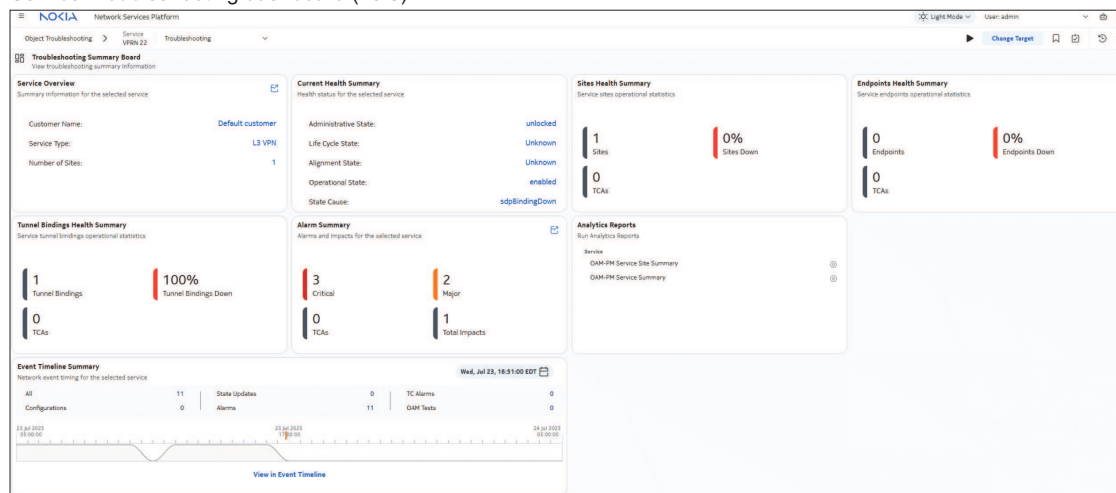
New navigation

Open **Object Troubleshooting, Service**

Service Supervision dashboard (pre 23.11)



Service Troubleshooting dashboard (25.8)



40443

A.3.3 How have supervision views and groups changed?

In NSP 23.11, Network and Service Supervision views and groups that were available in the former Group Manager application are not included in Map Layouts and Groups. In Map Layouts and Groups, you can configure a common map layout for use in all NSP map views, and you can configure resource group directories and resource groups for use in the Network Functions GUI views. For more information, see "Network resource groups" in the *NSP System Administrator Guide*.

New navigation

Open **Map Layouts and Groups, Map Layout**

Group Manager (pre 23.11)

The top screenshot shows the 'Network Supervision Views' tab in the Group Manager. It displays a table of groups with columns: Group Name, Description, and Number of NE(s). The table lists various groups such as 1830 VWM OSU Group, 7210 SAS D Group, 7210 SAS Dxp Group, 7210 SAS M Group, 7210 SAS T Group, 7450 ESS Group, 7701 CPAA Group, 7750 SR Group, 7950 XRS Group, 9471 WMM Group, AIM Group, Alcatel-Lucent-9774-L, DC VSAS Group, Nokia - 5G BTS Group, Nokia - 5G Classical BT, Nokia - 5G DU BTS Gro, Nokia - Airscale - BTS, and Nokia - DCAP Group.

The bottom screenshot shows the 'Service Supervision Views' tab. It displays a table of services with columns: Group Name, Description, and Number of Service(s). The table lists various services such as 1818_1818, 1_1, 2_2, 3_3, 4_4, Kivesi_3377, Svc Sup Reg Cust 2022_2022, routingLtd_12, and test_981.



May Layouts and Groups (25.8)

The screenshot shows the 'Map Layouts and Groups' section in the Network Services Platform. The 'Test' layout is selected, displaying a table of regions with columns: Region Name, Description, and Number of NE(s). The table lists two regions: Assurance MDM Regression - IBSF and Pre-Provisioned NEs.

40444

A.4 Combining views

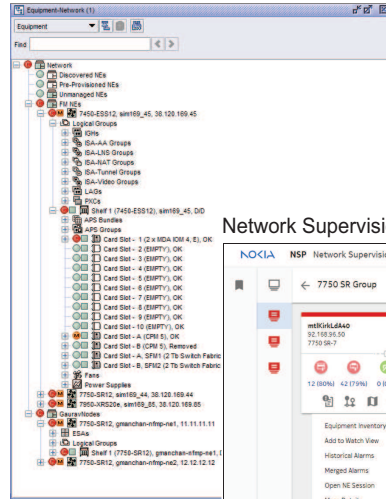
A.4.1 How do I view unhealthy NEs?

Before 23.11, you viewed top unhealthy NEs through the Top Unhealthy NEs matrix in Fault Management or Network Supervision. In NSP 23.11, the Fault Management and Network Supervision matrixes have been combined into one simplified and focused matrix as part of the Network Map and Health dashboard.

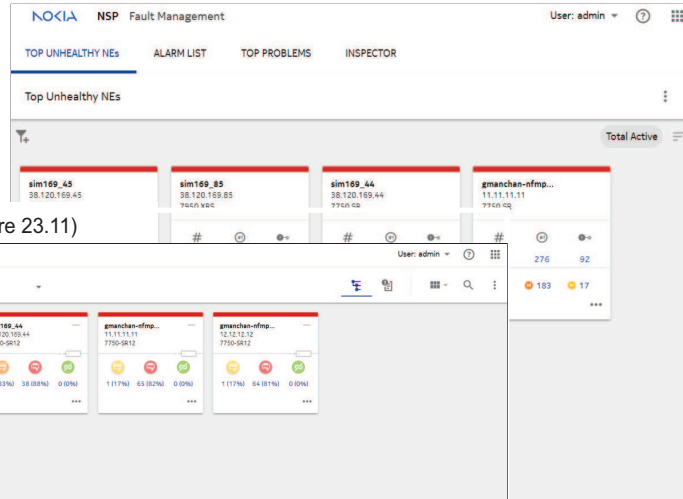
New navigation

Open **Network Map and Health, Unhealthy NEs**

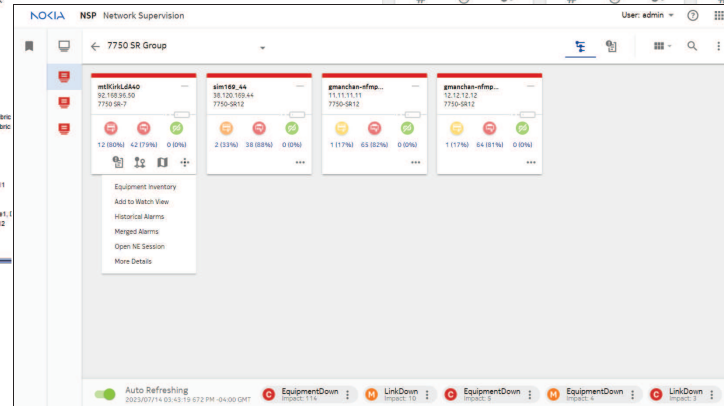
NFM-P GUI Navigation Tree



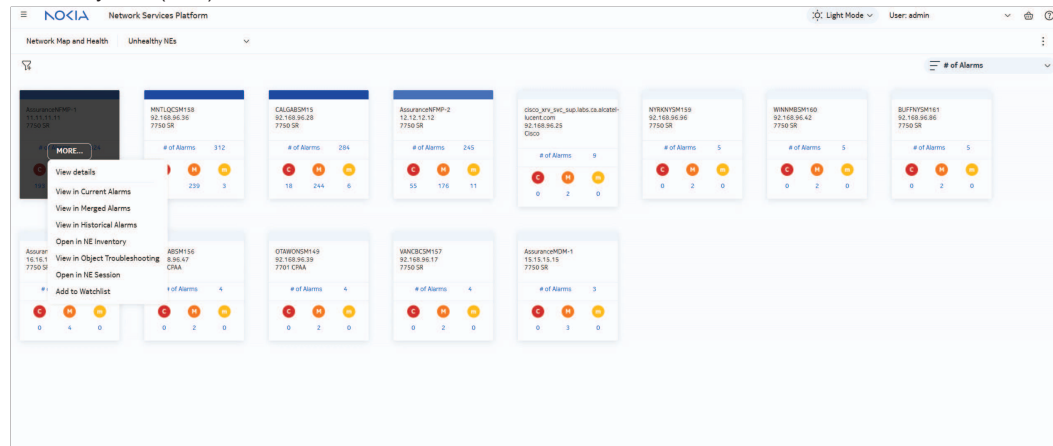
Fault Management Top Unhealthy NEs (pre 23.11)



Network Supervision (pre 23.11)



Unhealthy NEs (25.8)



40445

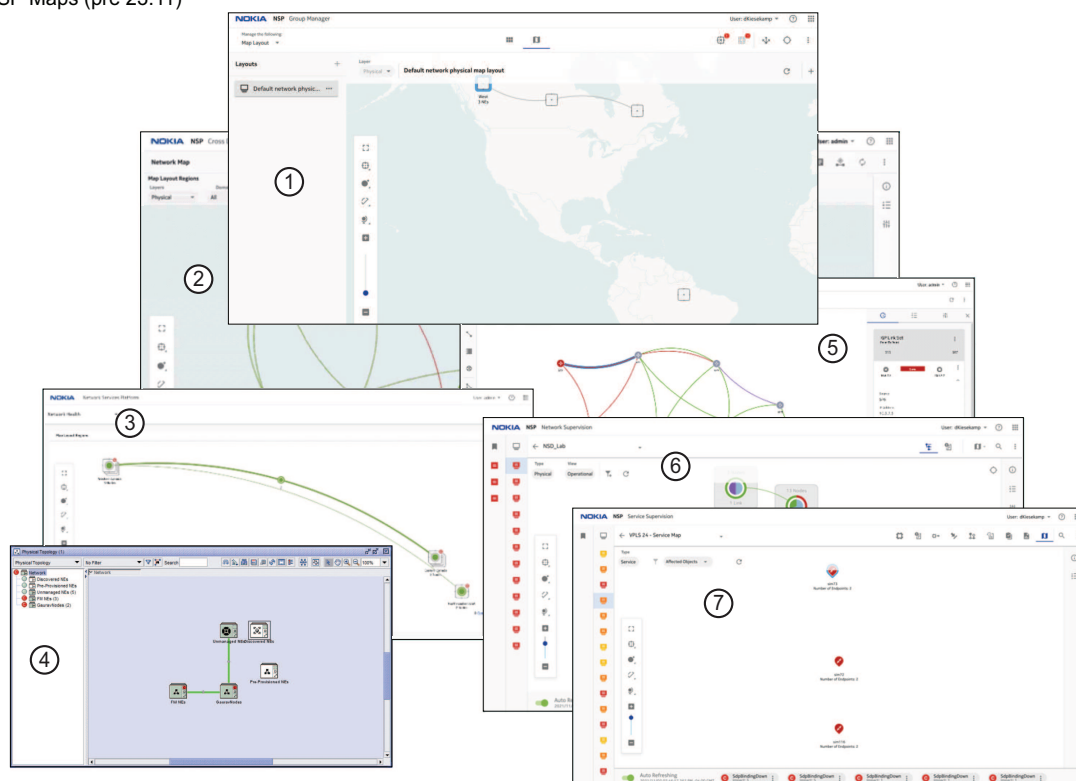
A.4.2 How have maps changed?

In NSP 23.11, all existing NSP maps have been consolidated to simplify and enhance users' experience through improved labeling and layout of information.

New navigation

Open Network Health View, Network Map View

NSP Maps (pre 23.11)



Network Map View (25.8)



40446

Figure legend: NSP Maps (pre 23.11)

1. Group Manager Map Layout
2. Cross Domain Coordinator Network Map
3. NSP Network Health
4. NFM-P GUI Physical Topology
5. IP/MPLS Optimization Network Map
6. Network Supervision Map
7. Service Supervision Service Map